Optimization

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Contents

Chapter 1

Existence of a minimizer

1.1 Hilbert spaces

Definition 1

A (real) **pre-Hilbertian space** $\mathcal{H}$ is a real vector space endowed with a function $\langle \cdot \mid \cdot \rangle : \mathcal{H}^2 \to \mathbb{R}$, called **inner product** such that

① $(\forall (x,y) \in \mathcal{H}^2)\ \langle x \mid y \rangle = \langle y \mid x \rangle$

② $(\forall (x,y,z) \in \mathcal{H}^3)\ \langle x + y \mid z \rangle = \langle x \mid z \rangle + \langle y \mid z \rangle$

③ $(\forall (x,y) \in \mathcal{H}^2)\ (\forall \alpha \in \mathbb{R})\ \langle \alpha x \mid y \rangle = \alpha \langle x \mid y \rangle$

④ $(\forall x \in \mathcal{H})\ \langle x \mid x \rangle \geq 0$ and $\langle x \mid x \rangle = 0 \Leftrightarrow x = 0$.

Property 1.

i) *The associated norm is*

$$(\forall x \in \mathcal{H}) \qquad \|x\| = \sqrt{\langle x \mid x \rangle}.$$

ii) *Parallelogram identity*

$$(\forall (x,y) \in \mathcal{H}^2) \qquad \|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|y\|^2).$$

Proof. We have, for every $(x,y) \in \mathcal{H}^2$,

$$\|x+y\|^2 + \|x-y\|^2 = \|x\|^2 + 2\langle x \mid y \rangle + \|y\|^2 + \|x\|^2 - 2\langle x \mid y \rangle + \|y\|^2.$$

□

iii) *Median identity*
Let $(a,b,c) \in \mathcal{H}^3$ and let m be the midpoint of $[b,c]$.

$$\|a-b\|^2 + \|a-c\|^2 = 2\|a-m\|^2 + \frac{1}{2}\|b-c\|^2.$$

Proof. We have, for every $(a, b, c) \in \mathcal{H}^3$,

$$\begin{aligned}&\|a-b\|^2 + \|a-c\|^2 \\ =&\|a-m\|^2 + 2\langle a-m \mid m-b\rangle + \|m-b\|^2 + \|a-m\|^2 + 2\langle a-m \mid m-c\rangle + \|m-c\|^2.\end{aligned}$$

Since $m = (b + c)/2$,

$$\|a-b\|^2+\|a-c\|^2 = 2\|a-m\|^2+\frac{1}{4}\|c-b\|^2+\|a-m\|^2+\frac{1}{4}\|b-c\|^2 = 2\|a-m\|^2+\frac{1}{2}\|b-c\|^2.$$

□

iv) *Cauchy-Schwarz inequality*

$$(\forall (x, y) \in \mathcal{H}^2) \quad |\langle x \mid y\rangle| \ \leq \|x\|\|y\|,$$

where the equality is reached if and only if $x = \alpha y$ with $\alpha \in \mathbb{R}$ or $y = 0$.

v) *$\langle \cdot \mid \cdot \rangle$ is a continuous function.*

Definition 2

A (real) **Hilbert space** $\mathcal{H}$ is a complete pre-Hilbertian space.

Example 1. *$\mathcal{H} = \mathbb{R}^N$ (Euclidean space with dimension N) equiped with the inner product*

$$(\forall x = (x^{(n)})_{1\leq n\leq N})(\forall y = (y^{(n)})_{1\leq n\leq N}) \quad \langle x \mid y\rangle = \sum_{n=1}^{N} x^{(n)} y^{(n)}.$$

Definition 3

Let $\mathcal{H}$ and $\mathcal{G}$ be two normed vector spaces. A linear operator $L\colon \mathcal{H} \to \mathcal{G}$ is **bounded** if

$$\|L\| = \sup_{\|x\|_{\mathcal{H}}\leq 1} \|Lx\|_{\mathcal{G}} < +\infty.$$

Remark 1.

i) *In the above definition, for simplicity sake, we often omit the subscript specifying the underlying Hilbert space involved in the used norms. We can equivalently define $\|L\|$ as*

$$\|L\| = \sup_{\|x\|\leq 1} \|Lx\| < +\infty$$

or

$$\|L\| = \sup_{\|x\|=1} \|Lx\| < +\infty$$

or

$$\|L\| = \sup_{x\in\mathcal{H}\setminus\{0\}} \frac{\|Lx\|}{\|x\|} < +\infty.$$

ii) *If $\mathcal{H}$ is finite dimensional, then L is bounded.*

iii) *A linear operator from $\mathcal{H}$ to $\mathcal{G}$ is continuous if and only if it is bounded.*

Notation 1

$\mathcal{B}(\mathcal{H}, \mathcal{G})$ is the Banach space of bounded linear operators from $\mathcal{H}$ to $\mathcal{G}$.

Definition 4

Let $\mathcal{H}$ and $\mathcal{G}$ be two Hilbert spaces. Let $L \in \mathcal{B}(\mathcal{H}, \mathcal{G})$. Its **adjoint** L^* is the operator in $\mathcal{B}(\mathcal{G}, \mathcal{H})$ defined as

$$(\forall (x, y) \in \mathcal{H} \times \mathcal{G}) \qquad \langle y \mid Lx \rangle_{\mathcal{G}} = \langle L^* y \mid x \rangle_{\mathcal{H}} .$$

Remark 2. *For simplicity sake, we often omit the subscript specifying the underlying Hilbert space and write the above equality as*

$$(\forall (x, y) \in \mathcal{H} \times \mathcal{G}) \qquad \langle y \mid Lx \rangle = \langle L^* y \mid x \rangle .$$

or, equivalently,

$$(\forall (x, y) \in \mathcal{H} \times \mathcal{G}) \qquad \langle Lx \mid y \rangle = \langle x \mid L^* y \rangle .$$

Example 2. *If*

$$L \colon \mathcal{H} \to \mathcal{H}^n \colon x \mapsto (x, \ldots, x),$$

then

$$L^* \colon \mathcal{H}^n \to \mathcal{H} \colon y = (y_1, \ldots, y_n) \mapsto \sum_{i=1}^n y_i.$$

Proof. For every $x \in \mathcal{H}$ and $y = (y_1, \ldots, y_n) \in \mathcal{H}^n$,

$$\langle Lx \mid y \rangle = \langle (x, \ldots, x) \mid (y_1, \ldots, y_n) \rangle = \sum_{i=1}^n \langle x \mid y_i \rangle = \left\langle x \mid \sum_{i=1}^n y_i \right\rangle .$$

□

Remark 3.

i) *We have $\|L^*\| = \|L\|$.*

ii) *According to the Banach inverse theorem, if L is bijective (i.e. an **isomorphism**) then $L^{-1} \in \mathcal{B}(\mathcal{G}, \mathcal{H})$ and $(L^{-1})^* = (L^*)^{-1}$.*

iii) *If $\mathcal{H} = \mathbb{R}^N$ and $\mathcal{G} = \mathbb{R}^M$, then $L^* = L^\top$.*

Proof. For every $(x, y) \in \mathcal{H} \times \mathcal{G}$,

$$\langle Lx \mid y \rangle = (Lx)^\top y = x^\top L^\top y = \langle x \mid L^* y \rangle = x^\top L^* y.$$

□

1.2 Notions of functional analysis

1.2.1 Some definitions

Definition 5

Let S be a nonempty set of a Hilbert space $\mathcal{H}$. Let $f\colon S \to [-\infty, +\infty]$.

- The (effective) **domain** of f is $\operatorname{dom} f = \{x \in S \mid f(x) < +\infty\}$.
- The function f is **proper** if $(\forall x \in S)\ f(x) \neq -\infty$ and $\operatorname{dom} f \neq \varnothing$.

Example 3.

i) *The domain of the function*

$$(\forall x \in \mathbb{R}) \quad f(x) = x^2$$

is $\mathbb{R}$. f *is thus proper.*

ii) *Let* $\delta \in \,]0, +\infty[$. *The domain of the function*

$$(\forall x \in \mathbb{R}) \quad f(x) = \begin{cases} \dfrac{1}{x} + x^2 & \textit{if } x \in]0, \delta] \\ +\infty & \textit{otherwise} \end{cases}$$

is $]0, \delta]$. f *is thus proper.*

Definition 6

Let $C \subset \mathcal{H}$. The **indicator function of** C is

$$(\forall x \in \mathcal{H}) \qquad \iota_C(x) = \begin{cases} 0 & \text{if } x \in C \\ +\infty & \text{otherwise.} \end{cases}$$

Remark 4. *We deduce from this definition that* $\operatorname{dom} \iota_C = C$, *and* ι_C *is proper if and only if* $C \neq \varnothing$.

1.2.2 Convergence in Hilbert spaces

Definition 7

Let $\mathcal{H}$ be a Hilbert space. Let $(x_n)_{n\in\mathbb{N}}$ be a sequence in $\mathcal{H}$ and $\widehat{x} \in \mathcal{H}$.

i) $(x_n)_{n\in\mathbb{N}}$ **converges strongly** to $\widehat{x}$ if

$$\lim_{n\to+\infty} \|x_n - \widehat{x}\| = 0.$$

It is denoted by $x_n \to \widehat{x}$.

ii) $(x_n)_{n\in\mathbb{N}}$ **converges weakly** to $\widehat{x}$ if

$$(\forall y \in \mathcal{H}) \qquad \lim_{n\to+\infty} \langle y \mid x_n - \widehat{x}\rangle = 0.$$

It is denoted by $x_n \rightharpoonup \widehat{x}$.

Remark 5.

i) $x_n \to \widehat{x} \Rightarrow x_n \rightharpoonup \widehat{x}$.

Proof. If $x_n \to \widehat{x}$ then it follows from Cauchy-Schwarz inequality that, for every $y \in \mathcal{H}$,

$$|\langle y \mid x_n - \widehat{x}\rangle| \le \|y\|\|x_n - \widehat{x}\| \to 0.$$

□

ii) *If $x_n \rightharpoonup \widehat{x}$ and $\|x_n\| \to \|\widehat{x}\|$, then $x_n \to \widehat{x}$.*

Proof. We have

$$\|x_n - \widehat{x}\|^2 = \|x_n\|^2 + \|x\|^2 - 2\langle x_n \mid \widehat{x}\rangle \to 0. \tag{1.1}$$

□

iii) *Every weakly convergent sequence is bounded (Banach-Steinhaus theorem).*

iv) *In a finite dimensional Hilbert space, strong and weak convergences are equivalent.*

Definition 8

Let S be a subset of a Hilbert space $\mathcal{H}$.

i) S is **bounded** if it is included in a ball.

ii) S is **closed** if the limit of every converging sequence of elements of S belongs to S.

iii) S is **compact** if, from every sequence $(x_n)_{n\in\mathbb{N}}$ of S, one can extract a subsequence $(x_{n_k})_{k\in\mathbb{N}}$ which converges to a point of S.

Remark 6.

i) *If S is compact, then it is closed and bounded.*

ii) *The converse property holds when $\mathcal{H}$ is finite dimensional.*

iii) *A subset of a compact set is compact if and only if it is closed.*

Definition 9

Let $(\xi_n)_{n\in\mathbb{N}}$ be a sequence of elements in $[-\infty,+\infty]$.
Its **infimum limit** is $\liminf \xi_n = \lim_{n\to+\infty} \inf\left\{\xi_k \mid k \geq n\right\} \in [-\infty,+\infty]$
and its **supremum limit** is $\limsup \xi_n = \lim_{n\to+\infty} \sup\left\{\xi_k \mid k \geq n\right\} \in [-\infty,+\infty]$.

Property 2.

i) $\limsup \xi_n = -\liminf(-\xi_n)$

ii) $\liminf \xi_n \leq \limsup \xi_n$

iii) $\lim_{n\to+\infty} \xi_n = \overline{\xi} \in [-\infty,+\infty]$ *if and only if* $\liminf \xi_n = \limsup \xi_n = \overline{\xi}$.

Lemma 1

Let $(\xi_n)_{n\in\mathbb{N}}$ be a sequence of elements in $[-\infty,+\infty]$.
Let $\mu = \liminf \xi_n \in [-\infty,+\infty]$.
Then there exists a subsequence $(\xi_{n_k})_{k\in\mathbb{N}}$ converging to μ.

Proof. Result from topology. □

Definition 10

Let $f : \mathcal{H} \to [-\infty,+\infty]$. The **epigraph** of f is

$$\operatorname{epi} f = \left\{(x,\zeta) \in \operatorname{dom} f \times \mathbb{R} \mid f(x) \leq \zeta\right\}$$

Example 4.

i) *Let*

$$(\forall x \in \mathbb{R}) \quad f(x) = |x|.$$

Then, its epigraph is the closed set

$$\operatorname{epi} f = \left\{(x,\zeta) \in \mathbb{R}^2 \mid -\zeta \leq x \leq \zeta\right\}.$$

ii) *The epigraph of $\iota_{]\delta_1,\delta_2]}$ with $(\delta_1,\delta_2) \in \mathbb{R}^2$, $\delta_1 < \delta_2$, is*

$$\operatorname{epi} \iota_{]\delta_1,\delta_2]} =]\delta_1,\delta_2] \times [0,+\infty[.$$

Definition 11

Let $f : \mathcal{H} \to]-\infty,+\infty]$.
f is a **lower semi-continuous** (l.s.c.) function at $x \in \mathcal{H}$ if, for every sequence $(x_n)_{n\in\mathbb{N}}$

of $\mathcal{H}$,

$$x_n \to x \quad \Rightarrow \quad \liminf f(x_n) \geq f(x).$$

Proposition 1

Let $f : \mathcal{H} \to\ ]-\infty, +\infty]$.
f is a lower semi-continuous function on $\mathcal{H}$ if and only if $\operatorname{epi} f$ *is closed*

Proof.

- Let us first show that, if f is a l.s.c. function on $\mathcal{H}$, then $\operatorname{epi} f$ is closed.
  Let $(x_n, \xi_n)_{n\in\mathbb{N}}$ be a sequence of $\operatorname{epi} f$. We have, for every $n \in \mathbb{N}$, $f(x_n) \leq \xi_n$.
  Assume that that $(x_n, \xi_n) \to (x, \xi)$. Then

  $$f(x) \leq \liminf f(x_n) \leq \liminf \xi_n = \lim_{n\to+\infty} \xi_n = \xi.$$

  This shows that $(x, \xi) \in \operatorname{epi} f$. Hence $\operatorname{epi} f$ is closed.

- Let us now prove the converse property.
  Let $x \in \mathcal{H}$, let $(x_n)_{n\in\mathbb{N}}$ be a sequence of $\mathcal{H}$ converging to x, and let $\mu = \liminf f(x_n)$. According to Lemma 1, there exists a subsequence $(x_{n_k})_{k\in\mathbb{N}}$ of $(x_n)_{n\in\mathbb{N}}$ such that $\lim_{k\to+\infty} f(x_{n_k}) = \mu$.
  Assume that $\mu \neq +\infty$ and let $\xi \in]\mu, +\infty[$. There exists $K \in \mathbb{N}$ such that, for every $k \geq K$, $f(x_{n_k}) \leq \xi$. Thus $(x_{n_k}, \xi)_{k\geq K}$ is a sequence of $\operatorname{epi} f$ converging to (x, ξ).
  Since $\operatorname{epi} f$ is closed, $(x, \xi) \in \operatorname{epi} f$, that is $\xi \geq f(x)$. Since the inequality holds for every $\xi > \mu$, $\mu \geq f(x)$.
  The last inequality obviously also holds when $\mu = +\infty$, which shows that f is l.s.c. at x.

□

Example 5. *Let $(\delta_1, \delta_2) \in \mathbb{R}^2$ with $\delta_1 < \delta_2$.*
$\iota_{]\delta_1,\delta_2]}$ is not lower semi-continuous, whereas $\iota_{[\delta_1,\delta_2]}$ is lower semi-continuous.
Indeed, $\operatorname{epi} \iota_{[\delta_1,\delta_2]} = [\delta_1, \delta_2] \times \{0\}$ *is closed, whereas* $\operatorname{epi} \iota_{]\delta_1,\delta_2]} =]\delta_1, \delta_2] \times \{0\}$ *is not.*

Property 3.

i) *Every continuous function on $\mathcal{H}$ is l.s.c.*

ii) *Every finite sum of l.s.c. functions is l.s.c.*

Proof. For every $j \in \{1, \dots, J\}$ with $J \in \mathbb{N}^*$, let $f_j : \mathcal{H} \to\ ]-\infty, +\infty]$ be l.s.c. Let $x \in \mathcal{H}$ and let $(x_k)_{k\in\mathbb{N}}$ be a sequence of $\mathcal{H}$ such that $x_k \to x$. Then

$$\liminf \sum_{j=1}^{J} f_j(x_k) \geq \sum_{j=1}^{J} \liminf f_j(x_k) \geq \sum_{j=1}^{J} f_j(x).$$

□

iii) *Let* $(f_i)_{i\in I}$ *be a family of l.s.c functions.*
Then, $\overline{f} = \sup_{i\in I} f_i$ *is l.s.c.*

Proof. $\operatorname{dom} \overline{f} = \big\{x \in \mathcal{H} \;\big|\; (\forall i \in I)\, f_i(x) < +\infty\big\} = \bigcap_{i\in I} \operatorname{dom} f_i$.
In addition,

$$\begin{aligned}
\operatorname{epi} \overline{f} &= \big\{(x,\zeta) \in \operatorname{dom} \overline{f} \times \mathbb{R} \;\big|\; \sup_{i\in I} f_i(x) \le \zeta\big\} \\
&= \big\{(x,\zeta) \in (\cap_{i\in I} \operatorname{dom} f_i) \times \mathbb{R} \;\big|\; (\forall i \in I)\, f_i(x) \le \zeta\big\} \\
&= \bigcap_{i\in I} \operatorname{epi} f_i.
\end{aligned}$$

Since $\operatorname{epi} \overline{f}$ is an intersection of closed sets, it is closed. □

Note that a property similar to iii) is satisfied by continuous functions when I is a discrete set.

1.3 Minimizers

Definition 12

Let S be a nonempty set of a Hilbert space $\mathcal{H}$.
Let $f \colon S \to \,]-\infty,+\infty]$ be a proper function and let $\widehat{x} \in S$.

i) $\widehat{x}$ is a **local minimizer** of f if $\widehat{x} \in \operatorname{dom} f$ and there exists an open neighborhood O of $\widehat{x}$ such that
$$(\forall x \in O \cap S) \qquad f(\widehat{x}) \le f(x).$$

ii) $\widehat{x}$ is a **(global) minimizer** of f if
$$(\forall x \in S) \qquad f(\widehat{x}) \le f(x).$$

iii) $\widehat{x}$ is a strict local minimizer of f if there exists an open neighborhood O of $\widehat{x}$ such that
$$(\forall x \in (O \cap S) \setminus \{\widehat{x}\}) \qquad f(\widehat{x}) < f(x).$$

iv) $\widehat{x}$ is a strict (global and unique) minimizer of f if
$$(\forall x \in S \setminus \{\widehat{x}\}) \qquad f(\widehat{x}) < f(x).$$

Theorem 1. (Weierstrass)

Let S *be a nonempty compact set of a Hilbert space* $\mathcal{H}$*. Let* $f : S \to \,]-\infty,+\infty]$ *be a proper l.s.c function.*

Then, there exists $\widehat{x} \in S$ *such that*

$$f(\widehat{x}) = \inf_{x \in S} f(x).$$

In addition, the set of minimizers of f *is a compact set.*

Proof. Let $\delta = \inf_{x \in S} f(x)$. We have $\delta \in [-\infty, +\infty[$ and there exists a sequence $(x_n)_{n\in\mathbb{N}}$ of elements of S such that $\lim_{n\to+\infty} f(x_n) = \delta$.
Since S is compact, there exists a subsequence $(x_{k_n})_{n\in\mathbb{N}}$ of $(x_n)_{n\in\mathbb{N}}$ such that $x_{k_n} \to \widehat{x} \in S$.
Then, $f(\widehat{x}) \leq \liminf f(x_{k_n}) = \delta$ and, consequently, $f(\widehat{x}) = \delta \in \mathbb{R}$.

Let $(\widehat{x}_n)_{n\in\mathbb{N}}$ be a sequence of minimizers of f such that $\widehat{x}_n \to \widehat{x}$. Then $\widehat{x} \in S$ and

$$f(\widehat{x}) \leq \liminf f(\widehat{x}_n) = \delta.$$

This shows that $\mathrm{Argmin} f$ is a closed subset of S, i.e. a compact subset of S. □

Definition 13

Let $\mathcal{H}$ be a Hilbert space. Let $f \colon \mathcal{H} \to\,]-\infty, +\infty]$.
f is **coercive** if $\lim_{\|x\|\to+\infty} f(x) = +\infty$.

Example 6.

i) *The function defined as*

$$(\forall x \in \mathbb{R}) \quad f(x) = \begin{cases} 1/x & \text{if } x > 0 \\ +\infty & \text{otherwise} \end{cases}$$

is not coercive.

ii) *The function defined as*

$$(\forall x \in \mathbb{R}) \quad f(x) = x^2$$

is coercive.

iii) $\iota_{[\delta_1,\delta_2]}$ *where* $(\delta_1, \delta_2) \in \mathbb{R}^2$ *with* $\delta_1 < \delta_2$ *is coercive*

Theorem 2

Let $\mathcal{H}$ *be a finite dimensional Hilbert space.*
Let $f : \mathcal{H} \to\,]-\infty, +\infty]$ *be a proper l.s.c. coercive function.*
Then, the set of minimizers of f *is a nonempty compact set.*

Proof. Since f is proper, there exists $x_0 \in \mathcal{H}$ such that $f(x_0) \in \mathbb{R}$. The coercivity of f implies that there exists $\eta \in \,]0, +\infty[$ such that, for every $x \in \mathcal{H}$ satisfying $\|x - x_0\| > \eta$, $f(x) > f(x_0)$. Let

$$S = \{x \in \mathcal{H} \mid \|x - x_0\| \le \eta\},$$

$S \cap \operatorname{dom} f \neq \varnothing$ and S is compact. Then, according to Weierstrass theorem, there exists $\widehat{x} \in S$ such that $f(\widehat{x}) = \inf_{x \in S} f(x) \le f(x_0)$. Thus, $f(\widehat{x}) = \inf_{x \in \mathcal{H}} f(x)$.
Since $\operatorname{Argmin} f \subset S$, it follows from the last part of Weierstrass theorem, that $\operatorname{Argmin} f$ is compact. □

Example 7. *When the conditions of Theorems 1 or 2 are not met, the existence of a minimizer of a cost function may not be guaranteed as shown by the following counter-example from the domain of statistics.*
Let $(x^{(k)})_{1 \le k \le K}$ be $K \in \mathbb{N} \setminus \{0\}$ observed real values, which are realizations of independent random variables identically distributed according to a Gaussian mixture model. The corresponding probability density function reads

$$(\forall \xi \in \mathbb{R}) \quad \mathsf{p}(\xi; \mu, \sigma) = \frac{1}{2\sqrt{2\pi}} \exp\Big(-\frac{\xi^2}{2}\Big) + \frac{1}{2\sqrt{2\pi}\sigma} \exp\Big(-\frac{(\xi - \mu)^2}{2\sigma^2}\Big)$$

where $\mu \in \mathbb{R}$ and $\sigma \in \,]0, +\infty[$ are unknown parameters.
The ***maximum likelihood estimator*** *of the parameters is typically defined by solving the optimization problem:*

$$\underset{\mu \in \mathbb{R}, \sigma \in]0,+\infty[}{\text{minimize}} \quad f(\mu, \sigma)$$

where

$$f(\mu, \sigma) = -\sum_{k=1}^{K} \log \mathsf{p}(x^{(k)}; \mu, \sigma).$$

However, there does not exist $(\widehat{\mu}, \widehat{\sigma}) \in \mathbb{R} \times \,]0, +\infty[$ minimizing f since $\lim_{\substack{\sigma \to 0 \\ \sigma > 0}} f(x^{(1)}, \sigma) = -\infty$.

1.4 Differential tools

1.4.1 Gâteaux differential

Definition 14

Let $\mathcal{H}$ and $\mathcal{K}$ be normed spaces. Let $C \subset \mathcal{H}$ and let x be an interior point of C. Let $T \colon C \to \mathcal{K}$.

- T is **Gâteaux differentiable** at x if there exists $T'(x) \in \mathcal{B}(\mathcal{H}, \mathcal{K})$ such that

$$(\forall y \in \mathcal{H}) \qquad T'(x)y = \lim_{\substack{\alpha \to 0 \\ \alpha > 0}} \frac{T(x + \alpha y) - T(x)}{\alpha}.$$

- $T'(x)$ is called the **Gâteaux derivative** of T at x
  $T'(x)y$ is called the **Gâteaux differential** of T at x in the direction y

Property 4.

i) *As a limit, the Gâteaux derivative is uniquely defined.*

ii) *If T is Gâteaux differentiable at x, then*

$$(\forall y \in \mathcal{H}) \qquad T'(x)y = \lim_{\substack{\alpha \to 0 \\ \alpha \neq 0}} \frac{T(x+\alpha y) - T(x)}{\alpha}$$

Proof. For every $y \in \mathcal{H}$,

$$\begin{aligned} \lim_{\substack{\alpha \to 0 \\ \alpha < 0}} \frac{T(x+\alpha y) - T(x)}{\alpha} &= \lim_{\substack{\alpha \to 0 \\ \alpha > 0}} \frac{T(x-\alpha y) - T(x)}{-\alpha} \\ &= -T'(x)(-y) = T'(x)y. \end{aligned}$$

□

iii) *If $\mathcal{H}$ is a Hilbert space, $\mathcal{K} = \mathbb{R}$, and T is Gâteaux differentiable at x, there exists a unique vector $\nabla T(x)$ in $\mathcal{H}$, called the* ***gradient*** *of T at x, such that $(\forall y \in \mathcal{H})\ T'(x)y = \langle \nabla T(x) \mid y \rangle$.*

Proof. Consequence of Riesz representation theorem. □

1.4.2 Fréchet differential

Definition 15

Let $\mathcal{H}$ and $\mathcal{K}$ be normed spaces. Let x be an interior point of $C \subset \mathcal{H}$. Let $T \colon C \to \mathcal{K}$.

- T is **Fréchet differentiable** at x if there exists $T'(x) \in \mathcal{B}(\mathcal{H}, \mathcal{K})$ such that

$$\lim_{\substack{y \to 0 \\ y \in \mathcal{H} \setminus \{0\}}} \frac{\|T(x+y) - T(x) - T'(x)y\|}{\|y\|} = 0.$$

- $T'(x)$ is called the **Fréchet derivative** of T at x.
  $T'(x)y$ is called the **Fréchet differential** of T at x with increment y.

Property 5.

i) *If T is Fréchet differentiable at x, then it is Gâteaux differentiable at x and its Gâteaux and Fréchet derivatives are equal.*

Proof. If T is Fréchet differentiable at x, then

$$
\begin{aligned}
(\forall y \in \mathcal{H} \setminus \{0\}) \qquad & \lim_{\substack{\alpha \to 0 \\ \alpha > 0}} \frac{\|T(x+\alpha y) - T(x) - T'(x)(\alpha y)\|}{\|\alpha y\|} = 0 \\
\Leftrightarrow\ & \lim_{\substack{\alpha \to 0 \\ \alpha > 0}} \frac{T(x+\alpha y) - T(x) - \alpha T'(x)(y)}{\alpha \|y\|} = 0 \\
\Leftrightarrow\ & \lim_{\substack{\alpha \to 0 \\ \alpha > 0}} \frac{T(x+\alpha y) - T(x)}{\alpha} - T'(x)(y) = 0.
\end{aligned}
$$

□

ii) *The Fréchet derivative is thus uniquely defined.*

iii) *If T is Fréchet differentiable at x, then T is continuous at x.*

Proof. For every $y \in \mathcal{H}$ such that $x + y \in C$,

$$
\begin{aligned}
\|T(x+y) - T(x)\| &\leq \|T(x+y) - T(x) - T'(x)y\| + \|T'(x)y\| \\
&\leq o(\|y\|) + \|T'(x)\|\|y\|.
\end{aligned}
$$

□

iv) ***(Mean value inequality)*** *Let $\mathcal{H}$ and $\mathcal{K}$ be normed spaces. Let $C \subset \mathcal{H}$ be an open set and let $T\colon C \to \mathcal{K}$ be continuous on $[x, y] \subset C$ and Gâteaux differentiable on $]x, y[\neq \varnothing$. Then there exists $\alpha \in]0, 1[$ such that*

$$\|T(y) - T(x)\| \leq \|y - x\|\, \|T'(x + \alpha(y - x))\|.$$

Proof. Let us prove the result when $\mathcal{K}$ is a Hilbert space. Let $z \in \mathcal{K}$ and define $\varphi\colon [0,1] \to \mathbb{R}\colon \beta \mapsto \langle z \mid T(x + \beta(y - x))\rangle$. Then φ is continuous on $[0, 1]$ and differentiable on $]0, 1[$ and the derivative of φ is

$$
\begin{aligned}
(\forall \beta \in]0,1[) \quad \varphi'(\beta) &= \lim_{\substack{\delta \to 0 \\ \delta \neq 0}} \frac{\varphi(\beta+\delta) - \varphi(\delta)}{\delta} \\
&= \lim_{\substack{\delta \to 0 \\ \delta \neq 0}} \left\langle z \mid \frac{T(x + (\beta+\delta)(y-x)) - T(x + \beta(y-x))}{\delta} \right\rangle \\
&= \langle z \mid T'(x + \beta(y-x))(y-x)\rangle.
\end{aligned}
$$

By applying the mean value theorem to φ, we know that there exists $\alpha \in]0, 1[$ such that

$$
\begin{aligned}
& \varphi(1) - \varphi(0) = \varphi'(\alpha) \\
\Leftrightarrow\quad & \langle z \mid T(y) - T(x)\rangle = \langle z \mid T'(x + \alpha(y-x))(y-x)\rangle.
\end{aligned}
$$

By choosing z with unit norm such that $z = \eta(T(y) - T(x))$ where $\eta > 0$,

$$
\begin{aligned}
& \|T(y) - T(x)\| = \langle z \mid T'(x + \alpha(y-x))(y-x)\rangle \leq \|T'(x + \alpha(y-x))(y-x)\| \\
\Rightarrow\quad & \|T(y) - T(x)\| \leq \|T'(x + \alpha(y-x))\|\ \|y - x\|.
\end{aligned}
$$

□

v) ***(Chain rule)*** *Let $\mathcal{H}$, $\mathcal{G}$, and $\mathcal{K}$ be normed spaces. Let $C \subset \mathcal{H}$ and $D \subset \mathcal{G}$. Let $T \colon C \to D$ and $S \colon D \to \mathcal{K}$. Let $x \in \operatorname{int}(C)$ be such that $T(x) \in \operatorname{int}(D)$. If T is Fréchet differentiable at x and S is Fréchet (resp. Gâteaux) differentiable at $T(x)$, then $S \circ T$ is Fréchet (resp. Gâteaux) differentiable at x and*

$$(S \circ T)'(x) = S'(T(x)) \circ T'(x).$$

Proof. Let us prove the result in the case of the Fréchet differentiability.
Let $h \in \mathcal{H}$ be such that $x + h \subset \operatorname{int}(C)$. It follows from the Fréchet differentiability of T at x that

$$T(x+h) = T(x) + T'(x)h + o(\|h\|).$$

So, when $\|h\|$ is small enough, $T(x+h) \in \operatorname{int}(D)$ and

$$\begin{aligned} S(T(x+h)) &= S\big(T(x) + T'(x)h + o(\|h\|)\big) \\ &= S(T(x)) + S'(T(x))(T'(x)h + o(\|h\|)) + o\big(T'(x)h + o(\|h\|)\big). \end{aligned}$$

Because of the linearity of $S'(T(x))$,

$$S(T(x+h)) = S(T(x)) + S'(T(x))(T'(x)h) + S'(T(x))(o(\|h\|)) + o\big(T'(x)h + o(\|h\|)\big).$$

Since both $S'(T(x))$ and $T'(x)$ are bounded linear operators,

$$S'(T(x))(o(\|h\|)) + o\big(T'(x)h + o(\|h\|)\big) = o(\|h\|).$$

Since $S'(T(x)) \circ T'(x)$ is a composition of bounded linear operators, it is a bounded linear operator. We conclude that $S'(T(x))(T'(x)h) = (S'(T(x)) \circ T'(x))h$ is the Fréchet differential of $S \circ T$ at x with increment h. ☐

Example 8.

i) *When $\mathcal{H} = \mathbb{R}$ and $\mathcal{K} = \mathbb{R}^M$, Fréchet/Gâteaux differentiability is equivalent to standard derivability.*

ii) *Let $C \subset \mathbb{R}^N$. Let $T \colon C \to \mathbb{R}^M \colon x = (x^{(i)})_{1 \le i \le N} \mapsto (T^{(j)}(x))_{1 \le j \le M}$ where, for every $j \in \{1, \ldots, M\}$, $T^{(j)} \colon C \to \mathbb{R}$. Let $\overline{x}$ be an interior point of C. If, for every $j \in \{1, \ldots, M\}$ $T^{(j)}$ admits continuous partial derivatives in a neighborhood of $\overline{x}$, then T is Fréchet differentiable at $\overline{x}$ and*

$$(\forall y \in \mathbb{R}^N) \qquad T'(\overline{x})y = \begin{bmatrix} \frac{\partial T^{(1)}}{\partial x^{(1)}}(\overline{x}) & \ldots & \frac{\partial T^{(1)}}{\partial x^{(N)}}(\overline{x}) \\ \vdots & & \vdots \\ \frac{\partial T^{(M)}}{\partial x^{(1)}}(\overline{x}) & \ldots & \frac{\partial T^{(M)}}{\partial x^{(N)}}(\overline{x}) \end{bmatrix} y.$$

iii) *Let $\mathcal{H}$ and $\mathcal{K}$ be normed spaces. Let $T \in \mathcal{B}(\mathcal{H}, \mathcal{K})$. Then T is Fréchet differentiable at every $x \in \mathcal{H}$ and $T'(x) = T$.*

Proof. For every $y \in \mathcal{H}$,

$$T(x+y) - T(x) - T(y) = 0.$$

In addition, $T \in \mathcal{B}(\mathcal{H}, \mathcal{K})$. ☐

iv) *Let $\mathcal{H}$ and $\mathcal{K}$ be a normed spaces. Let $B\colon \mathcal{H}^2 \to \mathcal{K}$ be a bilinear function such that*

$$\sup_{\substack{(x_1,x_2)\in\mathcal{H}^2\\ x_1\neq 0, x_2\neq 0}} \frac{\|B(x_1,x_2)\|}{\|x_1\|\|x_2\|} = M_B \in \mathbb{R}$$

and let $T\colon \mathcal{H} \to \mathcal{K}\colon x \mapsto B(x,x)$. Then T is Fréchet differentiable at every $x \in \mathcal{H}$ and

$$(\forall y \in \mathcal{H}) \quad T'(x)y = B(x,y) + B(y,x).$$

Proof. For every $y \in \mathcal{H}$,

$$\begin{aligned}&\|T(x+y) - T(x) - B(x,y) - B(y,x)\|\\ =&\|B(x+y,x+y) - B(x,x) - B(x,y) - B(y,x)\|\\ =&\|B(y,y)\| \le M_B\|y\|^2\end{aligned}$$

Thus,

$$\lim_{\substack{y\to 0\\ y\in\mathcal{H}\setminus\{0\}}} \frac{\|T(x+y) - T(x) - B(x,y) - B(y,x)\|}{\|y\|} = 0.$$

In addition, $y \mapsto B(x,y) + B(y,x)$ is linear and, for every $y \in \mathcal{H}$,

$$\|B(x,y) + B(y,x)\| \le \|B(x,y)\| + \|B(y,x)\| \le 2M_B\|x\|\|y\|,$$

which shows that it is bounded. □

v) *Let $\mathcal{H}$ be a real Hilbert spaces and let $L \in \mathcal{B}(\mathcal{H},\mathcal{H})$.*
The function defined as

$$(\forall x \in \mathcal{H}) \quad T(x) = \langle x \mid Lx\rangle$$

is Fréchet differentiable on $\mathcal{H}$ and

$$(\forall x \in \mathcal{H}) \quad \nabla T(x) = (L + L^*)x. \tag{1.2}$$

Proof. Let $B\colon \mathcal{H}^2 \to \mathbb{R}\colon (x_1,x_2) \mapsto \langle x_1 \mid Lx_2\rangle$.
B is bilinear and, according to the Cauchy-Schwarz inequality,

$$\begin{aligned}&(\forall (x_1,x_2) \in \mathcal{H}^2) \quad |B(x_1,x_2)| \le \|x_1\|\|Lx_2\| \le \|L\|\|x_1\|\|x_2\|\\ \Rightarrow\quad &\sup_{\substack{(x_1,x_2)\in\mathcal{H}^2\\ x_1\neq 0, x_2\neq 0}} \frac{|B(x_1,x_2)|}{\|x_1\|\|x_2\|} \le \|L\|.\end{aligned}$$

Since $(\forall x \in \mathcal{H})$ $T(x) = B(x,x)$, then T is Fréchet differentiable at every $x \in \mathcal{H}$ and, for every $y \in \mathcal{H}$,

$$\begin{aligned}&T'(x)y = B(x,y) + B(y,x) = \langle x \mid Ly\rangle + \langle y \mid Lx\rangle = \langle L^*x \mid y\rangle + \langle y \mid Lx\rangle\\ \Leftrightarrow\quad &\langle \nabla T(x) \mid y\rangle = \langle (L^* + L)x \mid y\rangle.\end{aligned}$$

□

1.4.3 Twice differentiable functions

Definition 16

Let $\mathcal{H}$ be a Hilbert space. Let $C \subset \mathcal{H}$ be an open set.
Let $T\colon C \to \mathbb{R}$ be Fréchet differentiable on C.

- T is **twice Fréchet differentiable** at $x \in C$ if ∇T is Fréchet differentiable at x, i.e., there exists $\nabla^2 T(x) \in \mathcal{B}(\mathcal{H},\mathcal{H})$ such that

$$\lim_{\substack{y\to 0\\ y\in\mathcal{H}\setminus\{0\}}} \frac{\|\nabla T(x+y) - \nabla T(x) - (\nabla^2 T(x))y\|}{\|y\|} = 0.$$

- $\nabla^2 T(x)$ is called the **Hessian** of T at x.

Example 9. *Let us come back to Example 8v). We have shown that T is Fréchet-differentiable on $\mathcal{H}$ and its gradient is the bounded linear operator given by (1.2). It thus follows from Example 8iii) that T is twice Fréchet-differentiable on $\mathcal{H}$ and its Hessian at every point $x \in \mathcal{H}$ is*

$$\nabla^2 T(x) = L + L^*.$$

Property 6.

i) *Let $C \subset \mathcal{H}$ be an open set.*
Let $T\colon C \to \mathbb{R}$ be Fréchet differentiable on C and twice Fréchet differentiable at $x \in C$. Then, for every $(y,z) \in \mathcal{H}^2$,

$$\begin{aligned}&\lim_{\substack{\alpha\to 0\\ \alpha\neq 0}} \frac{T(x+\alpha(y+z)) - T(x+\alpha y) - T(x+\alpha z) + T(x)}{\alpha^2}\\ &= \left\langle y \mid (\nabla^2 T(x))z \right\rangle. \end{aligned} \tag{1.3}$$

Proof. The property is obviously satisfied if $y = 0$ or $z = 0$.
Let us now assume that $y \neq 0$ and $z \neq 0$.
There exists $\rho > 0$ such that the open ball $\mathcal{B}(x,\rho)$ with center x and radius ρ is included in C. Let $\overline{\alpha} \in \,]0,+\infty[$ be such that

$$\overline{\alpha}(\|y\| + \|z\|) < \rho$$

and let $\alpha \in]-\overline{\alpha},\overline{\alpha}[$. Then $x+\alpha y$, $x+\alpha z$, and $x+\alpha(y+z)$ belong to $\mathcal{B}(x,\rho)$ and

$$\begin{aligned}&\frac{T(x+\alpha(y+z)) - T(x+\alpha y) - T(x+\alpha z) + T(x)}{\alpha^2} - \left\langle y \mid (\nabla^2 T(x))z \right\rangle\\ &= g_{\alpha,z}(y) - g_{\alpha,z}(0),\end{aligned}$$

where, for every $u \in \mathcal{H}$ such that $\|u\| \leq \|y\|$,

$$g_{\alpha,z}(u) = \frac{T(x+\alpha(u+z)) - T(x+\alpha u)}{\alpha^2} - \left\langle u \mid (\nabla^2 T(x))z \right\rangle.$$

Since T is Fréchet differentiable on C, $g_{\alpha,z}$ is continuous on $[0,y]$, Fréchet differentiable on $]0,y[$ and, for every $u \in]0,y[$,

$$\nabla g_{\alpha,z}(u) = \frac{\nabla T(x+\alpha(u+z)) - \nabla T(x+\alpha u)}{\alpha} - (\nabla^2 T(x))z.$$

According to the mean value inequality, there exists $\beta_\alpha \in]0,1[$ such that

$$|g_{\alpha,z}(y) - g_{\alpha,z}(0)| \leq \|y\| \left\| \frac{\nabla T(x+\alpha(\beta_\alpha y+z)) - \nabla T(x+\alpha\beta_\alpha y)}{\alpha} - (\nabla^2 T(x))z \right\|.$$

It follows from the definition of the Hessian that

$$\nabla T(x+\alpha\beta_\alpha y) - \nabla T(x) = \alpha(\nabla^2 T(x))\beta_\alpha y + \alpha\beta_\alpha \|y\|\, \epsilon_1(\alpha\beta_\alpha y)$$
$$\nabla T(x+\alpha(\beta_\alpha y+z)) - \nabla T(x) = \alpha(\nabla^2 T(x))(\beta_\alpha y+z) + \alpha\|\beta_\alpha y+z\|\, \epsilon_2\big(\alpha(\beta_\alpha y+z)\big).$$

where $\epsilon_i\colon \mathcal{H} \to \mathcal{H}$ such that $\epsilon_i(h) \to 0$ as $h \to 0$, for $i \in \{1,2\}$.
By combining the previous two relations, we deduce that

$$\alpha^{-1}\big(\nabla T(x+\alpha(\beta_\alpha y+z)) - \nabla T(x+\alpha\beta_\alpha y)\big) - (\nabla^2 T(x))z$$
$$= \|\beta_\alpha y+z\|\epsilon_2\big(\alpha(\beta_\alpha y+z)\big) - \beta_\alpha\|y\|\epsilon_1(\alpha\beta_\alpha y).$$

Since β_α lies in a bounded interval,

$$\lim_{\substack{\alpha\to 0\\ \alpha\neq 0}} \frac{\nabla T(x+\alpha(\beta_\alpha y+z)) - \nabla T(x+\alpha\beta_\alpha y)}{\alpha} - (\nabla^2 T(x))z = 0$$

which yields $\lim_{\substack{\alpha\to 0\\ \alpha\neq 0}} g_{\alpha,z}(y) - g_{\alpha,z}(0) = 0$.

□

ii) *If T is twice Fréchet differentiable at x, its Hessian at x is a self-adjoint operator.*

Proof. It follows from the symmetry of (1.3) that

$$(\forall (y,z) \in \mathcal{H}^2) \quad \big\langle y \mid (\nabla^2 T(x))z \big\rangle = \big\langle z \mid (\nabla^2 T(x))y \big\rangle,$$

which shows that $(\nabla^2 T(x))^* = \nabla^2 T(x)$. □

iii) ***Taylor-Young formula***
Let $\mathcal{H}$ be a Hilbert space. Let $C \subset \mathcal{H}$ be an open set, let $T\colon C \to \mathbb{R}$, let $x \in C$, and let $h \in \mathcal{H}$ be such that $x+h \in C$.
Assume that T is Fréchet differentiable on C and twice Fréchet differentiable at x. Then

$$T(x+h) = T(x) + \langle \nabla T(x) \mid h\rangle + \frac{1}{2}\big\langle h \mid (\nabla^2 T(x))h \big\rangle + \|h\|^2\epsilon(h)$$

where $\epsilon\colon \mathcal{H} \to \mathbb{R}$ is such that $\epsilon(h) \to 0$ as $h \to 0$.

iv) ***Taylor-Mc Laurin formula***
Let $\mathcal{H}$ be a Hilbert space. Let $C \subset \mathcal{H}$ be an open set and let $T\colon C \to \mathbb{R}$. Assume that T is Fréchet differentiable on C, ∇T is continuous on $[x, x+h] \subset C$, and T is twice Fréchet differentiable on $]x, x+h[$. Then

$$T(x+h) = T(x) + \langle \nabla T(x) \mid h\rangle + \frac{1}{2}\big\langle h \mid (\nabla^2 T(x+\alpha h))h \big\rangle$$

where $\alpha \in]0,1[$.

1.5 Existence of a local minimizer

Let us first state a necessary first-order optimality condition.

Theorem 3. (Euler's inequality)

Let D be an open subset of a normed space $\mathcal{H}$ and let $C \subset D$. Let $f\colon D \to]-\infty, +\infty]$ be Gâteaux differentiable at $\widehat{x} \in C$.
If $\widehat{x}$ is a local minimizer of f on C then, for every $y \in \mathcal{H}$ such that the segment $[\widehat{x}, y]$ is a subset of C,

$$f'(\widehat{x})(y - \widehat{x}) \geq 0.$$

If $\widehat{x} \in \operatorname{int}(C)$, then the condition reduces to

$$f'(\widehat{x}) = 0.$$

Proof. Let $y \in \mathcal{H}$ be such that $[\widehat{x}, y] \subset C$. There exists $\delta \in [0,1]$ such that, for every $\alpha \in]0, \delta]$,

$$(1-\alpha)\widehat{x} + \alpha y = \widehat{x} + \alpha(y - \widehat{x}) \in C$$

and $f\big(\widehat{x} + \alpha(y - \widehat{x})\big) \geq f(\widehat{x})$. Then,

$$f'(\widehat{x})(y - \widehat{x}) = \lim_{\substack{\alpha \to 0 \\ \alpha > 0}} \frac{f\big(\widehat{x} + \alpha(y - \widehat{x})\big) - f(\widehat{x})}{\alpha} \geq 0.$$

If $\widehat{x} \in \operatorname{int}(C)$ then, for every $z \in \mathcal{H}$, there exists $\delta \in]0, +\infty[$ such $\widehat{x} \in [\widehat{x} - \delta z, \widehat{x} + \delta z] \subset C$. Hence,

$$f'(\widehat{x})(\pm \delta z) \geq 0,$$

that is $f'(\widehat{x})z = 0$. Consequently, $f'(\widehat{x}) = 0$. □

Definition 17

A zero of f' is called a **critical (or stationary) point** of f.

Let us now give sufficient conditions for the existence of a minimizer.

Theorem 4

Let C be an open subset of a Hilbert space $\mathcal{H}$. Let $f\colon C \to \mathbb{R}$ be Fréchet differentiable on C.

i) *If f is twice Fréchet differentiable at $\widehat{x} \in C$, $\nabla f(\widehat{x}) = 0$, and*

$$(\exists \eta \in]0, +\infty[)(\forall h \in \mathcal{H}) \quad \big\langle h \mid (\nabla^2 f(\widehat{x}))h \big\rangle \geq \eta \|h\|^2, \tag{1.4}$$

then f has a strict local minimum at $\widehat{x}$.

ii) *If f is twice Fréchet differentiable on an open neighborhood $D \subset C$ of $\widehat{x}$, $\nabla f(\widehat{x}) = 0$,* *and*

$$(\forall x \in D)(\forall h \in \mathcal{H}) \quad \left\langle h \mid (\nabla^2 f(x))h \right\rangle \geq 0, \tag{1.5}$$

then f has a local minimum at $\widehat{x}$.

Proof.

i) Let $h \in \mathcal{H} \setminus \{0\}$. For $\|h\|$ small enough, $\widehat{x} + h \in C$ and, according to Taylor-Young formula,

$$f(\widehat{x} + h) = f(\widehat{x}) + \langle \nabla f(\widehat{x}) \mid h \rangle + \frac{1}{2} \left\langle h \mid (\nabla^2 f(\widehat{x}))h \right\rangle + \|h\|^2 \epsilon(h)$$

with $\epsilon(h) \to 0$ as $h \to 0$. We have then

$$f(\widehat{x} + h) - f(\widehat{x}) \geq \Big(\frac{\eta}{2} + \epsilon(h)\Big) \|h\|^2.$$

Hence, there exits $\delta \in \,]0, +\infty[$ such that, if $0 < \|h\| < \delta$, $\eta/2 + \epsilon(h) > 0$, hence $f(\widehat{x} + h) > f(\widehat{x})$.

ii) Let $B \subset D$ be an open ball centered at $\widehat{x}$.
According to Taylor-Mc Laurin formula,

$$(\forall x \in B) \qquad f(x) = f(\widehat{x}) + \langle \nabla f(\widehat{x}) \mid (x - \widehat{x}) \rangle + \frac{1}{2} \left\langle x - \widehat{x} \mid (\nabla^2 f(y))(x - \widehat{x}) \right\rangle,$$

where $y \in]\widehat{x}, x[$. We have thus

$$(\forall x \in B) \qquad f(x) = f(\widehat{x}) + \frac{1}{2} \left\langle x - \widehat{x} \mid (\nabla^2 f(y))(x - \widehat{x}) \right\rangle \geq f(\widehat{x}).$$

$\square$

Remark 7.

i) (1.4) *means that the Hessian at $\widehat{x}$ is a positive self-adjoint operator (symmetric positive definite in finite dimension),*

ii) (1.5) *means that, for every $x \in D$, the Hessian at x is a nonnegative self-adjoint operator (symmetric positive semidefinite in finite dimension).*

Chapter 2

Convexity

2.1 Convex sets

Definition 18

Let $\mathcal{H}$ be a Hilbert space. $C \subset \mathcal{H}$ is a **convex set** if

$$(\forall (x,y) \in C^2)(\forall \alpha \in]0,1[) \qquad \alpha x + (1-\alpha)y \in C.$$

Example 10.

i) *Let $\mathcal{H} = \mathbb{R}^N$ and let $M \in \mathbb{R}^{N\times N}$ be a definite positive matrix. The ellipsoid*

$$C = \left\{x \in \mathbb{R}^N \;\middle|\; x^\top M x \leq 1\right\}$$

is a convex set.

ii) *However, a union of ellipsoids is generally non convex.*

Property 7.

i) *$\varnothing$ is considered as a convex set.*

ii) *If C is a convex set, then $(\forall n \in \mathbb{N}^*)$ $(\forall (x_1,\ldots,x_n) \in C^n)$ $(\forall (\alpha_1,\ldots,\alpha_n) \in [0,+\infty[^n)$ with $\sum_{i=1}^{n} \alpha_i = 1$,*

$$\sum_{i=1}^{n} \alpha_i x_i \in C.$$

Proof. By induction. □

iii) *Every vector (affine) space is convex.*

iv) *If C is a convex set, then its interior* $\mathrm{int}\,(C)$ *and its closure* $\overline{C}$ *are convex sets.*

Proof.

a) Let $(x, y) \in \mathrm{int}\,(C)^2$ and $\alpha \in]0, 1[$. Then there is an open ball $B(x, \rho) \subset C$ centered at x, with radius $\rho \in \,]0, +\infty[$. We have

$$B(\alpha x + (1 - \alpha)y, \alpha\rho) = \{\alpha z + (1 - \alpha)y \mid z \in B(x, \rho)\} \subset C.$$

Thus $\alpha x + (1 - \alpha)y \in \mathrm{int}\,(C)$.

b) Let $(x, y) \in \overline{C}^2$ and $\alpha \in]0, 1[$. There exists a sequence $(z_n)_{n\in\mathbb{N}}$ of elements of C such that $z_n \to x$. Hence $(\alpha z_n + (1 - \alpha)y)_{n\in\mathbb{N}}$ is a sequence of elements of C converging to $\alpha x + (1 - \alpha)y$. This shows that $\alpha x + (1 - \alpha)y \in \overline{C}$.

□

Definition 19

A set S in a vector space is a **cone** (with vertex 0) if

$$(\forall x \in S)(\forall \alpha \in \,]0, +\infty[) \quad \alpha x \in S.$$

Property 8.

i) *The* ***nonnegative orthant*** $[0, +\infty[^N$ *is a convex cone of* $\mathbb{R}^N$.

ii) *If C is a convex set then, for every $\alpha \in \mathbb{R}$,*

$$\alpha C = \{\alpha x \mid x \in C\}$$

is a convex set.

iii) *If C_1 and C_2 are convex sets, then $C_1 \times C_2$ is a convex set.*

iv) *If C_1 and C_2 are convex sets of the same space*

$$C_1 + C_2 = \{x_1 + x_2 \mid (x_1, x_2) \in C_1 \times C_2\}$$

is a convex set.

v) *If $(C_i)_{i\in\mathcal{I}}$ is a family of convex sets of the given space, then* $\bigcap_{i\in I} C_i$ *is convex.*

Definition 20

A polyedron in a Hilbert space is the intersection of a finite number of closed half-space.

Example 11. *According to the above definition and Property 8iv), a polyhedron is a closed and convex set.*

Definition 21

Let $\mathcal{H}$ be a Hilbert space and $C \subset \mathcal{H}$. The **convex hull** of C is the smallest convex set including C. It is denoted by $\mathrm{conv}(C)$.

Property 9.

i) $\mathrm{conv}(C)$ *is the intersection of all the convex sets including* C.

ii) *Let* $x \in \mathcal{H}$. $x \in \mathrm{conv}(C)$ *if and only if there exist* $n \in \mathbb{N}^*$, $(x_1, \dots, x_n) \in C^n$, *and* $(\alpha_1, \dots, \alpha_n) \in\ ]0, +\infty[^n$ *with* $\sum_{i=1}^{n} \alpha_i = 1$ *such that*

$$x = \sum_{i=1}^{n} \alpha_i x_i.$$

Proof. The set of vectors satisfying this condition includes C and it is convex. By definition, it thus includes $\mathrm{conv}(C)$.
Conversely, assume that x is expressed as above. Since $(x_1, \dots, x_n) \in C^n \subset \mathrm{conv}(C)^n$ and $\mathrm{conv}(C)$ is convex, $x \in \mathrm{conv}(C)$. □

2.2 Convex functions

Definition 22

Let $\mathcal{H}$ be a Hilbert space.

i) $f : \mathcal{H} \to\ ]-\infty, +\infty]$ is a **convex function** if

$$\big(\forall (x, y) \in \mathcal{H}^2\big)(\forall \alpha \in]0, 1[) \quad f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha) f(y)$$

ii) $f : \mathcal{H} \to [-\infty, +\infty]$ is a **convex function** if

$$\big(\forall (x, y) \in (\mathrm{dom}\, f)^2\big)(\forall \alpha \in]0, 1[) \quad f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha) f(y)$$

Example 12.

i) *The function*

$$(\forall x \in \mathbb{R}) \quad f(x) = |x|$$

is convex.

ii) *The function*

$$(\forall x \in \mathbb{R}) \quad f(x) = \sqrt{|x|}$$

is non convex.

iii) *Let* $\delta \in \,]0,+\infty[$. *The function* $\iota_{[-\delta,\delta]}$ *is convex.*

Definition 23

$f\colon \mathcal{H} \to [-\infty,+\infty]$ is concave if $-f$ is convex.

Proposition 2

$f : \mathcal{H} \to [-\infty,+\infty]$ *is convex if and only if its epigraph is a convex set.*

Proof. Assume that $\operatorname{epi} f$ is a convex set. For every $(x,y) \in (\operatorname{dom} f)^2$ and $(\eta,\rho) \in \mathbb{R}^2$ such that $\eta \geq f(x)$ and $\rho \geq f(y)$, $(x,\eta) \in \operatorname{epi} f$ and $(y,\rho) \in \operatorname{epi} f$.
Then, for every $\alpha \in]0,1[$,

$$\alpha(x,\eta) + (1-\alpha)(y,\rho) = (\alpha x + (1-\alpha)y, \alpha\eta + (1-\alpha)\rho) \in \operatorname{epi} f,$$

which means that

$$f(\alpha x + (1-\alpha)y) \leq \alpha\eta + (1-\alpha)\rho.$$

By letting η tend to $f(x)$ and ρ tend to $f(y)$,

$$f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha)f(y).$$

Conversely, assume that f is convex. Let $(x,\eta) \in \operatorname{epi} f$ and $(y,\rho) \in \operatorname{epi} f$. Then, $f(x) \leq \eta$ and $f(y) \leq \rho$ and, for every $\alpha \in]0,1[$,

$$f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha)f(y) \leq \alpha\eta + (1-\alpha)\rho.$$

Therefore, $\alpha(x,\eta) + (1-\alpha)(y,\rho) \in \operatorname{epi} f$ and $\operatorname{epi} f$ is a convex set. □

Property 10.

i) *If* $f\colon \mathcal{H} \to [-\infty,+\infty]$ *is convex, then* $\operatorname{dom} f$ *is convex and the* ***lower level set*** *of* f *at height* $\eta \in \mathbb{R}$

$$\operatorname{lev}_{\leq \eta} f = \left\{ x \in \mathcal{H} \;\middle|\; f(x) \leq \eta \right\}$$

is a convex set.

Proof. Let $\eta \in \mathbb{R}$. For every $(x,y) \in (\operatorname{lev}_{\leq\eta} f)^2$,

$$f(x) \leq \eta$$
$$f(y) \leq \eta.$$

Thus, for every $\alpha \in]0,1[$,

$$f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha)f(y) \leq \eta,$$

which shows that $\alpha x + (1-\alpha)y \in \operatorname{lev}_{\leq\eta} f$.
$\operatorname{dom} f$ can be viewed as the limit case of $\operatorname{lev}_{\leq\eta} f$ when $\eta \to +\infty$. □

ii) *$f\colon \mathcal{H} \to]-\infty,+\infty]$ is convex if and only if $(\forall (x,y) \in (\operatorname{dom} f)^2)$ $\varphi_{x,y}\colon [0,1] \to]-\infty,+\infty]\colon \alpha \mapsto f(\alpha x + (1-\alpha)y)$ is convex.*

Proof. For every $(x,y) \in (\operatorname{dom} f)^2$, assume that $\varphi_{x,y}$ is convex. Then, for every $\alpha \in]0,1[$,

$$\begin{aligned} f(\alpha x + (1-\alpha)y) &= \varphi_{x,y}(\alpha) \\ &= \varphi_{x,y}(\alpha 1 + (1-\alpha)0) \\ &\leq \alpha\varphi_{x,y}(1) + (1-\alpha)\varphi_{x,y}(0) \\ &= \alpha f(x) + (1-\alpha) f(y) \end{aligned}$$

which shows that f is convex.
Conversely, assume that $f\colon \mathcal{H} \to]-\infty,+\infty]$ is convex. Then, for every $(x,y) \in (\operatorname{dom} f)^2$ and $(\alpha,\beta,\lambda) \in [0,1]^3$, we have $\lambda\alpha + (1-\lambda)\beta \in [0,1]$ and

$$\begin{aligned} \varphi_{x,y}(\lambda\alpha + (1-\lambda)\beta) &= f\big((\lambda\alpha + (1-\lambda)\beta)x + (1-\lambda\alpha-(1-\lambda)\beta)y\big) \\ &= f\big(\lambda(\alpha x + (1-\alpha)y) + (1-\lambda)(\beta x + (1-\beta)y)\big) \\ &\leq \lambda f(\alpha x + (1-\alpha)y) + (1-\lambda) f(\beta x + (1-\beta)y) \\ &= \lambda\varphi_{x,y}(\alpha) + (1-\lambda)\varphi_{x,y}(\beta), \end{aligned}$$

which shows that $\varphi_{x,y}$ is convex. □

iii) *Let $n \in \mathbb{N}^*$ and $(\lambda_i)_{1\leq i\leq n} \in [0,+\infty[^n$. For every $i \in \{1,\ldots,n\}$, let $f_i\colon \mathcal{H} \to]-\infty,+\infty]$ be convex. Then $\sum_{i=1}^n \lambda_i f_i$ is convex.*

iv) *Let $(f_i)_{i\in I}$ be a family of convex functions. Then, $\sup_{i\in I} f_i$ is convex.*

Proof. For every $i \in I$, $\operatorname{epi} f_i$ is a convex set. Therefore

$$\operatorname{epi} \sup_{i\in I} f_i = \bigcap_{i\in I} \operatorname{epi} f_i$$

is a convex set. □

Notation 2

Let $\mathcal{H}$ be a Hilbert space. $\Gamma_0(\mathcal{H})$ is the class of convex, l.s.c., and proper functions from $\mathcal{H}$ to $]-\infty,+\infty]$.

Property 11. *Let $C \subset \mathcal{H}$.*
$\iota_C \in \Gamma_0(\mathcal{H})$ if and only if C is a nonempty closed convex set.

Proof. $\operatorname{epi}_{\iota_C} = C \times [0,+\infty[$. □

Definition 24

Let $\mathcal{H}$ be a Hilbert space. Let $f\colon \mathcal{H} \to \,]-\infty, +\infty]$.
f is **strictly convex** if

$$(\forall x \in \operatorname{dom} f)(\forall y \in \operatorname{dom} f)(\forall \alpha \in]0,1[)$$
$$x \neq y \quad \Rightarrow \quad f(\alpha x + (1-\alpha)y) < \alpha f(x) + (1-\alpha) f(y).$$

Example 13.

i) *The functions defined in Example 6 are strictly convex.*

ii) *The functions defined in Example 12 are not strictly convex.*

2.3 Minimizers of a convex function

Theorem 5

Let $\mathcal{H}$ be a Hilbert space. Let $f\colon \mathcal{H} \to \,]-\infty, +\infty]$ be a proper convex function such that $\mu = \inf f > -\infty$.

i) $\{x \in \mathcal{H} \mid f(x) = \mu\}$ *is convex.*

ii) *Every local minimizer of f is a global minimizer.*

iii) *If f is strictly convex, then there exists at most one minimizer.*

Proof.

i) Let $\Omega = \{x \in \mathcal{H} \mid f(x) = \mu\}$. Let $(x,y) \in \Omega^2$ and let $\alpha \in]0,1[$. We have

$$f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha) f(y) = \mu$$

which shows that $\alpha x + (1-\alpha) y \in \Omega$.

ii) Let $\widehat{x}$ be a local minimizer of f. For every $y \in \mathcal{H} \setminus \{\widehat{x}\}$, there exists $\alpha \in]0,1[$ such that

$$\begin{aligned} & f(\widehat{x}) \leq f\big(\widehat{x} + \alpha(y - \widehat{x})\big) \leq (1-\alpha) f(\widehat{x}) + \alpha f(y) \\ \Rightarrow \quad & f(\widehat{x}) \leq f(y) \end{aligned}$$

iii) If f is strictly convex, the inequality is strict.

$\square$

Theorem 6

Let $\mathcal{H}$ be a Hilbert space and C a closed convex subset of $\mathcal{H}$. Let $f \in \Gamma_0(\mathcal{H})$ such that $\operatorname{dom} f \cap C \neq \varnothing$.
If f is coercive or C is bounded, then there exists $\widehat{x} \in C$ such that

$$f(\widehat{x}) = \inf_{x \in C} f(x).$$

If, moreover, f is strictly convex, this minimizer $\widehat{x}$ is unique.

Proof. Although the result is valid for infinite dimensional spaces, we will prove it only when $\mathcal{H}$ is finite dimensional.
Since C is a nonempty closed and convex set, $\iota_C \in \Gamma_0(\mathcal{H})$. Then, $f + \iota_C$ is l.s.c. and convex. Since $\operatorname{dom} f \cap C \neq \varnothing$, $f + \iota_C \in \Gamma_0(\mathcal{H})$. In addition, as f is coercive or C is bounded, $f + \iota_C$ is coercive. Theorem 2 thus guarantees that there exists $\widehat{x} \in C$ such that $f(\widehat{x}) = \inf_{x \in \mathcal{H}} f(x) + \iota_C(x) = \inf_{x \in C} f(x)$.
If f is strictly convex, then $f + \iota_C$ is strictly convex, and it follows from Theorem 5iii) that there is a unique minimizer. $\square$

2.4 Differentiable convex functions

Let us give some characterizations of convexity for differentiable functions.

Lemma 2

Let $\mathcal{H}$ be a Hilbert space. Let $f \colon \mathcal{H} \to \,]-\infty, +\infty]$ be Gâteaux differentiable at $x \in$ $\operatorname{int} \operatorname{dom} f$.
If f is convex, then

$$(\forall y \in \mathcal{H}) \quad f(y) \geq f(x) + \langle \nabla f(x) \mid y - x \rangle.$$

Proof. Assume that f is convex. For every $\alpha \in]0,1[$ and $y \in \mathcal{H}$,

$$\begin{aligned} & f\big(x + \alpha(y-x)\big) \leq (1-\alpha) f(x) + \alpha f(y) \\ \Rightarrow \quad & \langle \nabla f(x) \mid y - x \rangle = \lim_{\substack{\alpha \to 0 \\ \alpha > 0}} \frac{f\big(x + \alpha(y-x)\big) - f(x)}{\alpha} \leq f(y) - f(x). \end{aligned}$$

$\square$

Theorem 7

Let $\mathcal{H}$ be a Hilbert space. Let $f \colon \mathcal{H} \to \,]-\infty, +\infty]$ be Gâteaux differentiable on $\operatorname{dom} f$, *which is a nonempty open convex set.*

Then, f is convex if and only if

$$(\forall (x,y) \in (\operatorname{dom} f)^2) \quad f(y) \geq f(x) + \langle \nabla f(x) \mid y - x \rangle .$$

Proof. If f is convex, the inequality follows from Lemma 2.
Conversely, if the gradient inequality is satisfied, we have, for every $(x,y) \in (\operatorname{dom} f)^2$ and $\alpha \in]0,1[$, $\alpha x + (1-\alpha)y \in \operatorname{dom} f$, and

$$\begin{aligned} f(x) &\geq f(\alpha x + (1-\alpha)y) + (1-\alpha) \langle \nabla f(\alpha x + (1-\alpha)y) \mid x - y \rangle \\ f(y) &\geq f(\alpha x + (1-\alpha)y) + \alpha \langle \nabla f(\alpha x + (1-\alpha)y) \mid y - x \rangle . \end{aligned}$$

By multiplying the first inequality by α and the second one by $1-\alpha$ and summing them, we get

$$\alpha f(x) + (1-\alpha) f(y) \geq f(\alpha x + (1-\alpha)y).$$

□

Proposition 3

Let $f\colon \mathcal{H} \to]-\infty, +\infty]$ be Gâteaux differentiable on $\operatorname{dom} f$, which is a nonempty open convex set.
Then, f is strictly convex if and only if, for every $(x,y) \in (\operatorname{dom} f)^2$ with $x \neq y$,

$$f(y) > f(x) + \langle \nabla f(x) \mid y - x \rangle .$$

Proof. Assume that f is strictly convex. Let $(x,y) \in \operatorname{dom} f$ with $x \neq y$. Let $\alpha \in]0,1[$ and let $z = \alpha x + (1-\alpha)y \in \operatorname{dom} f$. Since f is convex,

$$\begin{aligned} & f(z) \geq f(x) + \langle \nabla f(x) \mid z - x \rangle \\ \Leftrightarrow \quad & f(\alpha x + (1-\alpha)y) \geq f(x) + (1-\alpha) \langle \nabla f(x) \mid y - x \rangle . \end{aligned}$$

By using the inequality $\alpha f(x) + (1-\alpha) f(y) > f(\alpha x + (1-\alpha)y)$, we deduce that

$$\begin{aligned} & \alpha f(x) + (1-\alpha) f(y) > f(x) + (1-\alpha) \langle \nabla f(x) \mid y - x \rangle \\ \Rightarrow \quad & f(y) > f(x) + \langle \nabla f(x) \mid y - x \rangle . \end{aligned}$$

Conversely, if the gradient inequality is satisfied, we have, for every $(x,y) \in (\operatorname{dom} f)^2$ with $x \neq y$ and $\alpha \in]0,1[$, $\alpha x + (1-\alpha)y \in \operatorname{dom} f \setminus \{x,y\}$, and

$$\begin{aligned} f(x) &> f(\alpha x + (1-\alpha)y) + (1-\alpha) \langle \nabla f(\alpha x + (1-\alpha)y) \mid x - y \rangle \\ f(y) &> f(\alpha x + (1-\alpha)y) + \alpha \langle \nabla f(\alpha x + (1-\alpha)y) \mid y - x \rangle . \end{aligned}$$

By multiplying the first inequality by α and the second one by $1-\alpha$ and summing them, we get

$$\alpha f(x) + (1-\alpha) f(y) > f(\alpha x + (1-\alpha)y).$$

□

Remark 8. *Let f be a function satisfying the assumptions of Proposition 3. Then, we define the **Bregman divergence** associated with f as*

$$(\forall (x,y) \in (\operatorname{dom} f)^2) \quad \mathsf{D}_f(y,x) = f(y) - f(x) - \langle \nabla f(x) \mid y - x \rangle . \tag{2.1}$$

As shown by Theorem 7,

$$(\forall (x,y) \in (\operatorname{dom} f)^2) \quad \mathsf{D}_f(y,x) \geq 0. \tag{2.2}$$

In addition, if f is strictly convex, it follows from Proposition 3 that

$$(\forall (x,y) \in (\operatorname{dom} f)^2) \qquad \mathsf{D}_f(y,x) = 0 \quad \Leftrightarrow \quad x = y. \tag{2.3}$$

These properties allow D_f to be used as a metric on $\operatorname{dom} f$. In particular, if $f = \|\cdot\|^2$, then $\mathsf{D}_f(y,x) = \|y-x\|^2$. Furthermore, if $\mathcal{H} = \mathbb{R}^N$ and

$$(\forall x = (x^{(i)})_{1 \leq i \leq N} \in \mathbb{R}^N) \quad f(x) = \begin{cases} \sum_{i=1}^N x^{(i)} \ln(x^{(i)}) & \text{if } x \in]0,+\infty[^N \\ +\infty & \text{otherwise,} \end{cases} \tag{2.4}$$

then the Kullback-Leibler divergence is obtained:

$$(\forall x = (x^{(i)})_{1 \leq i \leq N} \in]0,+\infty[^N)(\forall y = (y^{(i)})_{1 \leq i \leq N} \in]0,+\infty[^N)$$
$$\mathsf{D}_f(y,x) = \sum_{i=1}^N y^{(i)} \ln\left(\frac{y^{(i)}}{x^{(i)}}\right) - y^{(i)} + x^{(i)}. \tag{2.5}$$

Proposition 4

Let $f\colon \mathcal{H} \to]-\infty,+\infty]$ be Gâteaux differentiable on $\operatorname{dom} f$, which is a nonempty open convex set.
Then, f is convex if and only if ∇f is monotone on $\operatorname{dom} f$, i.e.

$$(\forall (x,y) \in (\operatorname{dom} f)^2) \quad \langle \nabla f(y) - \nabla f(x) \mid y - x \rangle \geq 0.$$

Proof. Assume that f is convex. For every $(x,y) \in (\operatorname{dom} f)^2$,

$$f(x) - f(y) \geq \langle \nabla f(y) \mid x - y \rangle$$
$$f(y) - f(x) \geq \langle \nabla f(x) \mid y - x \rangle .$$

By summing, $\langle \nabla f(y) - \nabla f(x) \mid x - y \rangle \leq 0$.
Conversely, assume that ∇f is monotone on $\operatorname{dom} f$. For every $(x,y) \in (\operatorname{dom} f)^2$, let

$$\varphi\colon [0,1] \to \mathbb{R}\colon \alpha \mapsto f(x + \alpha(y-x)).$$

φ is continuous on $[0,1]$, differentiable on $]0,1[$ and its derivative is

$$(\forall \alpha \in]0,1[) \qquad \dot{\varphi}(\alpha) = \langle \nabla f(x + \alpha(y-x)) \mid y - x \rangle .$$

According to the mean value theorem, there exists $\alpha \in]0,1[$ such that

$$\begin{aligned} & \varphi(1) - \varphi(0) = \dot{\varphi}(\alpha) \\ \Rightarrow \quad & f(y) - f(x) = \langle \nabla f(x + \alpha(y-x)) \mid y - x \rangle . \end{aligned}$$

On the other hand, as $x + \alpha(y-x) \in \operatorname{dom} f$,

$$\langle \nabla f(x + \alpha(y-x)) - \nabla f(x) \mid y - x \rangle \geq 0$$

which leads to $f(y) - f(x) \geq \langle \nabla f(x) \mid y - x \rangle$. □

Proposition 5

Let $f\colon \mathcal{H} \to]-\infty, +\infty]$ be Gâteaux differentiable on $\operatorname{dom} f$, which is a nonempty open convex set.
Then, f is strictly convex if and only if ∇f is strictly monotone on $\operatorname{dom} f$, i.e. for every $(x,y) \in (\operatorname{dom} f)^2$ with $x \neq y$,

$$\langle \nabla f(y) - \nabla f(x) \mid y - x \rangle > 0.$$

Proof. For every $(x,y) \in (\operatorname{dom} f)^2$ with $x \neq y$, we can follow the same proof as in the previous proposition, by noticing that all inequalities are now strict. □

Theorem 8

Let $\mathcal{H}$ be a Hilbert space. Let $f\colon \mathcal{H} \to]-\infty, +\infty]$ be a twice Fréchet differentiable function on $\operatorname{dom} f$, which is a nonempty open convex set.

i) *f is convex if and only if, for every $x \in \operatorname{dom} f$,*

$$(\forall z \in \mathcal{H}) \qquad \left\langle z \mid \nabla^2 f(x) z \right\rangle \geq 0.$$

ii) *If, for every $x \in \operatorname{dom} f$,*

$$(\forall z \in \mathcal{H} \setminus \{0\}) \qquad \left\langle z \mid \nabla^2 f(x) z \right\rangle > 0,$$

then f is strictly convex.

Proof. Let $(x,y) \in (\operatorname{dom} f)^2$ with $x \neq y$. According to Taylor-Mc Laurin formula, there exists $u \in]x,y[\subset \operatorname{dom} f$ such that

$$f(y) = f(x) + \langle \nabla f(x) \mid y - x \rangle + \frac{1}{2} \left\langle y - x \mid \nabla^2 f(u)(y-x) \right\rangle .$$

If $\langle y - x \mid \nabla^2 f(u)(y-x) \rangle \geq 0$, then $f(y) \geq f(x) + \langle \nabla f(x) \mid y - x \rangle$.
Theorem 7 allows us to conclude that f is convex.
If $\langle y - x \mid \nabla^2 f(u)(y-x) \rangle > 0$, then $f(y) > f(x) + \langle \nabla f(x) \mid y - x \rangle$. Proposition 3 allows us to

conclude that f is strictly convex.
Conversely, if f is convex. Since $\operatorname{dom} f$ is open, for every $x \in \operatorname{dom} f$ and $z \in \mathcal{H}$, there exists $\delta \in \,]0,+\infty[$ such that $(\forall \alpha \in]0,\delta[)$ $x+\alpha z \in \operatorname{dom} f$ and

$$\begin{aligned} &\langle \nabla f(x+\alpha z) - \nabla f(x) \mid \alpha z\rangle \geq 0 \\ \Leftrightarrow &\left\langle \alpha^{-1}\big(\nabla f(x+\alpha z) - \nabla f(x)\big) \mid z\right\rangle \geq 0 \end{aligned}$$

By letting $\alpha \to 0$, $\langle \nabla^2 f(x) z \mid z\rangle \geq 0$. □

Let us now provide a necessary and sufficient condition for the existence of a minimizer of a differentiable convex function.

Theorem 9

Let $\mathcal{H}$ be Hilbert space. Let $f\colon \mathcal{H} \to \,]-\infty,+\infty]$ be a Gâteaux differentiable convex function on $\operatorname{dom} f$, which is an open set. Let $C \subset \operatorname{dom} f$ be a convex set.
$\widehat{x} \in C$ is a (global) minimizer of f on C if and only if

$$(\forall y \in C) \qquad \langle \nabla f(\widehat{x}) \mid y - \widehat{x}\rangle \geq 0.$$

If $\widehat{x} \in \operatorname{int}(C)$, then the condition reduces to

$$\nabla f(\widehat{x}) = 0.$$

Proof. We have already seen in Theorem 3 that the inequality is a necessary condition for $\widehat{x}$ to be a local minimizer of f on C and that it reduces to the vanishing condition on the gradient if $\widehat{x} \in \operatorname{int}(C)$.
Conversely, assume that the inequality holds. Let $y \in C$. Since f is convex and Gâteaux differentiable, it follows from Theorem 7 that

$$f(y) \geq f(\widehat{x}) + \langle \nabla f(\widehat{x}) \mid y - \widehat{x}\rangle \geq f(\widehat{x}).$$

Hence, $\widehat{x}$ is a minimizer of f on C. □

2.5 Projection onto a closed convex set

2.5.1 Definition

Theorem 10

Let C be a nonempty ***closed convex set*** *of a Hilbert space $\mathcal{H}$.*

i) *For every $x \in \mathcal{H}$, there exists a unique point $\widehat{x}$ in C which lies at minimum distance of x.*
The application $P_C\colon \mathcal{H} \to C$ which maps every $x \in \mathcal{H}$ to its associated point $\widehat{x}$ is called the ***projection onto*** *C.*

ii) *For every $x \in \mathcal{H}$, $\widehat{x} = P_C(x)$ if and only if $\widehat{x} \in C$ and*

$$(\forall y \in C) \qquad \langle x - \widehat{x} \mid y - \widehat{x} \rangle \leq 0.$$

Proof.

i) Let $x \in \mathcal{H}$. The function $f \colon \mathcal{H} \to \mathbb{R} \colon y \mapsto \|y - x\|^2$ is continuous, strictly convex and coercive. Since C is nonempty $\operatorname{dom} f \cap C = C \neq \varnothing$, according to Theorem 6, there exists a unique minimizer $\widehat{x}$ of f on C.

ii) f is Gâteaux differentiable and

$$(\forall y \in \mathcal{H}) \qquad \nabla f(y) = 2(y - x).$$

Therefore, according to Theorem 9, $\widehat{x}$ is the minimizer of f onto C if and only if

$$(\forall y \in C) \qquad \langle \nabla f(\widehat{x}) \mid y - \widehat{x} \rangle = 2 \langle \widehat{x} - x \mid y - \widehat{x} \rangle \geq 0.$$

□

2.5.2 Geometrical interpretation

Definition 25

Let C be a nonempty subset of $\mathcal{H}$.
For every $x \in \mathcal{H}$, the **normal cone** to C at x is defined as

$$N_C(x) = \begin{cases} \big\{ u \in \mathcal{H} \;\big|\; (\forall y \in C) \;\; \langle u \mid y - x \rangle \leq 0 \big\} & \text{if } x \in C \\ \varnothing & \text{otherwise.} \end{cases}$$

Remark 9. *For every $x \in C$, $N_C(x)$ is a closed convex cone with vertex 0 since*

- $0 \in C$
- *For every $\alpha \in \;]0, +\infty[$ and $u \in N_C(x)$,*

$$(\forall y \in C) \quad \langle \alpha u \mid y - x \rangle \leq 0 \quad \Leftrightarrow \quad \alpha u \in N_C(x).$$

- *For every $\alpha \in]0, 1[$ and $(u, v) \in N_C(x)^2$,*

$$\begin{aligned} &(\forall y \in C) \;\; \langle \alpha u + (1 - \alpha) v \mid y - x \rangle = \alpha \langle u \mid y - x \rangle + (1 - \alpha) \langle v \mid y - x \rangle \leq 0 \\ \Leftrightarrow \;\; &\alpha u + (1 - \alpha) v \in N_C(x) \end{aligned}$$

- *For every sequence $(u_n)_{n\geq 0}$ of $N_C(x)$, it follows from the continuity of the scalar product that $u_n \to u \in \mathcal{H} \Rightarrow u \in N_C(x)$.*

Property 12.

i) *If C is a vector space, then for every $x \in C$, $N_C(x) = C^\perp$.*

Proof.

$$
\begin{aligned}
u \in N_C(x) \quad &\Leftrightarrow \quad (\forall y \in C) \;\; \langle u \mid y - x \rangle \leq 0 \\
&\Leftrightarrow \quad (\forall z \in C) \;\; \langle u \mid z \rangle \leq 0 \\
&\Leftrightarrow \quad (\forall z \in C) \;\; \begin{cases} \langle u \mid z \rangle \leq 0 \\ \langle u \mid -z \rangle \leq 0 \end{cases} \\
&\Leftrightarrow \quad (\forall z \in C) \;\; \langle u \mid z \rangle = 0.
\end{aligned}
$$

□

ii) *If $x \in \operatorname{int} C$, then $N_C(x) = \{0\}$.*

Proof. If $x \in \operatorname{int} C$ then, for every $z \in \mathcal{H}$, there exists $\rho \in \,]0, +\infty[$ such that $\rho z + x \in C$. We have then, for every $u \in N_C(x)$,

$$
\begin{aligned}
&\langle u \mid \rho z + x - x \rangle = \rho \langle u \mid z \rangle \leq 0 \\
\Leftrightarrow \quad &\langle u \mid z \rangle \leq 0
\end{aligned}
$$

As previously, we deduce that, for every $z \in \mathcal{H}$, $\langle u \mid z \rangle = 0$, i.e. $u = 0$. □

iii) *Let C be nonempty closed convex set of a Hilbert space $\mathcal{H}$. For every $x \in \mathcal{H}$,*

$$\widehat{x} = P_C(x) \quad \Leftrightarrow \quad x - \widehat{x} \in N_C(\widehat{x}).$$

Proof. This follows from Theorem 10ii). □

2.5.3 Examples

i) If C is a closed vector space of a Hilbert space $\mathcal{H}$, then P_C is the orthogonal projection onto C. Then, for every $x \in \mathcal{H}$,

$$\widehat{x} = P_C(x) \quad \Leftrightarrow \quad \begin{cases} \widehat{x} \in C \\ x - \widehat{x} \in C^\perp. \end{cases} \tag{2.6}$$

Proof. The equivalence follows from Properties 12i) and iii). □

ii) If C is a nonempty closed convex set of a Hilbert space $\mathcal{H}$, P_C is a linear operator if and only C is a vector space.

Proof. If P_C is linear, for every $(x, y) \in C^2$ and $(\alpha, \beta) \in \mathbb{R}^2$,

$$\alpha x + \beta y = \alpha P_C(x) + \beta P_C(y) = P_C(\alpha x + \beta y) \in C,$$

which shows that C is a vector subspace of $\mathcal{H}$.
Conversely, if C is a vector space, the linearity of P_C follows from (2.6). □

iii) Let $\mathcal{H}$ be a Hilbert space, let $u \in \mathcal{H} \setminus \{0\}$, let $\delta \in \mathbb{R}$, and let C be the closed affine hyperplane defined as

$$C = \big\{x \in \mathcal{H} \;\big|\; \langle u \mid x\rangle = \delta\big\}.$$

The projection onto C is

$$(\forall x \in \mathcal{H}) \qquad P_C(x) = x + \frac{\delta - \langle u \mid x\rangle}{\|u\|^2} u.$$

Proof. C is a nonempty closed convex set.
For every $x \in \mathcal{H}$,

$$\langle u \mid P_C(x)\rangle = \left\langle u \mid x + \frac{\delta - \langle u \mid x\rangle}{\|u\|^2} u \right\rangle = \delta.$$

Then $P_C(x) \in C$.
For every $x \in \mathcal{H}$ and $y \in C$,

$$\begin{aligned}
\langle x - P_C(x) \mid y - P_C(x)\rangle =& \frac{\langle u \mid x\rangle - \delta}{\|u\|^2} \langle u \mid y - P_C(x)\rangle \\
=& \frac{\langle u \mid x\rangle - \delta}{\|u\|^2} (\langle u \mid y\rangle - \langle u \mid P_C(x)\rangle) \\
=& \frac{\langle u \mid x\rangle - \delta}{\|u\|^2} (\langle u \mid y\rangle - \delta) = 0.
\end{aligned}$$

According to Theorem 10ii), this shows that P_C is the projection onto C. □

iv) Let $\mathcal{H}$ be a Hilbert space, let $u \in \mathcal{H} \setminus \{0\}$, let $\delta \in \mathbb{R}$, and let C be the closed half space defined as

$$C = \big\{x \in \mathcal{H} \;\big|\; \langle u \mid x\rangle \leq \delta\big\}.$$

The projection onto C is

$$(\forall x \in \mathcal{H}) \qquad P_C(x) = \begin{cases} x + \dfrac{\delta - \langle u \mid x\rangle}{\|u\|^2} u & \text{if } \langle u \mid x\rangle > \delta \\ x & \text{otherwise.} \end{cases}$$

Proof. Similar to the previous one. □

2.5.4 Properties of the projection

Definition 26

Let $\mathcal{H}$ be a Hilbert space and let $T\colon \mathcal{H} \to \mathcal{H}$.

i) T is **nonexpansive** (or 1-Lipschitz) if

$$(\forall (x,y) \in \mathcal{H}^2) \qquad \|T(x) - T(y)\| \leq \|x - y\|.$$

ii) T is **firmly nonexpansive** if

$$(\forall (x,y) \in \mathcal{H}^2) \qquad \|T(x) - T(y)\|^2 \leq \langle x - y \mid T(x) - T(y) \rangle .$$

Remark 10.

i) *It follows from the Cauchy-Schwarz inequality that, if T is firmly nonexpansive, then it is nonexpansive. In addition, any nonexpansive operator is uniformly continuous.*

ii) *It follows from straightforward calculations that T is firmly nonexpansive if and only if*

$$(\forall (x,y) \in \mathcal{H}^2) \qquad \|T(x) - T(y)\|^2 + \|x - T(x) - y + T(y)\|^2 \leq \|x - y\|^2.$$

Property 13.

i) *Let C be a nonempty closed convex set of a Hilbert space $\mathcal{H}$. The projection onto C is a firmly nonexpansive operator.*

Proof. For every $(x,y) \in \mathcal{H}^2$,

$$\begin{aligned}\langle x - y \mid P_C(x) - P_C(y) \rangle = &\langle x - P_C(x) \mid P_C(x) - P_C(y) \rangle \\ &+ \|P_C(x) - P_C(y)\|^2 \\ &+ \langle P_C(y) - y \mid P_C(x) - P_C(y) \rangle .\end{aligned}$$

Since $P_C(x) \in C$ and $P_C(y) \in C$

$$\begin{aligned}\langle x - P_C(x) \mid P_C(y) - P_C(x) \rangle \leq 0 \\ \langle y - P_C(y) \mid P_C(x) - P_C(y) \rangle \leq 0.\end{aligned}$$

Then, $\langle x - y \mid P_C(x) - P_C(y) \rangle \geq \|P_C(x) - P_C(y)\|^2$. □

ii) *The projection onto C is a nonexpansive operator.*

iii) *The projection onto C is uniformly continuous.*

iv) *The* ***distance to*** *C defined as*

$$(\forall x \in \mathcal{H}) \qquad d_C(x) = \inf_{y \in C} \|x - y\| = \|x - P_C(x)\|$$

is continuous.

2.5.5 Convex feasibility problems

Problem 1

Let $\mathcal{H}$ be a Hilbert space. Let $m \in \mathbb{N} \setminus \{0, 1\}$.

Let $(C_i)_{1 \le i \le m}$ be closed convex subsets of $\mathcal{H}$ such that $\bigcap_{i=1}^{m} C_i \neq \varnothing$.

We want to

$$\text{Find } \widehat{x} \in \bigcap_{i=1}^{m} C_i.$$

Algorithm 1. (POCS: Projections Onto Convex Sets)

Let $(\lambda_n)_{n \ge 0}$ be a sequence of $[\epsilon_1, 2 - \epsilon_2]$ with $(\epsilon_1, \epsilon_2) \in \,]0, +\infty[^2$ such that $\epsilon_1 + \epsilon_2 < 2$. For every $n \in \mathbb{N}$, let $i_n - 1$ denote the remainder after division of n by m.

$$\begin{array}{l} \text{Set } x_0 \in \mathcal{H} \\ \text{For } n = 0, \ldots \\ \lfloor \; x_{n+1} = x_n + \lambda_n (P_{C_{i_n}}(x_n) - x_n). \end{array}$$

Remark 11. *For every $n \in \mathbb{N}$, λ_n is an over-relaxation parameter. If $\lambda_n = 1$, the n-th iteration simplifies to*

$$x_{n+1} = P_{C_{i_n}}(x_n).$$

Theorem 11

The sequence $(x_n)_{n \in \mathbb{N}}$ generated by the POCS algorithm converges weakly to a point in $\bigcap_{i=1}^{m} C_i$.

Remark 12. *The result remains valid if the* ***cyclic rule*** *for activating the projections is replaced by a* ***quasi-cyclic rule****, i.e. the indice $(i_n)_{n \ge 0}$ are now such that*

$$(\exists K \in \mathbb{N})(\forall n \in \mathbb{N}) \qquad \{1, \ldots, m\} \subset \{i_n, \ldots, i_{n+K}\}.$$

2.5.6 Convergence proof of POCS

Let us first introduce the notion of Fejér-monotonicity.

Definition 27

Let D be a nonempty subset of a Hilbert space $\mathcal{H}$.
Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in $\mathcal{H}$.
$(x_n)_{n \in \mathbb{N}}$ is **Fejér-monotone** with respect to D if

$$(\forall x \in D)(\forall n \in \mathbb{N}) \qquad \|x_{n+1} - x\| \le \|x_n - x\|.$$

Property 14. *Let $\varnothing \neq D \subset \mathcal{H}$.*
Let $(x_n)_{n\in\mathbb{N}}$ be Fejér-monotone with respect to D.

i) *For every $x \in D$, $(\|x_n - x\|)_{n\in\mathbb{N}}$ converges.*

ii) *$(x_n)_{n\in\mathbb{N}}$ is bounded.*

Proof.

i) $(\|x_n - x\|)_{n\in\mathbb{N}}$ is a decaying sequence which is lower bounded by 0. Hence, it converges.

ii) Let $x \in D$. Since $(\|x_n - x\|)_{n\in\mathbb{N}}$ converges, it is bounded. Then, there exists $\mu \in [0, +\infty[$ such that
$$\|x_n\| - \|x\| \leq \|x_n - x\| \leq \mu,$$
which shows that $(x_n)_{n\in\mathbb{N}}$ is bounded.

□

Theorem 12. (Fejér-monotone convergence)

Let D be a nonempty subset of a Hilbert space $\mathcal{H}$.
Let $(x_n)_{n\in\mathbb{N}}$ be a sequence in $\mathcal{H}$. $(x_n)_{n\in\mathbb{N}}$ converges weakly to a point in D if

i) *$(x_n)_{n\in\mathbb{N}}$ is Fejér-monotone with respect to D*

and

ii) *every weak sequential cluster point of $(x_n)_{n\in\mathbb{N}}$ (i.e. weak limit of a subsequence of $(x_n)_{n\in\mathbb{N}}$) lies in D.*

Proof. We will only prove this result when $\mathcal{H}$ is finite dimensional.
Since $(x_n)_{n\in\mathbb{N}}$ is Fejér-monotone with respect to D, it is a bounded sequence. There thus exists a subsequence $(x_{n_k})_{k\in\mathbb{N}}$ of $(x_n)_{n\in\mathbb{N}}$ such that $x_{n_k} \to \widehat{x}$ and, by assumption, $\widehat{x} \in D$.
On the other hand, there exists $\alpha \in \mathbb{R}$ such that $\|x_n - \widehat{x}\| \to \alpha$. Since $\|x_{n_k} - \widehat{x}\| \to 0$, $\alpha = 0$.
Hence, $x_n \to \widehat{x}$. □

We are now ready to prove Theorem 11 by proceeding in four steps.

Step 1:

Let $D = \bigcap_{i=1}^{m} C_i$. For every $x \in D$ and for every $n \in \mathbb{N}$,

$$\begin{aligned}
\|x_{n+1} - x\|^2 &= \|x_n - x\|^2 + 2\lambda_n \left\langle x_n - x \mid P_{C_{i_n}}(x_n) - x_n \right\rangle + \lambda_n^2 \|P_{C_{i_n}}(x_n) - x_n\|^2 \\
&= \|x_n - x\|^2 + 2\lambda_n \left\langle x_n - P_{C_{i_n}}(x_n) \mid P_{C_{i_n}}(x_n) - x_n \right\rangle \\
&\quad + 2\lambda_n \left\langle P_{C_{i_n}}(x_n) - x \mid P_{C_{i_n}}(x_n) - x_n \right\rangle + \lambda_n^2 \|P_{C_{i_n}}(x_n) - x_n\|^2.
\end{aligned}$$

Since $x \in D \subset C_{i_n}$, $\left\langle P_{C_{i_n}}(x_n) - x \mid P_{C_{i_n}}(x_n) - x_n \right\rangle \leq 0$. By using the fact that $\lambda_n \in [\epsilon_1, 2-\epsilon_2]$, it can be deduced that

$$\begin{aligned}\|x_{n+1} - x\|^2 &\leq \|x_n - x\|^2 - (2 - \lambda_n)\lambda_n \|P_{C_{i_n}}(x_n) - x_n\|^2 \\ &\leq \|x_n - x\|^2 - \epsilon_1\epsilon_2 \|P_{C_{i_n}}(x_n) - x_n\|^2 \leq \|x_n - x\|^2.\end{aligned}$$

This shows that $(x_n)_{n\in\mathbb{N}}$ is Fejér-monotone with respect to D.

Step 2:
Since $(x_n)_{n\in\mathbb{N}}$ is Fejér-monotone with respect to D, for every $x \in D$, $(\|x_n - x\|)_{n\in\mathbb{N}}$ converges . In addition, since we have proved that

$$(\forall n \in \mathbb{N}) \quad \|x_{n+1} - x\|^2 \leq \|x_n - x\|^2 - \epsilon_1\epsilon_2 \|P_{C_{i_n}}(x_n) - x_n\|^2,$$

$P_{C_{i_n}}(x_n) - x_n \to 0$.
By using now the fact that

$$x_{n+1} - x_n = \lambda_n(P_{C_{i_n}}(x_n) - x_n)$$

where $(\lambda_n)_{n\in\mathbb{N}}$ is bounded, we deduce that $x_{n+1} - x_n \to 0$.

Step 3:
Let $j \in \{1, \ldots, m\}$ and let m_n be the smallest integer greater than or equal to n such that $i_{m_n} = j$. We have then $n \leq m_n < n + m$. In addition, since P_{C_j} is nonexpansive,

$$\begin{aligned}\|P_{C_j}(x_n) - x_n\| &\leq \|P_{C_j}(x_n) - P_{C_{i_{m_n}}}(x_{m_n}) + P_{C_{i_{m_n}}}(x_{m_n}) - x_{m_n} + x_{m_n} - x_n\| \\ &\leq \|P_{C_j}(x_n) - P_{C_j}(x_{m_n})\| + \|P_{C_{i_{m_n}}}(x_{m_n}) - x_{m_n}\| + \|x_{m_n} - x_n\| \\ &\leq \|P_{C_{i_{m_n}}}(x_{m_n}) - x_{m_n}\| + 2\|x_{m_n} - x_n\|.\end{aligned}$$

As $n \to +\infty$, $m_n \to +\infty$ and, according to Step 2, $P_{C_{i_{m_n}}}(x_{m_n}) - x_{m_n} \to 0$. In addition,

$$\|x_{m_n} - x_n\| \leq \sum_{i=n}^{m_n-1} \|x_{i+1} - x_i\|.$$

As there are at most $m - 1$ terms in the summation and we have proved that $x_{n+1} - x_n \to 0$, it follows that $\|x_{m_n} - x_n\| \to 0$ and $P_{C_j}(x_n) - x_n \to 0$.

Step 4: (assuming $\mathcal{H}$ finite dimensional)
Let x be a cluster point of $(x_n)_{n\in\mathbb{N}}$. There exists a subsequence $(x_{n_k})_{k\in\mathbb{N}}$ of $(x_n)_{n\in\mathbb{N}}$ such that $x_{n_k} \to x$.
For every $j \in \{1, \ldots, m\}$, since P_{C_j} is continuous and $P_{C_j}(x_{n_k}) - x_{n_k} \to 0$, we have $P_{C_j}(x) = x$.
This shows that $x \in D$.
It follows from Theorem 12 that $x_n \to x$.

Chapter 3

Duality

3.1 Motivation

Suppose that we want to minimize a proper lower-semicontinuous function $f\colon \mathbb{R}^N \to \,]-\infty,+\infty]$ such that $\operatorname{dom} f = C$ is a compact set. When f is nonconvex, a main difficulty is that there might exist several local extrema.

A possible strategy in this context consists of finding a convex surrogate function $\check{f}$ for f. For example, one may choose $\check{f}$ as the largest l.s,c. convex function lower bounding f. The question however is: how to build $\check{f}$ for an arbitrary function f defined on $\mathbb{R}^N$? This chapter will provide an answer to this question as well as many other fundamental ones.

3.2 Conjugate function

Definition 28

Let $\mathcal{H}$ be a Hilbert space and $f\colon \mathcal{H} \to [-\infty,+\infty]$.
The Fenchel-Legendre **conjugate** of f is the function $f^*\colon \mathcal{H} \to [-\infty,+\infty]$ such that

$$(\forall u \in \mathcal{H}) \qquad f^*(u) = \sup_{x\in\mathcal{H}} \big(\langle x \mid u\rangle - f(x)\big).$$

Remark 13.

i) *The supremum in the above definition can be calculated on* $\operatorname{dom} f$.

ii) $(\forall u \in \mathcal{H})$ $f^*(u) \neq -\infty$ *if and only if* $\operatorname{dom} f \neq \varnothing$.

Proof. Assume that there exists $u \in \mathcal{H}$ such that $f^*(u) = -\infty$. This means that

$$\sup_{x\in\operatorname{dom} f} \big(\langle x \mid u\rangle - f(x)\big) = -\infty.$$

This happens if and only if $\operatorname{dom} f = \varnothing$. $\square$

iii) *If* $(\exists x \in \mathcal{H})\ f(x) = -\infty$, *then* $(\forall u \in \mathcal{H})\ f^*(u) = +\infty$.

Example 14.

i) *If*

$$(\forall x \in \mathcal{H}) \quad f(x) = \langle v \mid x \rangle$$

with $v \in \mathcal{H}$, *then* $f^* = \iota_{\{v\}}$.

Proof. Exercise $\square$

ii) $f = \frac{1}{2}\|\cdot\|^2 \Rightarrow f^* = \frac{1}{2}\|\cdot\|^2$.

Proof. For every $(x,u) \in \mathcal{H}^2$, $\langle x \mid u \rangle - \frac{1}{2}\|x\|^2 = \frac{1}{2}\|u\|^2 - \frac{1}{2}\|u-x\|^2$ is maximum at $x = u$. Consequently, $f^*(u) = \frac{1}{2}\|u\|^2$. $\square$

iii) *If*

$$(\forall x \in \mathbb{R}) \qquad f(x) = \frac{1}{q}|x|^q$$

with $q \in]1,+\infty[$, *then*

$$(\forall u \in \mathbb{R}) \qquad f^*(u) = \frac{1}{q^*}|u|^{q^*}$$

with $\frac{1}{q} + \frac{1}{q^*} = 1$.

Proof. For every $u \in \mathbb{R}$,

$$f^*(u) = \sup_{x \in \mathbb{R}} \varphi_u(x)$$

where

$$(\forall x \in \mathbb{R}) \quad \varphi_u(x) = xu - \frac{1}{q}|x|^q.$$

We have

$$(\forall x \in \mathbb{R}) \quad \varphi_u'(x) = \begin{cases} u - x^{q-1} & \text{if } x \geq 0 \\ u + (-x)^{q-1} & \text{otherwise.} \end{cases}$$

If $u \geq 0$, φ_u is increasing on $]-\infty, u^{1/(q-1)}[$ and decreasing on $]u^{1/(q-1)}, +\infty[$, and, if $u < 0$, φ_u is increasing on $]-\infty, -(-u)^{1/(q-1)}[$ and decreasing on $]-(-u)^{1/(q-1)}, +\infty[$. Thus, the maximum value taken by φ_u is

$$f^*(u) = |u||u|^{1/(q-1)} - \frac{1}{q}|u|^{q/(q-1)} = \frac{1}{q^*}|u|^{q^*}.$$

$\square$

Property 15.

i) *If* f *is even, then* f^* *is even.*

Proof. For every $u \in \mathcal{H}$,

$$\begin{aligned} f^*(-u) &= \sup_{x\in\mathcal{H}} \big(\langle x \mid -u\rangle - f(x)\big) \\ &= \sup_{x\in\mathcal{H}} \big(\langle -x \mid -u\rangle - f(-x)\big) \\ &= \sup_{x\in\mathcal{H}} \big(\langle x \mid u\rangle - f(x)\big) \\ &= f^*(u). \end{aligned}$$

□

ii) *For every $\alpha \in \,]0,+\infty[$, $(\alpha f)^* = \alpha f^*(\cdot/\alpha)$.*

Proof. For every $u \in \mathcal{H}$,

$$\begin{aligned} (\alpha f)^*(u) &= \sup_{x\in\mathcal{H}} \big(\langle x \mid u\rangle - \alpha f(x)\big) \\ &= \alpha \sup_{x\in\mathcal{H}} \Big(\Big\langle x \mid \frac{u}{\alpha}\Big\rangle - f(x)\Big) \\ &= \alpha f^*\Big(\frac{u}{\alpha}\Big). \end{aligned}$$

□

iii) *For every $(y,v) \in \mathcal{H}^2$ and $\alpha \in \mathbb{R}$, $\big(f(\cdot - y) + \langle \cdot \mid v\rangle + \alpha\big)^* = f^*(\cdot - v) + \langle y \mid \cdot - v\rangle - \alpha$.*

Proof. For every $u \in \mathcal{H}$,

$$\begin{aligned} \big(f(\cdot - y) + \langle \cdot \mid v\rangle + \alpha\big)^*(u) &= \sup_{x\in\mathcal{H}} \big(\langle x \mid u\rangle - f(x-y) - \langle x \mid v\rangle - \alpha\big) \\ &= \sup_{x\in\mathcal{H}} \big(\langle x - y \mid u - v\rangle - f(x-y)\big) + \langle y \mid u - v\rangle - \alpha \\ &= f^*(u-v) + \langle y \mid u - v\rangle - \alpha. \end{aligned}$$

□

iv) *Let $\mathcal{G}$ be a Hilbert space and $L \in \mathcal{B}(\mathcal{G},\mathcal{H})$ be an isomorphism. $(f\circ L)^* = f^* \circ (L^{-1})^*$.*

Proof. For every $u \in \mathcal{H}$,

$$(f\circ L)^*(u) = \sup_{x\in\mathcal{H}} \big(\langle x \mid u\rangle - f(Lx)\big).$$

By making the change of variable $x' = Lx$,

$$\begin{aligned} (f\circ L)^*(u) &= \sup_{x'\in\mathcal{H}} \big(\big\langle L^{-1}x' \mid u\big\rangle - f(x')\big) \\ &= \sup_{x'\in\mathcal{H}} \big(\big\langle x' \mid (L^{-1})^* u\big\rangle - f(x')\big) \\ &= f^*\big((L^{-1})^* u\big). \end{aligned}$$

□

v) *Let* $L \in \mathcal{B}(\mathcal{G}, \mathcal{H})$ *and let*

$$(\forall x \in \mathcal{H}) \quad (L \rhd f)(x) = \inf_{Ly=x} f(y).$$

Then $(L \rhd f)^* = f^* \circ L^*$.
($L \rhd f$ *is called the* ***infimal postcomposition*** *of* f *by* L*.)*

Proof. For every $u \in \mathcal{H}$,

$$\begin{aligned}
(L \rhd f)^*(u) &= \sup_{x \in \mathcal{H}} \big(\langle x \mid u \rangle - (L \rhd f)(x) \big) \\
&= \sup_{x \in \mathcal{H}} \big(\langle x \mid u \rangle - \inf_{Ly=x} f(y) \big) \\
&= \sup_{x \in \mathcal{H}, Ly=x} \big(\langle x \mid u \rangle - f(y) \big) \\
&= \sup_{y \in \mathcal{G}} \big(\langle Ly \mid u \rangle - f(y) \big) \\
&= \sup_{y \in \mathcal{G}} \big(\langle y \mid L^* u \rangle - f(y) \big) \\
&= f^*(L^* u).
\end{aligned}$$

□

vi) f^* ***is l.s.c. and convex.***

vii) $f^*(0) = -\inf f$

Proposition 6. (Conjugate of separable functions)

Let I *is a finite nonempty subset of* $\mathbb{N}$. *Let* $(\mathcal{H}_i)_{i \in I}$ *be Hilbert spaces and let* $\mathcal{H} = \bigtimes_{i \in I} \mathcal{H}_i$. *For every* $i \in I$, *let* $f_i \colon \mathcal{H}_i \to \left]-\infty, +\infty\right]$ *be a proper function. Let*

$$f \colon \mathcal{H} \to \left]-\infty, +\infty\right] : x = (x_i)_{i \in I} \mapsto \sum_{i \in I} f_i(x_i)$$

Then,

$$\big(\forall u = (u_i)_{i \in I} \in \mathcal{H}\big) \qquad f^*(u) = \sum_{i \in I} f_i^*(u_i).$$

Proof. Let $u = (u_i)_{i \in I} \in \mathcal{H}$. We have

$$\begin{aligned}
f^*(u) &= \sup_{x \in \mathcal{H}} \langle x \mid u \rangle - f(x) \\
&= \sup_{x = (x_i)_{i \in I} \in \mathcal{H}} \sum_{i \in I} \langle x_i \mid u_i \rangle - f_i(x_i) \\
&= \sum_{i \in I} \sup_{x_i \in \mathcal{H}_i} \langle x_i \mid u_i \rangle - f_i(x_i) \\
&= \sum_{i \in I} f_i^*(u_i).
\end{aligned}$$

□

Example 15. *Let* $q \in]1, +\infty[$. *Recall that the* ℓ_q *norm of* $\mathbb{R}^N$ *is defined as*

$$(\forall x = (x^{(n)})_{1 \le n \le N} \in \mathbb{R}^N) \quad \|x\|_q = \left(\sum_{n=1} |x^{(n)}|^q \right)^{1/q}.$$

Let us now consider the function defined as

$$(\forall x \in \mathbb{R}^N) \quad f(x) = \frac{1}{q}\|x\|_q^q.$$

Since this function is separable, it follows from Proposition 6 and Example 14iii) that

$$(\forall u \in \mathbb{R}^N) \quad f^*(u) = \frac{1}{q^*}\|u\|_{q^*}^{q^*}.$$

with $\frac{1}{q} + \frac{1}{q^*} = 1$

Proposition 7. (Fenchel-Young inequality)

If $f : \mathcal{H} \to]-\infty, +\infty]$ *is proper, then*

$$\big(\forall (x, u) \in \mathcal{H}^2\big) \qquad f(x) + f^*(u) \ge \langle x \mid u \rangle .$$

Proof. We have

$$(\forall u \in \mathcal{H})(\forall x \in \mathcal{H}) \qquad f^*(u) \ge \langle x \mid u \rangle - f(x).$$

□

One may wonder when Fenchel-Young inequality becomes an inequality. The following result provides a partial answer to this question.

Proposition 8

Assume that $f : \mathcal{H} \to]-\infty, +\infty]$ *is a proper convex function which is Gâteaux differentiable at* $x \in \operatorname{int} \operatorname{dom} f$. *Let* $u \in \mathcal{H}$.
Then

$$f(x) + f^*(u) = \langle x \mid u \rangle .$$

if and only if $u = \nabla f(x)$.

Proof. We have

$$\begin{aligned} & f(x) + f^*(u) = \langle x \mid u \rangle \\ \Leftrightarrow \quad & f^*(u) = \langle x \mid u \rangle - f(x). \end{aligned}$$

By using the definition of the conjugate, this is also equivalent to

$$\begin{aligned} & (\forall y \in \mathcal{H}) \quad \langle y \mid u\rangle - f(y) \leq \langle x \mid u\rangle - f(x) \\ \Leftrightarrow\ & (\forall y \in \mathcal{H}) \quad f(y) \geq f(x) + \langle y - x \mid u\rangle\,. \end{aligned} \tag{3.1}$$

From Lemma 2, this inequality is satisfied if $u = \nabla f(x)$.
On the other hand, if u satisfies (3.1) then, for every $\alpha \in\ ]0,+\infty[$,

$$f(x+\alpha(y-x)) \geq f(x) + \alpha\,\langle y - x \mid u\rangle\,.$$

Therefore

$$\begin{aligned} & \langle y - x \mid \nabla f(x)\rangle = \lim_{\substack{\alpha\to 0\\ \alpha>0}} \frac{f(x+\alpha(y-x)) - f(x)}{\alpha} \geq \langle y - x \mid u\rangle \\ \Leftrightarrow\quad & \langle y - x \mid u - \nabla f(x)\rangle \leq 0. \end{aligned}$$

By choosing $y = x + u - \nabla f(x)$, we deduce that $\|u - \nabla f(x)\|^2 = 0$, that is $u = \nabla f(x)$. □

3.3 Biconjugate

Definition 29

Let $\mathcal{H}$ be a Hilbert space and $f\colon \mathcal{H} \to [-\infty,+\infty]$.
The **biconjugate** of f is $f^{**} = (f^*)^*$.

Property 16. *Let $f\colon \mathcal{H} \to [-\infty,+\infty]$.*

i) $f^{**} \leq f$

Proof. If $(\exists x \in \mathcal{H})\ f(x) = -\infty$, then $f^* = +\infty \Rightarrow f^{**} = -\infty$.
We can thus assume that $f\colon \mathcal{H} \to\ ]-\infty,+\infty]$ and that it is proper (otherwise the inequality trivially holds).
According to Fenchel-Young inequality,

$$\begin{aligned} (\forall x \in \mathcal{H}) \qquad & (\forall u \in \mathcal{H}) \quad \langle x \mid u\rangle - f^*(u) \leq f(x) \\ & \Rightarrow f^{**}(x) = \sup_{u\in\mathcal{H}} \langle x \mid u\rangle - f^*(u) \leq f(x). \end{aligned}$$

□

ii) *If $f\colon \mathcal{H} \to [-\infty,+\infty]$ and $g\colon \mathcal{H} \to [-\infty,+\infty]$ with $f \leq g$, then $f^* \geq g^*$, and $f^{**} \leq g^{**}$.*

Proof. If, for every $x \in \mathcal{H}$, $f(x) \leq g(x)$ then, for every $u \in \mathcal{H}$,

$$f^*(u) = \sup_{x\in\mathcal{H}} \langle x \mid u\rangle - f(x) \geq \sup_{x\in\mathcal{H}} \langle x \mid u\rangle - g(x) = g^*(u).$$

□

iii) $f^{***} = f^*$

Proof. We have $f^{***} = (f^*)^{**} \leq f^*$.
In addition, $f^{**} \leq f \Rightarrow f^{***} = (f^{**})^* \geq f^*$ □

Lemma 3

Let $\mathcal{H}$ be a Hilbert space.

i) *If $f \in \Gamma_0(\mathcal{H})$, then the set $\mathcal{A}_f$ of continuous affine functions from $\mathcal{H}$ to $\mathbb{R}$ which lower-bound f is nonempty and*

$$(\forall x \in \mathcal{H}) \quad f(x) = \sup_{a \in \mathcal{A}_f} a(x).$$

ii) *If $\mathcal{H}$ is finite dimensional, $f \colon \mathcal{H} \to \,]-\infty, +\infty]$ is convex, and $\overline{x} \in \operatorname{int}(\operatorname{dom} f)$, then there exists $a \in \mathcal{A}_f$ such that $f(\overline{x}) = a(\overline{x})$.*

Proof. The proof being rather technical, it is deferred to Appendix B. □

Theorem 13. (Moreau-Fenchel)

Let $\mathcal{H}$ be a Hilbert space and let $f \colon \mathcal{H} \to \,]-\infty, +\infty]$ be a proper function.
If f is l.s.c. and convex, then $f^{**} = f$.

Proof. If $f^{**} = f$, then $f \in \Gamma_0(\mathcal{H})$.
Conversely, assume that $f \in \Gamma_0(\mathcal{H})$.
Let us first consider the case when f is a continuous affine function, i.e. there exists $(v, \alpha) \in \mathcal{H} \times \mathbb{R}$ such that $f = \langle \cdot \mid v \rangle + \alpha$. We have then

$$(\forall u \in \mathcal{H}) \quad f^*(u) = \iota_{\{v\}}(u) - \alpha.$$

and

$$(\forall x \in \mathcal{H}) \quad f^{**}(x) = \sup_{u \in \mathcal{H}} \langle u \mid x \rangle - \iota_{\{v\}}(u) + \alpha = \langle v \mid x \rangle + \alpha = f(x).$$

Let $a \colon \mathcal{H} \to \mathbb{R}$ be a continuous affine function such that $a \leq f$. Then

$$a = a^{**} \leq f^{**}.$$

Hence, according to Lemma 3i),

$$f = \sup_{a \in \mathcal{A}_f} a \leq f^{**}$$

and, since we already know that $f^{**} \leq f$, we deduce that $f = f^{**}$. □

Corollary 1

If $f \in \Gamma_0(\mathcal{H})$, then $f^* \in \Gamma_0(\mathcal{H})$.

Proof. As a consequence of Fenchel-Moreau theorem, if $f \in \Gamma_0(\mathcal{H})$, then

$$\begin{cases} \operatorname{dom} f \neq \varnothing \Leftrightarrow f^* > -\infty \\ f^{**} = f > -\infty \ \Leftrightarrow \operatorname{dom} f^* \neq \varnothing \end{cases}$$

which shows that f^* is proper. Since we already know that f^* is l.s.c. and convex, $f^* \in \Gamma_0(\mathcal{H})$. □

Proposition 9

Let $\mathcal{H}$ be a Hilbert space and let $f\colon \mathcal{H} \to \,]-\infty, +\infty]$ be a proper function such that $\operatorname{dom} f^* \neq \varnothing$.
*Then, f^{**} is the **lower semicontinuous convex envelope** of f, i.e. the largest l.s.c. convex function $g\colon \mathcal{H} \to \,]-\infty, +\infty]$ such that*

$$(\forall x \in \mathcal{H}) \qquad g(x) \leq f(x).$$

Proof. Let F be the set of l.s.c convex functions $g\colon \mathcal{H} \to \,]-\infty, +\infty]$ such that $(\forall x \in \mathcal{H})$ $g(x) \leq f(x)$.
Since $\operatorname{dom} f^* \neq \varnothing$, $f^{**} > -\infty$. In addition $f^{**} \leq f$ and f^{**} is convex and l.s.c. Hence $f^{**} \in F$, which is a nonempty set.
The function $\check{f}\colon x \mapsto \sup \big\{ g(x) \bigm| g \in F \big\}$ is the l.s.c. convex envelope of f. Since $\check{f} \leq f$, it can be deduced that $\check{f} \in \Gamma_0(\mathcal{H})$.
In addition, $f^{**} \in F \Rightarrow f^{**} \leq \check{f}$ and, according to Moreau-Fenchel theorem,

$$\check{f} \leq f \quad \Rightarrow \quad f^{**} \geq \check{f}^{**} = \check{f}.$$

In conclusion, $\check{f} = f^{**}$. □

3.4 Support functions

Definition 30

Let $\mathcal{H}$ be a Hilbert space and $C \subset \mathcal{H}$. σ_C is the **support function** of C if

$$\begin{aligned} (\forall u \in \mathcal{H}) \qquad \sigma_C(u) &= \sup_{x \in C} \langle x \mid u \rangle \\ &= \iota_C^*(u). \end{aligned}$$

Remark 14. *It follows from Property 11 and Theorem 13 that, if C is a nonempty closed convex set, then $\sigma_C^* = \iota_C^{**} = \iota_C$.*

Example 16.

i) *Let* $f\colon \mathbb{R} \to \,]-\infty,+\infty] : x \mapsto \begin{cases} \delta_1 x & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ \delta_2 x & \text{if } x > 0 \end{cases}$

with $-\infty \le \delta_1 < \delta_2 \le +\infty$.
Then, $f = \sigma_C$ *where* C *is the closed real interval such that* $\inf C = \delta_1$ *et* $\sup C = \delta_2$.

Proof. For every $u \in \mathbb{R}$,

$$\sigma_C(u) = \begin{cases} \sup_{x\in[\delta_1,\delta_2]} xu & \text{if } (\delta_1,\delta_2) \in \mathbb{R}^2 \\ \sup_{x\in]-\infty,\delta_2]} xu & \text{if } \delta_1 = -\infty \text{ and } \delta_2 \in \mathbb{R} \\ \sup_{x\in[\delta_1,+\infty[} xu & \text{if } \delta_1 \in \mathbb{R} \text{ and } \delta_2 = +\infty \\ \sup_{x\in]-\infty,+\infty[} xu & \text{if } \delta_1 = -\infty \text{ and } \delta_2 = +\infty. \end{cases}$$

We deduce that $\sigma_C(0) = 0$ and, when $u > 0$ (resp. $u < 0$), the maximum is reached when $x \to \delta_2$ (resp. $x \to \delta_1$) and we have then

$$\sigma_C(u) = \begin{cases} \delta_2 u & \text{if } \delta_2 \in \mathbb{R} \\ +\infty & \text{if } \delta_2 = +\infty \end{cases}$$

$$\Big(\text{rep. } \sigma_C(u) = \begin{cases} \delta_1 u & \text{if } \delta_1 \in \mathbb{R} \\ +\infty & \text{if } \delta_1 = -\infty \end{cases}\Big).$$

□

ii) *Let f be the ℓ^q norm of $\mathbb{R}^N$ with $q \in [1,+\infty]$.*
We have $f = \sigma_C$ where

$$C = \big\{ y \in \mathbb{R}^N \bigm| \|y\|_{q^*} \le 1 \big\}$$

with $\frac{1}{q} + \frac{1}{q^*} = 1$.
In particular, the ℓ_1 norm is the support function of the hypercube $C = [-1,1]^N$.

Proof. According to Hölder's inequality,

$$(\forall u \in \mathbb{R}^N) \qquad \sigma_C(u) = \sup_{x\in C} \langle u \mid x\rangle = \sup_{x\in C} \|u\|_q \|x\|_{q^*} = \|u\|_q.$$

□

3.5 Fenchel-Rockafellar duality

3.5.1 Main results

In the remainder of this chapter, we will be interested in the two following optimization problems.

Problem 2. (Primal)

Let $\mathcal{H}$ and $\mathcal{G}$ be two real Hilbert spaces.
Let $f\colon \mathcal{H} \to]-\infty,+\infty]$, $g\colon \mathcal{G} \to]-\infty,+\infty]$. Let $L \in \mathcal{B}(\mathcal{H},\mathcal{G})$.
We want to

$$\underset{x\in\mathcal{H}}{\text{minimize}}\;\; f(x)+g(Lx).$$

Problem 3. (Dual)

Let $\mathcal{H}$ and $\mathcal{G}$ be two real Hilbert spaces.
Let $f\colon \mathcal{H} \to]-\infty,+\infty]$, $g\colon \mathcal{G} \to]-\infty,+\infty]$. Let $L \in \mathcal{B}(\mathcal{H},\mathcal{G})$.
We want to

$$\underset{v\in\mathcal{G}}{\text{minimize}}\;\; f^*(-L^*v)+g^*(v).$$

Remark 15.

i) *As a consequence of Property 15vi), the dual problem is a convex optimization problem.*

ii) *If $f \in \Gamma_0(\mathcal{H})$ and $g \in \Gamma_0(\mathcal{G})$, it follows from Theorem 13 that the dual of Problem 3 is Problem 2.*

Theorem 14. (Weak duality)

Let $\mathcal{H}$ and $\mathcal{G}$ be two real Hilbert spaces.
Let f be a proper fonction from $\mathcal{H}$ to $]-\infty,+\infty]$, g be a proper function from $\mathcal{G}$ to $]-\infty,+\infty]$, and $L \in \mathcal{B}(\mathcal{H},\mathcal{G})$. Let

$$\mu = \inf_{x\in\mathcal{H}} f(x)+g(Lx)$$

and

$$\mu^* = \inf_{v\in\mathcal{G}} f^*(-L^*v)+g^*(v).$$

We have $\mu \geq -\mu^*$.
If $\mu \in \mathbb{R}$, $\mu+\mu^*$ is called the ***duality gap****.*

Proof. According to Fenchel-Young inequality, for every $x \in \mathcal{H}$ and $v \in \mathcal{G}$,

$$\begin{aligned}
&\begin{cases} f(x)+f^*(-L^*v) \geq \langle x \mid -L^*v\rangle \\ g(Lx)+g^*(v) \geq \langle Lx \mid v\rangle \end{cases}\\
\Rightarrow\quad & f(x)+g(Lx)+f^*(-L^*v)+g^*(v) \geq \langle x \mid -L^*v\rangle + \langle Lx \mid v\rangle = 0\\
\Rightarrow\quad & \inf_{x\in\mathcal{H}}\big(f(x)+g(Lx)\big) + \inf_{v\in\mathcal{G}}\big(f^*(-L^*v)+g^*(v)\big) \geq 0.
\end{aligned}$$

$\square$

Theorem 15. (Strong duality)

Let $\mathcal{H}$ and $\mathcal{G}$ be two real Hilbert spaces.
Let $f \in \Gamma_0(\mathcal{H})$, $g \in \Gamma_0(\mathcal{G})$, and $L \in \mathcal{B}(\mathcal{H}, \mathcal{G})$.
If $0 \in \operatorname{int}(\operatorname{dom} g - L(\operatorname{dom} f))$, then

$$\mu = \inf_{x \in \mathcal{H}} f(x) + g(Lx) = -\min_{v \in \mathcal{G}} f^*(-L^*v) + g^*(v) = -\mu^*.$$

Remark 16.

i) *Hereabove*

$$\operatorname{dom} g - L(\operatorname{dom} f) = \big\{ y - Lx \;\big|\; (x, y) \in \operatorname{dom} f \times \operatorname{dom} g \big\}.$$

ii) *A sufficient condition for $0 \in \operatorname{int}(\operatorname{dom} g - L(\operatorname{dom} f))$ is*

$$\operatorname{int}(\operatorname{dom} g) \cap L(\operatorname{dom} f) \neq \varnothing \quad \textit{or} \quad \operatorname{dom} g \cap \operatorname{int}\big(L(\operatorname{dom} f)\big) \neq \varnothing.$$

Proof. If $x \in \operatorname{int}(\operatorname{dom} g) \cap L(\operatorname{dom} f)$, then $\operatorname{dom} g$ includes a ball $B(x, \rho)$ centered at x with radius $\rho \in\ ]0, +\infty[$.
Then $B(x, \rho) - x = B(0, \rho) \subset \operatorname{dom} g - L(\operatorname{dom} f)$,
which shows that 0 is an interior point of $\operatorname{dom} g - L(\operatorname{dom} f)$.
The second sufficient condition is similarly obtained. □

iii) *Suppose that the assumptions of Theorem 15 hold.*
Let $\widehat{v} \in \mathcal{G}$ be such that $\mu^* = f^*(-L^*\widehat{v}) + g^*(\widehat{v})$.
If there exists $\widehat{x} \in \mathcal{H}$ such that $\mu = f(\widehat{x}) + g(L\widehat{x})$, then

$$f^*(-L^*\widehat{v}) = \langle -L^*\widehat{v} \mid \widehat{x} \rangle - f(\widehat{x}) \quad \textit{and} \quad g^*(\widehat{v}) = \langle \widehat{v} \mid L\widehat{x} \rangle - g(L\widehat{x}).$$

Proof. According to Fenchel-Young inequality,

$$f^*(-L^*\widehat{v}) \geq \langle -L^*\widehat{v} \mid \widehat{x} \rangle - f(\widehat{x}) \quad \text{and} \quad g^*(\widehat{v}) \geq \langle \widehat{v} \mid L\widehat{x} \rangle - g(L\widehat{x}).$$

If one of the inequalities is strict,

$$\mu^* > \langle -L^*\widehat{v} \mid \widehat{x} \rangle - f(\widehat{x}) + \langle \widehat{v} \mid L\widehat{x} \rangle - g(L\widehat{x}) = -\mu,$$

which contradicts the strong duality result. □

The latter result might be useful to deduce a solution $\widehat{x}$ of the primal problem knowing a solution $\widehat{v}$ to the dual one.

3.5.2 Proof of strong duality

Let us first state some properties of the so-called value function.

Proposition 10

Let $\mathcal{H}$ and $\mathcal{G}$ be Hilbert spaces. Let $f \in \Gamma_0(\mathcal{H})$, let $g \in \Gamma_0(\mathcal{G})$, and let $L \in \mathcal{B}(\mathcal{H}, \mathcal{G})$. Let ϑ be the ***value function*** *defined as*

$$(\forall y \in \mathcal{G}) \quad \vartheta(y) = \inf_{x \in \mathcal{H}} f(x) + g(Lx + y).$$

Then

i) $\operatorname{dom} \vartheta = \operatorname{dom} g - L(\operatorname{dom} f)$

ii) *ϑ is convex*

iii) $(\forall v \in \mathcal{G})$ $\vartheta^*(v) = f^*(-L^*v) + g^*(v)$.

Proof.

i) Let $y \in \mathcal{G}$.

$$\begin{aligned}
\vartheta(y) < +\infty &\Leftrightarrow (\exists x \in \mathcal{H}) \ \begin{cases} f(x) < +\infty \\ g(Lx + y) < +\infty \end{cases} \\
&\Leftrightarrow (\exists x \in \operatorname{dom} f) \ \ Lx + y \in \operatorname{dom} g \\
&\Leftrightarrow y \in \operatorname{dom} g - L(\operatorname{dom} f).
\end{aligned}$$

ii) For every $(y_1, y_2) \in \mathcal{G}^2$ and $\alpha \in]0, 1[$,

$$\begin{aligned}
&\vartheta(\alpha y_1 + (1-\alpha) y_2) \\
&= \inf_{(x_1,x_2) \in \mathcal{H}^2} f(\alpha x_1 + (1-\alpha) x_2) + g(L(\alpha x_1 + (1-\alpha) x_2) + \alpha y_1 + (1-\alpha) y_2) \\
&= \inf_{(x_1,x_2) \in \mathcal{H}^2} f(\alpha x_1 + (1-\alpha) x_2) + g(\alpha(L x_1 + y_1) + (1-\alpha)(L x_2 + y_2)) \\
&\leq \inf_{(x_1,x_2) \in \mathcal{H}^2} \alpha(f(x_1) + g(L x_1 + y_1)) + (1-\alpha)(f(x_2) + g(L x_2 + y_2)) \\
&= \alpha \vartheta(y_1) + (1-\alpha) \vartheta(y_2).
\end{aligned}$$

iii) For every $v \in \mathcal{G}$,

$$\begin{aligned}
\vartheta^*(v) &= \sup_{y \in \mathcal{G}} \langle y \mid v \rangle - \vartheta(y) \\
&= \sup_{x \in \mathcal{H}, y \in \mathcal{G}} \langle y \mid v \rangle - f(x) - g(Lx + y) \\
&= \sup_{x \in \mathcal{H}, y \in \mathcal{G}} - \langle Lx \mid v \rangle - f(x) + \langle Lx + y \mid v \rangle - g(Lx + y) \\
&= \sup_{x \in \mathcal{H}} (\langle x \mid -L^* v \rangle - f(x) + \sup_{y \in \mathcal{H}} \langle Lx + y \mid v \rangle - g(Lx + y)) \\
&= \sup_{x \in \mathcal{H}} \langle x \mid -L^* v \rangle - f(x) + g^*(v) \\
&= f^*(-L^* v) + g^*(v).
\end{aligned}$$

$\square$

Let us now provide a proof of Theorem 15 when $\mathcal{G}$ is finite dimensional.
First note that $\vartheta(0) = \mu$.
If there exists $y \in \mathcal{G}$ such that $\vartheta(y) = -\infty$, then

$$(\forall v \in \mathcal{H}) \quad \vartheta^*(v) = +\infty.$$

In addition since $0 \in \operatorname{int}(\operatorname{dom} g - L(\operatorname{dom} f)) = \operatorname{int}(\operatorname{dom} \vartheta)$, there exists $\alpha \in]0,1[$ such that $z = -\alpha y/(1-\alpha) \in \operatorname{dom} \vartheta$.
As ϑ is convex, we have then

$$\vartheta(0) = \vartheta((1-\alpha)z + \alpha y) \leq (1-\alpha)\vartheta(z) + \alpha\vartheta(y) = -\infty$$

Hence, by using Proposition 10iii),

$$(\forall v \in \mathcal{G}) \quad \mu = -\infty = -\vartheta^*(v) = -f^*(-Lv) - g^*(v) = -\mu^*.$$

Assume now that $\vartheta \colon \mathcal{G} \to]-\infty, +\infty]$. According to Lemma 3ii), there exists $\widehat{v} \in \mathcal{G}$ such that

$$(\forall y \in \mathcal{G}) \quad \vartheta(y) \geq \vartheta(0) + \langle \widehat{v} \mid y \rangle$$

which implies that

$$\begin{aligned} & \vartheta^*(\widehat{v}) = \sup_{y \in \mathcal{G}} \langle \widehat{v} \mid y \rangle - \vartheta(y) \leq -\vartheta(0) \\ \Leftrightarrow \quad & \mu \leq -f^*(-L\widehat{v}) - g^*(\widehat{v}). \end{aligned}$$

On the other hand, by weak duality,

$$-\mu^* = \sup_{v \in \mathcal{G}} -f^*(-Lv) - g^*(v) \leq \mu.$$

We conclude that $\mu = -\mu^* = -f^*(-L\widehat{v}) - g^*(\widehat{v})$.

3.6 Minimax problem

Theorem 16. (Minimax)

Let $\mathcal{H}$ and $\mathcal{G}$ be two Hilbert spaces. Let $L \in \mathcal{B}(\mathcal{H}, \mathcal{G})$.
Let C be a nonempty closed convex set of $\mathcal{H}$.
Let D be a nonempty closed and bounded convex set of $\mathcal{G}$.
Then,

$$\inf_{x \in C} \max_{v \in D} \langle Lx \mid v \rangle = \max_{v \in D} \inf_{x \in C} \langle Lx \mid v \rangle.$$

Proof. Since D is bounded, closed, and convex,

$$(\forall y \in \mathcal{G}) \qquad \sigma_D(y) = \sup_{z\in D} \langle y \mid z\rangle = \max_{z\in D} \langle y \mid z\rangle \le \|y\| \sup_{z\in D} \|z\| < +\infty. \tag{3.2}$$

We have thus

$$\mu = \inf_{x\in C} \max_{v\in D} \langle Lx \mid v\rangle = \inf_{x\in \mathcal{H}} \iota_C(x) + \sigma_D(Lx).$$

According to (3.2), $\operatorname{dom}\sigma_D = \mathcal{G}$. As $C \neq \varnothing$, it follows that $\operatorname{int}(\operatorname{dom}\sigma_D) \cap L(C) \neq \varnothing$. Since $\iota_C \in \Gamma_0(\mathcal{H})$ and $\sigma_D = \iota_D^* \in \Gamma_0(\mathcal{G})$, strong duality holds and

$$\begin{aligned}\mu = -\min_{v\in\mathcal{G}} \iota_C^*(-L^*v) + \sigma_D^*(v) &= -\min_{v\in\mathcal{G}} \sigma_C(-L^*v) + \iota_D(v)\\ &= -\min_{v\in D} \sup_{x\in C} \langle -L^*v \mid x\rangle\\ &= -\min_{v\in D}(-\inf_{x\in C} \langle v \mid Lx\rangle) = \max_{v\in D} \inf_{x\in C} \langle Lx \mid v\rangle .\end{aligned}$$

□

Example 17. *Two players have N possible actions numbered by $\{1,\ldots,N\}$ in their game. Assume that they play independently (e.g. rock-paper-scissor).*
For every $i \in \{1,\ldots,N\}$, $x^{(i)}$ is the probability that the first one plays the i-th action and $v^{(i)}$ is the probability that the second one plays the i-th action[1]*. We assume that $x = (x^{(i)})_{1\le i\le N}$ (resp. $v = (v^{(i)})_{1\le i\le N}$) is constrained to belong to a closed convex subset C (resp. D) of $\big\{p = (p^{(i)})_{1\le i\le N} \in [0,1]^N \;\big|\; \sum_{i=1}^N p^{(i)} = 1\big\}$.*
*For every $(i,j) \in \{1,\ldots,N\}^2$, let $L_{i,j}$ be the reward of the second player when playing action i, while the first one plays action j, and let $-L_{i,j}$ be the reward of the first player when playing action j, while the second one plays action i (**zero-sum game**). The average reward for the second player is thus*

$$\sum_{i=1}^N \sum_{j=1}^N L_{i,j} x_j v_i = \langle Lx \mid v\rangle$$

*where $L = (L_{i,j})_{1\le i,j\le N}$ is the **pay-off matrix**, whereas $-\langle Lx \mid v\rangle$ is the average reward for the first player.*
Player 1 thinks that player 2 is clever and will choose v so as to maximize his/her own reward $\langle Lx \mid v\rangle$ for any given choice for x. His/her strategy is thus to minimize the best reward of player 2, i.e.

$$\operatorname*{minimize}_{x\in C} \; \max_{v\in D} \langle Lx \mid v\rangle .$$

If player 2 has a similar strategy, he/she will

$$\operatorname*{maximize}_{v\in D} \; \min_{x\in C} \langle Lx \mid v\rangle .$$

Let players 1 and 2 play their strategies $\widehat{x}$ and $\widehat{v}$.
We have

$$\begin{aligned}\langle L\widehat{x} \mid \widehat{v}\rangle &\le \max_{v\in D} \langle L\widehat{x} \mid v\rangle = \min_{x\in C} \max_{v\in D} \langle Lx \mid v\rangle\\ \langle L\widehat{x} \mid \widehat{v}\rangle &\ge \min_{x\in C} \langle Lx \mid \widehat{v}\rangle = \max_{v\in D} \min_{x\in C} \langle Lx \mid v\rangle .\end{aligned}$$

[1]Games where the strategies of the players are defined in a probabilistic manner are called **mixed games**.

The minimax theorem states that

$$\langle L\widehat{x} \mid \widehat{v}\rangle = \max_{v\in D} \langle L\widehat{x} \mid v\rangle$$

and

$$\langle L\widehat{x} \mid \widehat{v}\rangle = \min_{x\in C} \langle Lx \mid \widehat{v}\rangle ,$$

*that is $\widehat{v}$ is the **best response** of player 2 to player 1 and $\widehat{x}$ is the **best response** of player 1 to player 2. We have obtained a **Nash equilibrium**.*

Chapter 4

Linear programming

4.1 Motivation example

We want to purchase quantities of two kinds of food so as to maximize protein intake, while fulfilling the following diet requirements.

	food 1 (per unit)	food 2 (per unit)	limits (per meal)
kcal	200	100	400
fat (g)	2	4	8
protein (g)	3	2	

Table 4.1: Diet requirements.

Let $x^{(1)} \in \mathbb{R}$ and $x^{(2)} \in \mathbb{R}$ be the quantities of purchased products 1 and 2.
The diet constraints are

$$\begin{aligned} x^{(1)} &\geq 0 \\ x^{(2)} &\geq 0 \\ 2x^{(1)} + x^{(2)} &\leq 4 \\ x^{(1)} + 2x^{(2)} &\leq 4 \end{aligned}$$

The function to maximize is

$$f(x^{(1)}, x^{(2)}) = 3x^{(1)} + 2x^{(2)}.$$

A graphical solution of the problem (see Figure 4.1) yields the optimal choice:

$$S = \begin{bmatrix} x^{(1)} \\ x^{(2)} \end{bmatrix} = \frac{4}{3} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$f(x^{(1)}, x^{(2)}) = \frac{20}{3}$$

Such a graphical resolution is only possible when we have few food products and the number of constraints is small. We will see in this chapter how to deal with more complicated scenarios.

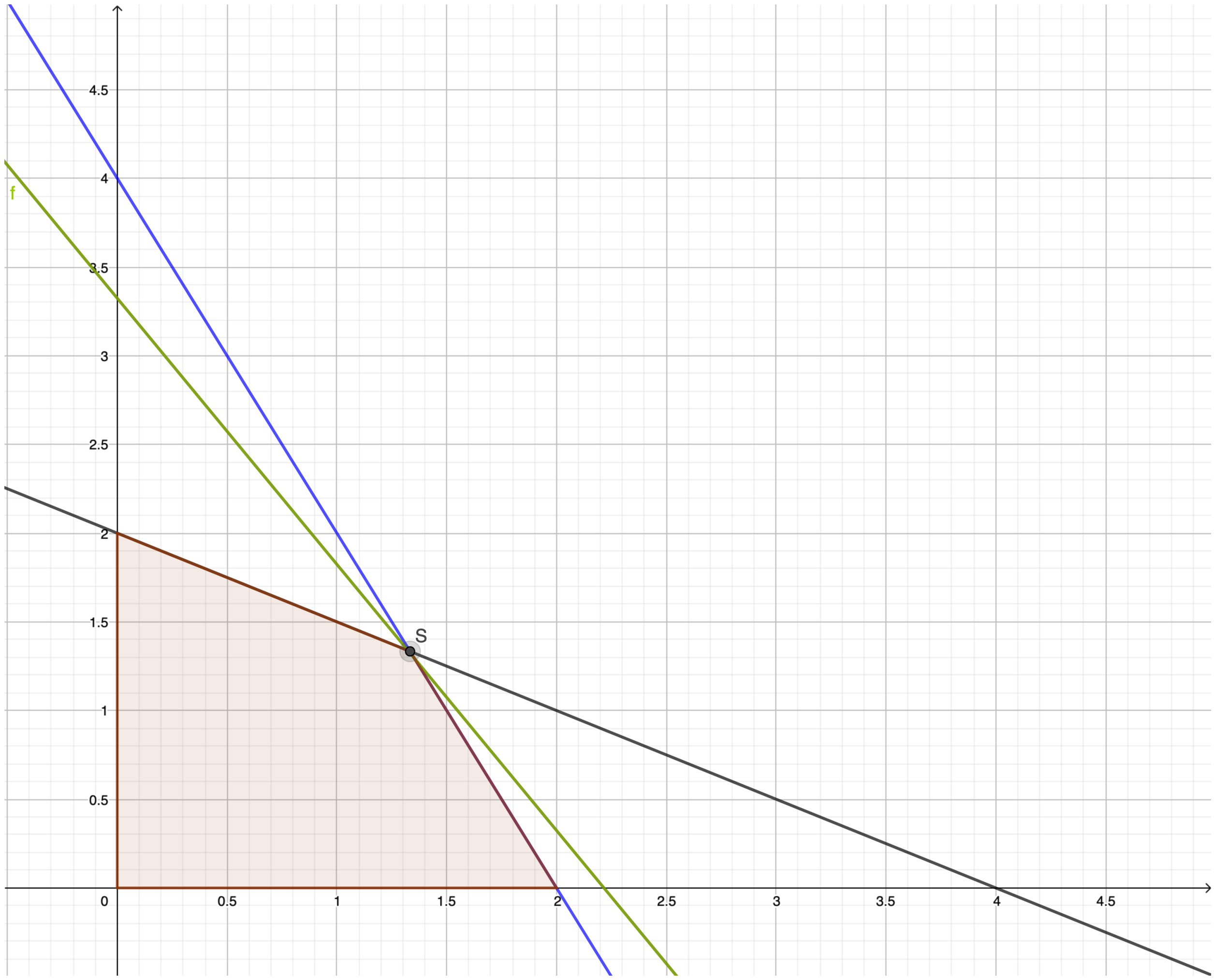

Figure 4.1: Graphical solution of the diet problem.

4.2 Canonical problem

Let us introduce the canonical formulation of **linear programming** problems.

Problem 4

Let $L \in \mathbb{R}^{K\times N}$, $b \in \mathbb{R}^K$, and $c \in \mathbb{R}^N$.
We consider the following problem:

$$\textbf{Primal-LP}: \qquad \underset{x\in[0,+\infty[^N}{\text{minimize}} \;\; \langle c \mid x\rangle \quad \text{s.t.} \quad Lx \geq b$$

and the dual problem:

$$\textbf{Dual-LP}: \qquad \underset{y \in [0,+\infty[^K}{\text{maximize}} \ \langle b \mid y \rangle \quad \text{s.t.} \quad L^\top y \le c.$$

Example 18. *The diet problem can be modeled as Primal-LP with*

$$\begin{aligned} &N = 2, \quad K = 2 \\ &c = \begin{bmatrix} -3 \\ -2 \end{bmatrix} \\ &L = -\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad b = -\begin{bmatrix} 4 \\ 4 \end{bmatrix}, \end{aligned}$$

although Dual-LP may appear a more natural model.

Remark 17.

i) *Primal-LP and Dual-LP are dual problems as defined in Section 3.5.*

Proof. Left as exercise. □

ii) *An equality constraint $Lx = b$ can be transformed into the inequality constraint*

$$\begin{bmatrix} L \\ -L \end{bmatrix} x \ge \begin{bmatrix} b \\ -b \end{bmatrix}.$$

iii) *Variables x with arbitrary signs can be handled by setting $x = x_+ - x_-$ where*

$$\begin{bmatrix} x_+ \\ x_- \end{bmatrix} \ge 0.$$

Definition 31

i) The **primal feasibility set** is

$$\mathcal{A} = \big\{ x \in [0,+\infty[^N \bigm| Lx \ge b \big\}.$$

ii) The primal problem is **feasible** if $\mathcal{A} \neq \varnothing$.

iii) The primal problem is (lower) **bounded** if $\mu = \inf_{x \in \mathcal{A}} \langle c \mid x \rangle > -\infty$.

iv) The **dual feasibility set** is

$$\mathcal{B} = \big\{ y \in [0,+\infty[^K \bigm| L^\top y \le c \big\}.$$

v) The dual problem is **feasible** if $\mathcal{B} \neq \varnothing$.

vi) The dual problem is (upper) **bounded** if $-\mu^* = \sup_{y \in \mathcal{B}} \langle b \mid y \rangle < +\infty$.

Remark 18. *$\mathcal{A}$ and $\mathcal{B}$ are (closed convex) polyhedra (see Definition 20), so that Primal-LP and Dual-LP are convex optimization problems.*

Proof. Let $(b^{(j)})_{1\le j\le K}$ be the components of b and let $(\ell_j)_{1\le j\le K}$ be the lines of L. Without loss of generality, we can assume that none of the line vectors is null (otherwise it would define an uninformative constraint).
We have then

$$x = (x^{(i)})_{1\le i\le N} \in \mathcal{A} \quad \Leftrightarrow \quad \begin{cases} (\forall i \in \{1,\dots,N\}) & x^{(i)} \ge 0 \\ (\forall j \in \{1,\dots,K\}) & \ell_j^\top x \ge b^{(j)} \end{cases}$$

This shows that $\mathcal{A}$ is the intersection of $N+K$ closed half-spaces, i.e. it is a polyhedron. The same argument applies to $\mathcal{B}$. $\square$

4.3 Farkas lemma

Here is the most classical form of this famous result.

Proposition 11. (Farkas)

Let $c \in \mathbb{R}^N$ and $A \in \mathbb{R}^{K\times N}$. Only one of the following two cases arises:

i) *there exists $x \in \mathbb{R}^N$ such that $Ax \ge 0$ and $\langle c \mid x\rangle < 0$*

ii) *there exists $y \in [0,+\infty[^K$ such that $A^\top y = c$.*

Proof. Let $(e_j)_{1\le k\le K}$ be the canonical basis of $\mathbb{R}^K$ and let C is the nonempty closed and convex cone of $\mathbb{R}^N$ generated by the vectors $(A^\top e_j)_{1\le j\le K}$. We have

$$c \in C \quad \Leftrightarrow \quad (\exists (y^{(j)})_{1\le j\le K} \in [0,+\infty[^K)\;\; c = \sum_{j=1}^K y^{(j)} A^\top e_j \quad \Leftrightarrow \quad ii).$$

From Theorem 25i), either $c \in C$ or (exclusively) there exists a closed half-space including C and not containing c. The latter case means that

$$(\exists x \in \mathbb{R}^N \setminus \{0\})(\exists \eta \in \mathbb{R})(\forall z \in C) \quad \langle z \mid x\rangle \ge \eta > \langle c \mid x\rangle\,. \tag{4.1}$$

We can choose $\eta = \inf_{z\in C}\langle z \mid x\rangle$. Since $0 \in C$, $\eta \le 0$.
Ìn addition, if $\eta < 0$, then there exists $z \in C$ such that $\langle z \mid x\rangle < 0$. Since C is a cone, this implies that $(\forall \alpha \in\,]0,+\infty[)$ $\alpha z \in C$ and $\alpha\langle z \mid x\rangle \ge \eta$, which is inconsistent when $\alpha \to +\infty$.

Therefore $\eta = 0$ and (4.1) is equivalent to

$$
\begin{aligned}
& (\exists x \in \mathbb{R}^N \setminus \{0\}) \begin{cases} (\forall z \in C)\, \langle z \mid x \rangle \geq 0 \\ \langle c \mid x \rangle < 0 \end{cases} \\
\Leftrightarrow \quad & (\exists x \in \mathbb{R}^N \setminus \{0\}) \begin{cases} (\forall y \in [0,+\infty[^K)\, \langle A^\top y \mid x \rangle \geq 0 \\ \langle c \mid x \rangle < 0 \end{cases} \\
\Leftrightarrow \quad & (\exists x \in \mathbb{R}^N) \begin{cases} (\forall y \in [0,+\infty[^K)\, \langle y \mid Ax \rangle \geq 0 \\ \langle c \mid x \rangle < 0 \end{cases} \\
\Leftrightarrow \quad & i).
\end{aligned}
$$

□

A second formulation of the Farkas lemma can be viewed as a necessary condition for the inequalities $Ax \geq b$ to have a solution.

Corollary 2

Let $b \in \mathbb{R}^K$ and $A \in \mathbb{R}^{K \times N}$. Only one of the following two cases arises:

i) *there exists $x \in \mathbb{R}^N$ such that $Ax \geq b$*

ii) *there exists $y \in [0,+\infty[^K$ such that $A^\top y = 0$ and $\langle b \mid y \rangle > 0$.*

Proof. Without loss of generality, in ii) it can be assumed that $\langle b \mid y \rangle = 1$. We have then

$$
ii) \quad \Leftrightarrow \quad (\exists y \in [0,+\infty[^K) \;\; \widetilde{A}^\top y = \widetilde{c}
$$

where $\widetilde{A} = [A \;\; b]$ and $\widetilde{c} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.
According to Farkas lemma, either this situation happens or (exclusively)

$$
\begin{aligned}
& (\exists \widetilde{x} \in \mathbb{R}^{N+1}) \quad \widetilde{A}\widetilde{x} \geq 0 \text{ and } \langle \widetilde{c} \mid \widetilde{x} \rangle = -1 \\
\Leftrightarrow \quad & (\exists (x,\xi) \in \mathbb{R}^N \times \mathbb{R}) \quad Ax + b\xi \geq 0 \text{ and } \xi = -1 \\
\Leftrightarrow \quad & i).
\end{aligned}
$$

□

4.4 Duality

4.4.1 Weak duality

Proposition 12

Let $x \in \mathbb{R}^N$ and $y \in \mathbb{R}^K$. If x is feasible for the primal problem and y is feasible for the dual one, then

$$\langle b \mid y \rangle \leq \langle c \mid x \rangle .$$

Proof. Since $y \in \mathcal{B}$, $L^\top y \leq c$.
Since $x \in \mathcal{A} \subset [0, +\infty[^N$, we deduce that

$$\begin{aligned} & \left\langle x \mid L^\top y \right\rangle \leq \langle x \mid c \rangle \\ \Leftrightarrow \quad & \langle Lx \mid y \rangle \leq \langle x \mid c \rangle . \end{aligned}$$

Similarly by using the fact that $Lx \geq b$ and $y \in [0, +\infty[^K$,

$$\langle b \mid y \rangle \leq \langle Lx \mid y \rangle .$$

Combining the two latter inequalities yields the result. □

Remark 19.

i) *Alternatively, we could use general weak duality results.*

ii) *As a consequence of Proposition 12,*
if the primal problem is feasible, then the dual one is bounded.
If the dual problem is feasible, then the primal one is bounded.

4.4.2 Strong duality

Proposition 13

If the primal and dual problems are feasible, then $\mu = -\mu^* \in \mathbb{R}$.

Proof. Since the dual is feasible, the primal problem is bounded, i.e. $\mu = \inf_{x \in \mathcal{A}} \langle c \mid x \rangle > -\infty$.
As $\mathcal{A} \neq \varnothing$, $\mu \in \mathbb{R}$.
Therefore, for every $\epsilon > 0$, there does not exist $x \in \mathbb{R}^N$ such that

$$\begin{cases} Lx \geq b \\ x \geq 0 \\ \langle c \mid x \rangle \leq \mu - \epsilon \end{cases} \quad \Leftrightarrow \quad \underbrace{\begin{bmatrix} L \\ \mathrm{Id} \\ -c^\top \end{bmatrix}}_{A} x \geq \underbrace{\begin{bmatrix} b \\ 0 \\ \epsilon - \mu \end{bmatrix}}_{\widetilde{b}}$$

According to Corollary 2,

$$
\begin{aligned}
&(\exists z \in \mathbb{R}^{K+N+1}) \quad \begin{cases} z \geq 0 \\ A^\top z = 0 \\ \left\langle \widetilde{b} \mid z \right\rangle = 1 \end{cases} \\
\Leftrightarrow \quad &(\exists (y, \widetilde{y}, \nu) \in \mathbb{R}^K \times \mathbb{R}^N \times \mathbb{R}) \quad \begin{cases} y \geq 0,\ \widetilde{y} \geq 0,\ \nu \geq 0 \\ L^\top y + \widetilde{y} - \nu c = 0 \\ \langle b \mid y \rangle + (\epsilon - \mu)\nu = 1. \end{cases}
\end{aligned}
$$

If $\nu = 0$, this reduces to $(\exists (y, \widetilde{y}) \in \mathbb{R}^K \times \mathbb{R}^N) \begin{cases} y \geq 0,\ \widetilde{y} \geq 0 \\ L^\top y + \widetilde{y} = 0 \\ \langle b \mid y \rangle = 1. \end{cases}$

Since $\mathcal{B} \neq \varnothing$, there exists $\overline{y} \in [0, +\infty[^K$ such that $L^\top \overline{y} \leq c$.
For every $\lambda \in [0, +\infty[$, $\overline{y} + \lambda y \in \mathcal{B}$ since $\overline{y} + \lambda y \in [0, +\infty[^K$ and $L^\top(\overline{y} + \lambda y) \leq c - \lambda \widetilde{y} \leq c$.
In addition, $\langle b \mid \overline{y} + \lambda y \rangle = \langle b \mid \overline{y} \rangle + \lambda$, which tends to $+\infty$ as $\lambda \to +\infty$.
This shows that the dual problem is unbounded, which is impossible as the primal one is feasible.
Therefore, $\nu > 0$ and

$$
\begin{aligned}
&(\exists (y, \widetilde{y}) \in \mathbb{R}^K \times \mathbb{R}^N) \quad \begin{cases} y \geq 0,\ \widetilde{y} \geq 0 \\ L^\top \frac{y}{\nu} + \frac{\widetilde{y}}{\nu} = c \\ \left\langle b \mid \frac{y}{\nu} \right\rangle = \frac{1}{\nu} + \mu - \epsilon. \end{cases} \\
\Rightarrow \quad &(\exists y \in \mathbb{R}^K) \quad \begin{cases} y \geq 0 \\ L^\top \frac{y}{\nu} \leq c \\ \left\langle b \mid \frac{y}{\nu} \right\rangle \geq \mu - \epsilon \end{cases} \\
\Rightarrow \quad &-\mu^* = \sup_{y' \in \mathcal{B}} \langle b \mid y' \rangle \geq \mu - \epsilon.
\end{aligned}
$$

By letting ϵ tend to 0, we conclude that $-\mu^* \geq \mu$. The reverse inequality follows from weak duality. □

We are now able to characterize primal-dual solutions.

Proposition 14

Let $\widehat{x} = (\widehat{x}^{(i)})_{1 \leq i \leq N} \in \mathcal{A}$ and let $\widehat{y} = (\widehat{y}^{(j)})_{1 \leq j \leq K} \in \mathcal{B}$.
$(\widehat{x}, \widehat{y})$ is a pair of primal/dual solutions if and only if one of the following two equivalent conditions holds:

i) $\langle c \mid \widehat{x} \rangle = \langle b \mid \widehat{y} \rangle$ (no duality gap)

ii) $\begin{cases} (\forall j \in \{1, \ldots, K\}) \quad [b - L\widehat{x}]^{(j)}\, \widehat{y}^{(j)} = 0 \\ (\forall i \in \{1, \ldots, N\}) \quad [L^\top \widehat{y} - c]^{(i)}\, \widehat{x}^{(i)} = 0 \end{cases}$ (complementarity).

Proof.

i) The primal and dual being feasible, by strong duality

$$(\forall (x,y) \in \mathcal{A} \times \mathcal{B}) \quad \langle b \mid y \rangle \leq -\mu^* = \mu \leq \langle c \mid x \rangle \,.$$

Therefore i) is equivalent to

$$\begin{cases} \langle b \mid \widehat{y} \rangle = -\mu^* \\ \langle c \mid \widehat{x} \rangle = \mu. \end{cases}$$

ii) Assume that $(\widehat{x}, \widehat{y})$ is a pair of primal/dual solutions. Then $b - L\widehat{x} \leq 0$ and $\widehat{y} \geq 0$, which implies that

$$(\forall j \in \{1, \dots, K\}) \quad [b - L\widehat{x}]^{(j)} \, \widehat{y}^{(j)} \leq 0. \tag{4.2}$$

In addition

$$\sum_{j=1}^{K} [b - L\widehat{x}]^{(j)} \, \widehat{y}^{(j)} = \langle b - L\widehat{x} \mid \widehat{y} \rangle = \langle b \mid \widehat{y} \rangle - \langle L\widehat{x} \mid \widehat{y} \rangle = \langle b \mid \widehat{y} \rangle - \left\langle \widehat{x} \mid L^\top \widehat{y} \right\rangle .$$

Since $\widehat{x} \geq 0$ and $L^\top \widehat{y} \leq c$,

$$\sum_{j=1}^{K} [b - L\widehat{x}]^{(j)} \, \widehat{y}^{(j)} \geq \langle b \mid \widehat{y} \rangle - \langle \widehat{x} \mid c \rangle = 0.$$

This allows us to conclude that equality is reached in (4.2).
Similarly, it can be shown that a necessary condition for $(\widehat{x}, \widehat{y})$ to be a pair of primal/dual solutions is

$$(\forall i \in \{1, \dots, N\}) \;\; [L^\top \widehat{y} - c]^{(i)} \, \widehat{x}^{(i)} = 0.$$

Conversely, assume that ii) holds. Then

$$\begin{aligned} & \sum_{j=1}^{K} [b - L\widehat{x}]^{(j)} \, \widehat{y}^{(j)} + \sum_{i=1}^{N} [L^\top \widehat{y} - c]^{(i)} \, \widehat{x}^{(i)} = 0 \\ \Leftrightarrow \quad & \langle b - L\widehat{x} \mid \widehat{y} \rangle + \left\langle L^\top \widehat{y} - c \mid \widehat{x} \right\rangle = \langle b \mid \widehat{y} \rangle - \langle L\widehat{x} \mid \widehat{y} \rangle + \left\langle L^\top \widehat{y} \mid \widehat{x} \right\rangle - \langle c \mid \widehat{x} \rangle = 0 \\ \Leftrightarrow \quad & \langle b \mid \widehat{y} \rangle = \langle c \mid \widehat{x} \rangle \quad \Leftrightarrow \quad i). \end{aligned}$$

□

4.4.3 Standard form

Let us consider the primal problem:

$$\underset{x \in [0,+\infty[^N}{\text{minimize}} \;\; \langle c \mid x \rangle \quad \text{s.t.} \quad Lx \geq b.$$

Let us introduce the **slack variable**

$$s = Lx - b \geq 0.$$

Set

$$\begin{aligned} M &= N + K, \\ z &= \begin{bmatrix} x \\ s \end{bmatrix} \in \mathbb{R}^M, \\ A &= [L \quad -\mathrm{Id}] \in \mathbb{R}^{K \times M}, \\ d &= \begin{bmatrix} c \\ 0 \end{bmatrix} \in \mathbb{R}^M. \end{aligned} \tag{4.3}$$

The problem can be recast as

$$\underset{z \in [0,+\infty[^M}{\text{minimize}} \quad \langle d \mid z \rangle \quad \text{s.t.} \quad Az = b.$$

Example 19. *The standard form of the diet problem is given by*

$$\begin{aligned} M &= 4 \\ d &= \begin{bmatrix} -3 \\ -2 \\ 0 \\ 0 \end{bmatrix} \\ A &= -\begin{bmatrix} 2 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}, \quad b = -\begin{bmatrix} 4 \\ 4 \end{bmatrix}. \end{aligned}$$

Remark 20. *Let A be given by (4.3) and let*

$$\widetilde{\mathcal{A}} = \left\{ z \in [0, +\infty[^M \;\middle|\; Az = b \right\} \tag{4.4}$$

be the feasibility set for the standard problem. The affine function

$$\begin{aligned} \Phi \colon \mathbb{R}^N &\to \mathbb{R}^M \\ x &\mapsto z = \begin{bmatrix} x \\ s \end{bmatrix} = \begin{bmatrix} x \\ Lx - b \end{bmatrix} \end{aligned} \tag{4.5}$$

defines a bijection from $\mathcal{A}$ to $\widetilde{\mathcal{A}}$

4.4.4 Conditions for strong duality

Theorem 17

The following conditions are equivalent:

i) *the primal problem is feasible and bounded*

ii) *the dual problem is feasible and bounded*

iii) *the primal and dual problems are feasible*

iv) *there exists a solution to the primal problem*

v) *there exists a solution to the dual problem.*

Proof.

ii) $\Rightarrow$ i): Let us give a proof by contraposition by assuming that the primal problem is unfeasible or unbounded.
According to Remark 19ii), if it is unbounded, then the dual one is unfeasible.
If it is unfeasible, then

$$(\nexists x \in \mathbb{R}^N) \quad \begin{cases} Lx \geq b \\ x \geq 0 \end{cases} \quad \Leftrightarrow \quad \begin{bmatrix} L \\ \mathrm{Id} \end{bmatrix} x \geq \begin{bmatrix} b \\ 0 \end{bmatrix}.$$

According to Corollary 2, ,

$$(\exists (y, \widetilde{y}) \in [0, +\infty[^K \times [0, +\infty[^N) \quad \begin{cases} L^\top y + \widetilde{y} = 0 \\ \langle b \mid y \rangle = 1. \end{cases}$$

Assume that the dual is feasible. Then $(\exists \overline{y} \in [0, +\infty[^K)$ $L^\top \overline{y} \leq c$ and

$$(\forall \lambda \geq 0) \quad \begin{cases} \overline{y} + \lambda y \geq 0 \\ L^\top (\overline{y} + \lambda y) \leq c - \lambda \widetilde{y} \leq c \\ \langle b \mid \overline{y} + \lambda y \rangle = \langle b \mid \overline{y} \rangle + \lambda. \end{cases}$$

Hence, by letting $\lambda \to +\infty$, we see that the dual is unbounded.
In conclusion, we have proved the dual problem is unfeasible or unbounded.

i) $\Rightarrow$ ii): Proved in a symmetric manner.

i) $\Leftrightarrow$ iii): We have

$$\text{(i)} \Rightarrow \begin{cases} \text{primal feasible} \\ \text{(ii)} \end{cases} \quad \Rightarrow \begin{cases} \text{primal feasible} \\ \text{dual feasible} \end{cases}$$

and, according to Remark 19ii),

$$\text{primal and dual feasible} \Rightarrow \text{primal feasible and bounded.}$$

i) $\Rightarrow$ iv): Let us focus on the standard form of the LP problem introduced in Section 4.4.3. If the primal problem is bounded and feasible, then there exists a sequence $(z_n)_{n \in \mathbb{N}}$ of elements of $[0, +\infty[^M$ such that $A z_n = b$ and $\langle d \mid z_n \rangle \to \mu \in \mathbb{R}$.
Let $\widetilde{A} = \begin{bmatrix} A \\ d^\top \end{bmatrix}$, let $(e_i)_{1 \leq i \leq M}$ is the canonical basis of $\mathbb{R}^M$, and let

$$C = \big\{ \widetilde{A} z \;\big|\; z \in [0, +\infty[^M \big\} = \Big\{ \sum_{i=1}^{L} z^{(i)} \widetilde{A} e_i \;\Big|\; (\forall i \in \{1, \ldots, M\})\ z^{(i)} \geq 0 \Big\}.$$

$(\widetilde{A}z_n)_{n\in\mathbb{N}}$ is a convergent sequence of C since, for every $n \in \mathbb{N}$, $\widetilde{A}z_n \in C$ and

$$\widetilde{A}z_n = \begin{bmatrix} Az_n \\ \langle d \mid z_n \rangle \end{bmatrix} = \begin{bmatrix} b \\ \langle d \mid z_n \rangle \end{bmatrix} \to \begin{bmatrix} b \\ \mu \end{bmatrix}.$$

As C is a **closed** convex cone of $\mathbb{R}^{K+1}$, it follows that there exists $\widehat{z} \in [0,+\infty[^M$ such that $\widetilde{A}\widehat{z} = \begin{bmatrix} b \\ \mu \end{bmatrix}$, that is $A\widehat{z} = b$ and $\langle d \mid \widehat{z} \rangle = \mu$. This shows that $\widehat{z}$ is a solution to the standard problem.

iv) $\Rightarrow$ i): Obvious.

ii) $\Leftrightarrow$ v): Similar to i) $\Leftrightarrow$ iv).

□

4.5 Basic solutions to linear programming problems

In the rest of this chapter, our objective will be to find a solution to the LP problems expressed in its standard form as recalled below.

Problem 5

Let $A \in \mathbb{R}^{K\times M}$ such that $\operatorname{rank} A = K < M$, let $b \in \mathbb{R}^K$, and let $d \in \mathbb{R}^M$. We want to

$$\underset{z\in[0,+\infty[^M}{\text{minimize}} \quad \langle d \mid z \rangle \quad \text{s.t.} \quad Az = b.$$

Remark 21. *Assuming that* $\operatorname{rank} A = K$ *is not restrictive since, if this condition is not met, some lines of* A *correspond to either redundant or incompatible equality constraints.*
In addition, if $M = K$ *then there exists a unique solution to the equation* $Az = b$*, which makes the problem trivial.*

Definition 32

Let C be a convex set.
$z \in C$ is an **extreme point** of C if

$$(\exists (u,v) \in C^2) \quad z = \frac{u+v}{2}$$
$$\Rightarrow \quad u = v.$$

Remark 22. *If* z *is an extreme point of a convex* C *in a Hilbert space, then* $z \in \operatorname{bd}(C)$ *(*$=\overline{C} \setminus \operatorname{int}(C)$*). Otherwise,* $z \in C \setminus \operatorname{bd}(C) \Rightarrow z \in \operatorname{int}(C)$ *and there exists an open ball centered at* z *included in* C

Definition 33

The extreme points of a polyhedron are called its **vertices**.

Theorem 18

Assume that the standard problem admits a solution.
Then one of the vertices of the feasible set is a solution.

Proof. There exists a solution $\overline{z}$ with a maximum of zero components.
Assume that $\overline{z}$ is not a vertex of the feasible set $\widetilde{\mathcal{A}}$ as defined in (4.4). Then

$$(\exists(u,v) \in \widetilde{\mathcal{A}}^2) \quad u \neq v \quad \text{and} \quad \overline{z} = \frac{u+v}{2}. \tag{4.6}$$

Since $\overline{z}$ is a solution to the standard problem,

$$\langle d \mid u \rangle \geq \langle d \mid \overline{z} \rangle \quad \text{and} \quad \langle d \mid v \rangle \geq \langle d \mid \overline{z} \rangle \,.$$

On the other hand, since

$$\langle d \mid \overline{z} \rangle = \frac{1}{2}(\langle d \mid u \rangle + \langle d \mid v \rangle),$$

we have $\langle d \mid u \rangle = \langle d \mid \overline{z} \rangle = \langle d \mid v \rangle$.
For every $\lambda \in \mathbb{R}$, let

$$z_\lambda = \overline{z} + \lambda(u - v).$$

Then

$$\begin{aligned}
&\langle d \mid z_\lambda \rangle = \langle d \mid \overline{z} \rangle + \lambda(\langle d \mid u \rangle - \langle d \mid v \rangle) = \langle d \mid \overline{z} \rangle \\
&A z_\lambda = A\overline{z} + \lambda(Au - Av) = b.
\end{aligned}$$

Let us now see how to choose λ such that $z_\lambda \geq 0$, so ensuring that $z_\lambda \in \widetilde{\mathcal{A}}$.

- Let $\mathbb{K} = \big\{ i \in \{1, \dots, M\} \bigm| \overline{z}^{(i)} = 0 \big\}$.
  Then, it follows from (4.6) and the nonnegativity of the components of u and v that, for every $i \in \mathbb{K}$, $u^{(i)} = v^{(i)} = 0$, which implies that $z_\lambda^{(i)} = 0$.
- Let $\mathbb{J} = \big\{ j \in \{1, \dots, M\} \bigm| u^{(j)} \neq v^{(j)} \big\}$.
  For every $j \in \{1, \dots, M\} \setminus \mathbb{J})$ such that $j \notin \mathbb{K}$, $z_\lambda^{(j)} = \overline{z}^{(j)} > 0$
- Let us now consider indices in $\mathbb{J}$.
  We know that $\mathbb{J} \neq \varnothing$ and $\mathbb{J} \cap \mathbb{K} = \varnothing$.
  Suppose, for example, that $(\exists j \in \mathbb{J})\ v^{(j)} > u^{(j)}$ (otherwise, we substitute $(-\lambda)$ for λ and proceed symetrically).
  Let $\lambda = \min_{\substack{j \in \mathbb{J} \\ v^{(j)} > u^{(j)}}} \frac{\overline{z}^{(j)}}{v^{(j)} - u^{(j)}} = \frac{\overline{z}^{(j_0)}}{v^{(j_0)} - u^{(j_0)}} > 0.$
  Then, for every $j \in \mathbb{J} \setminus \{j_0\}$, either $u^{(j))} > v^{(j)}$, and $z_\lambda^{(j)} > 0$,
  or $z_\lambda^{(j)} \geq (v^{(j)} - u^{(j)})(\frac{\overline{z}^{(j)}}{v^{(j)} - u^{(j)}} - \lambda) \geq 0$.
  In addition, $z_\lambda^{(j_0)} = 0$.

In conclusion $z_\lambda \geq 0$ is a solution to the standard problem with at least one more zero component than $\overline{z}$ (for component index j_0), which is impossible. $\square$

Remark 23. *There exists a bijective mapping between the vertices of the feasible set $\widetilde{A}$ of the standard problem and the vertices of of the feasible set $\mathcal{A}$ of the associated canonical problem.*

Proof. Let Φ be the function defined in (4.5).

Let z be a vertex of $\widetilde{\mathcal{A}}$ and let $x = \Phi^{-1}(z)$. Let $(u, v) \in \mathcal{A}^2$ be such that $x = (u + v)/2$. Since Φ is affine, $z = (\Phi(u) + \Phi(v))/2$ with $(\Phi(u), \Phi(v)) \in \widetilde{\mathcal{A}}^2$. We deduce that $\Phi(u) = \Phi(v)$, hence $u = v$. This shows that x is vertex of $\mathcal{A}$.

Conversely, let x be a vertex of $\mathcal{A}$ and let $z = \Phi(x)$. Let $(\widetilde{u}, \widetilde{v}) \in \widetilde{\mathcal{A}}^2$ such that $z = (\widetilde{u}+\widetilde{v})/2$. Then $x = (\Phi^{-1}(\widetilde{u}) + \Phi^{-1}(\widetilde{v}))/2$ and $(\Phi^{-1}(\widetilde{u}), \Phi^{-1}(\widetilde{v})) \in \mathcal{A}^2$. We deduce that $\Phi^{-1}(\widetilde{u}) = \Phi^{-1}(\widetilde{v})$ $\Rightarrow \widetilde{u} = \widetilde{v}$, which shows that z is vertex of $\widetilde{\mathcal{A}}$. $\square$

Definition 34

Let $(a_i)_{1\leq i\leq M}$ be the column vectors of A. A solution $z = (z^{(i)})_{1\leq i\leq M}$ to the equation $Az = b$ is called a **basic solution** to this equation if $z = 0$ or if $\left\{a_i \;\middle|\; i \in \mathbb{I}_0\right\}$ is a family of linearly independent vectors where $\mathbb{I}_0 = \left\{i \in \{1, \dots, M\} \;\middle|\; z^{(i)} \neq 0\right\}$.
If $\operatorname{card} \mathbb{I}_0 = K$ (resp. $\operatorname{card} \mathbb{I}_0 \neq K$), then z is said to be **non degenerate** (resp. **degenerate**).

Definition 35

Let $\mathbb{I} \subset \{1, \dots, M\}$ be such that $A_\mathbb{I} = [a_i]_{i\in\mathbb{I}}$ is invertible. $\mathbb{I}$ is called a **basic index set**.

Remark 24. *If $\mathbb{I}$ is a basic index set, then $\operatorname{card} \mathbb{I} = K$ (since $A_\mathbb{I}$ has K rows). Consequently $A_\mathbb{I} \in \mathbb{R}^{K\times K}$.*

Proposition 15

i) *If $z = (z^{(i)})_{1\leq i\leq M}$ is a basic solution, then there exists a basic index set $\mathbb{I} \subset \{1, \dots, M\}$ such that*

$$(\forall i \in \{1, \dots, M\} \setminus \mathbb{I}) \quad z^{(i)} = 0. \tag{4.7}$$

ii) *If $\mathbb{I}$ is a basic index set, then there exists a basic solution $z = (z^{(i)})_{1\leq i\leq M}$ satisfying (4.7).*

iii) *A basic solution always exists.*

Proof.

i) Since $\operatorname{rank} A = K < M$, if z is a basic solution, then it suffices to complete $(a_i)_{i\in\mathbb{I}_0}$ with columns of A, $(a_i)_{i\in\mathbb{I}\setminus\mathbb{I}_0}$, which correspond to zero components of z and are linearly independent.

ii) Let $z \neq 0$ satisfy (4.7). Let $z_{\mathbb{I}} = (z^{(i)})_{i\in\mathbb{I}}$ be such that

$$z_{\mathbb{I}} = A_{\mathbb{I}}^{-1} b.$$

Then

$$Az = \sum_{i=1}^{M} z^{(i)} a_i = \sum_{i\in\mathbb{I}} z^{(i)} a_i = A_{\mathbb{I}} z_{\mathbb{I}} = b.$$

In addition, $\mathbb{I}_0 = \big\{ i \in \{1,\ldots,M\} \;\big|\; z^{(i)} \neq 0 \big\} \subset \mathbb{I}$, hence $(a_i)_{i\in\mathbb{I}_0}$ is a family of linearly independent vectors.

iii) Since $\operatorname{rank} A = K$, a basic index set always exists and it follows from Proposition 15ii) that a basic solution always exists.

$\square$

Theorem 19

z is a vertex of the feasible set of the standard problem if and only if $z \in [0,+\infty[^M$ and z is a basic solution.

Proof. Assume that $z \in [0,+\infty[^M$ is a basic solution.
Suppose that there exist $u = (u^{(i)})_{1\le i\le M}$ and $v = (v^{(i)})_{1\le i\le M}$ in the feasible set $\widetilde{\mathcal{A}}$ such that $z = (u+v)/2$. Let $\mathbb{I}_0 = \big\{ i \in \{1,\ldots,M\} \;\big|\; z^{(i)} > 0 \big\}$.
If $i \in \{1,\ldots,M\} \setminus \mathbb{I}_0$, then $z^{(i)} = 0 \Rightarrow u^{(i)} = v^{(i)} = 0$.
In addition, when $\mathbb{I}_0 \neq \varnothing$,

$$\begin{aligned} Au = Av = b \quad &\Rightarrow \quad A(u-v) = 0 \\ &\Leftrightarrow \quad \sum_{i=1}^{M} (u^{(i)} - v^{(i)}) a_i = \sum_{i\in\mathbb{I}_0} (u^{(i)} - v^{(i)}) a_i = 0. \end{aligned}$$

Since $\big\{ a_i \;\big|\; i \in \mathbb{I}_0 \big\}$ is a family of linearly independent vectors, for every $i \in \mathbb{I}_0$, $u^{(i)} = v^{(i)}$.
Hence, $u = v$ and z is a vertex of $\widetilde{\mathcal{A}}$.
Conversely, assume that z is a vertex of $\widetilde{\mathcal{A}}$ and it is not a basic solution. There thus exists $w = (w^{(i)})_{1\le i\le M} \in \mathbb{R}^M \setminus \{0\}$ such that $(\forall i \in \{1,\ldots,M\} \setminus \mathbb{I}_0)\; w^{(i)} = 0$ and $Aw = \sum_{i\in\mathbb{I}_0} w^{(i)} a_i = 0$.
Let $\epsilon > 0$. First note that $A(z \pm \epsilon w) = Az = b$.
Furthermore,

$$\begin{aligned} &(\forall i \in \{1,\ldots,M\} \setminus \mathbb{I}_0) \quad z^{(i)} \pm \epsilon w^{(i)} = 0, \\ &(\forall i \in \mathbb{I}_0) \quad z^{(i)} > 0. \end{aligned}$$

Let $i \in \mathbb{I}_0$. If $w^{(i)} = 0$, then $z^{(i)} \pm \epsilon w^{(i)} > 0$. If $w^{(i)} \neq 0$, then

$$z^{(i)} \pm \epsilon w^{(i)} \geq z^{(i)} - \epsilon |w^{(i)}|$$

and

$$z^{(i)} - \epsilon |w^{(i)}| > 0 \quad \Leftrightarrow \quad \epsilon < z^{(i)}/|w^{(i)}|.$$

In summary, provided that ϵ is small enough, $z \pm \epsilon w \in \widetilde{\mathcal{A}}$. In addition $z = \frac{1}{2}\big((z+\epsilon w)+(z-\epsilon w)\big)$, which contradicts the fact that z is a vertex of $\widetilde{\mathcal{A}}$. □

Theorems 18 and 19 suggest a naive algorithm for finding a solution to the standard problem, when such a solution exists. The algorithm proceeds by looking for all the possible feasible basic solutions.

Algorithm 2

i) *We extract all the possible invertible matrices $A_{\mathbb{I}}$ where $\mathbb{I} \subset \{1, \ldots, M\}$ and* $\operatorname{card} \mathbb{I} = K$.

ii) *We check that the associated basic solution has nonnegative components.*

iii) *We look for the vector with minimum cost among those.*

Although effective, this algorithm has a high computational cost of the order of C_M^K.

4.6 Simplex algorithm

4.6.1 Principle

The basic idea of the simplex algorithm is to modify a basic index set $\mathbb{I}$ just by changing one index at each iteration so that we move to a neighboring vertex with lower cost value. The algorithm is based on a reparametrization of the standard problem as expressed below.

Proposition 16

Consider Problem 5. Let $\mathbb{I}$ be a basic index set associated with a basic solution $w \in \mathbb{R}^M$.
Let $\mathbb{J} = \{1, \ldots, M\} \setminus \mathbb{I}$.
For every $z = (z^{(i)})_{1 \leq i \leq M} \in \mathbb{R}^M$, let $z_{\mathbb{I}} = (z^{(i)})_{i \in \mathbb{I}}$ (resp. $z_{\mathbb{J}} = (z^{(i)})_{i \in \mathbb{J}}$).
Let $A_{\mathbb{I}}$ (resp. $A_{\mathbb{J}}$) be the matrix of columns of A with index in $\mathbb{I}$ (resp. $\mathbb{J}$).
The standard LP problem can be recast as

$$\underset{z \in \mathbb{R}^M}{\text{minimize}} \;\; f(z) = \langle d_{\mathbb{I}} \mid w_{\mathbb{I}} \rangle + \langle r \mid z_{\mathbb{J}} \rangle \quad s.t. \quad \begin{cases} z_{\mathbb{J}} \geq 0 \\ z_{\mathbb{I}} = w_{\mathbb{I}} - A_{\mathbb{I}}^{-1} A_{\mathbb{J}} z_{\mathbb{J}} \geq 0 \end{cases} \tag{4.8}$$

where $r = d_{\mathbb{J}} - A_{\mathbb{J}}^\top (A_{\mathbb{I}}^{-1})^\top d_{\mathbb{I}}$ is called the ***reduced cost****.*

Proof. Let $z \in \mathbb{R}^M$. We have

$$Az = b \quad \Leftrightarrow \quad A_{\mathbb{I}} z_{\mathbb{I}} + A_{\mathbb{J}} z_{\mathbb{J}} = b \tag{4.9}$$

Since w is a basic solution associated to the basic index set $\mathbb{I}$,

$$w_{\mathbb{J}} = 0 \quad \Rightarrow \quad A_{\mathbb{I}} w_{\mathbb{I}} = b.$$

As $\mathbb{I}$ is a basic index set, $A_{\mathbb{I}}$ is invertible and, by inverting it, we deduce from (4.9) that

$$z_{\mathbb{I}} = w_{\mathbb{I}} - A_{\mathbb{I}}^{-1} A_{\mathbb{J}} z_{\mathbb{J}}.$$

In addition,

$$\begin{aligned} f(z) = \langle d \mid z \rangle &= \langle d_{\mathbb{I}} \mid z_{\mathbb{I}} \rangle + \langle d_{\mathbb{J}} \mid z_{\mathbb{J}} \rangle \\ &= \left\langle d_{\mathbb{I}} \mid w_{\mathbb{I}} - A_{\mathbb{I}}^{-1} A_{\mathbb{J}} z_{\mathbb{J}} \right\rangle + \langle d_{\mathbb{J}} \mid z_{\mathbb{J}} \rangle \\ &= \langle d_{\mathbb{I}} \mid w_{\mathbb{I}} \rangle - \left\langle A_{\mathbb{J}}^{\top} (A_{\mathbb{I}}^{-1})^{\top} d_{\mathbb{I}} \mid z_{\mathbb{J}} \right\rangle + \langle d_{\mathbb{J}} \mid z_{\mathbb{J}} \rangle \\ &= \langle d_{\mathbb{I}} \mid w_{\mathbb{I}} \rangle + \langle r \mid z_{\mathbb{J}} \rangle . \end{aligned}$$

□

Remark 25. *The reduced cost is the gradient of function f with respect to $z_{\mathbb{J}}$.*

Starting from a basic solution $w \in [0, +\infty[^M$, our objective will be now to solve Problem (4.9). The following cases can arise.

- Case 1: Assume that $r \geq 0$. Then $(\forall z_{\mathbb{J}} \geq 0)$ $f(z) \geq \langle d_{\mathbb{I}} \mid w_{\mathbb{I}} \rangle = f(w)$.
  This shows that w *is a solution to the standard LP problem.*

- Case 2: Let $r = (r^{(j')})_{j' \in \mathbb{J}}$ and assume that $(\exists j \in \mathbb{J})$ such that $r^{(j)} < 0$.
  Suppose that the only possibly nonzero component of variable $z_{\mathbb{J}}$ is $z^{(j)} \geq 0$. This choice makes the cost function decay since

  $$f(z) = f(w) + r^{(j)} z^{(j)} \leq f(w). \tag{4.10}$$

  To know whether z is feasible, we have however to look at the sign of the components of

  $$z_{\mathbb{I}} = w_{\mathbb{I}} - z^{(j)} \Delta_j,$$

  where

  $$\Delta_j = A_{\mathbb{I}}^{-1} a_j.$$

  - Case 2.1: If $\Delta_j \leq 0$ then, for every $z^{(j)} \geq 0$, $z_{\mathbb{I}} \geq 0$ and
    $f(z) = f(w) + r^{(j)} z^{(j)} \to -\infty$ as $z^{(j)} \to +\infty$.
    Therefore, *the LP problem is unbounded.*

– Case 2.2: Let $(\Delta_j^{(i)})_{i\in\mathbb{I}}$ be the components of Δ_j and let

$$\widetilde{\mathbb{I}}_j = \left\{i \in \mathbb{I} \mid \Delta_j^{(i)} > 0\right\} \neq \varnothing.$$

Let us first observe that $(\forall i \in \mathbb{I} \setminus \widetilde{\mathbb{I}}_j)\ z^{(i)} \geq 0$.
In addition, as $z^{(j)}$ increases, for every $i \in \widetilde{\mathbb{I}}_j$, $z^{(i)} = w^{(i)} - z^{(j)}\Delta_j^{(i)}$ and $f(z) = f(w) + r^{(j)}z^{(j)}$ both decrease. It is thus advantageous to choose the greatest value of $z^{(j)}$ such that

$$(\forall i \in \widetilde{\mathbb{I}}_j) \quad z^{(i)} \geq 0 \quad \Leftrightarrow \quad z^{(j)} \leq w^{(i)}/\Delta_j^{(i)},$$

that is

$$z^{(j)} = w^{(i_j)}/\Delta_j^{(i_j)},$$

where

$$i_j \in \underset{i\in\widetilde{\mathbb{I}}_j}{\operatorname{Argmin}}\ \frac{w^{(i)}}{\Delta_j^{(i)}}.$$

We have then $z^{(i_j)} = 0$.
This suggest replacing $\mathbb{I}$ by $(\mathbb{I} \setminus \{i_j\}) \cup \{j\}$ (**pivoting**). Iterating the process to further decay the cost function requires however that this pivoting operation leads to a basic index set. It is shown next that this indeed happens.

Proposition 17

Consider Case 2.2.
If $\mathbb{I}$ be a basic index set, then $(\mathbb{I} \setminus \{i_j\}) \cup \{j\}$ is a basic index set.

Proof. Without loss of generality, assume that $\mathbb{I} = \{1, \dots, K\}$ and $i_j = K$. Let $(\lambda_1, \dots, \lambda_{K-1}, \lambda_j) \in \mathbb{R}^K$ be such that

$$\begin{aligned} &\sum_{i=1}^{K-1} \lambda_i a_i + \lambda_j a_j = 0 \\ \Rightarrow \quad &\sum_{i=1}^{K-1} \lambda_i A_{\mathbb{I}}^{-1} a_i + \lambda_j \underbrace{A_{\mathbb{I}}^{-1} a_j}_{\Delta_j} = 0 \end{aligned}$$

Since $A_{\mathbb{I}}^{-1}A_{\mathbb{I}} = \mathrm{Id}$, for every $i \in \{1, \dots, K-1\}$ $A_{\mathbb{I}}^{-1}a_i = e_i$ where $(e_i)_{1\leq i\leq K}$ is the canonical basis of $\mathbb{R}^K$. The previous equality can thus be rewritten under matrix form as

$$\begin{bmatrix} 1 & 0 & \dots & 0 & \Delta_j^{(1)} \\ 0 & \ddots & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & 0 & \vdots \\ \vdots & & \ddots & 1 & \Delta_j^{(K-1)} \\ 0 & \dots & \dots & 0 & \Delta_j^{(K)} \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_{K-1} \\ \lambda_j \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ \vdots \\ \vdots \\ 0 \end{bmatrix}.$$

Since $i_j = K \in \widetilde{\mathbb{I}}_j$, $\Delta_j^{(K)} > 0 \Rightarrow \lambda_j = 0$ and $\lambda_1 = \dots = \lambda_{K-1} = 0$.
This shows that $\{a_1, \dots, a_{K-1}, a_j\}$ is a family of linearly independent vectors of $\mathbb{R}^K$. □

We can now summarize the simplex algorithm.

Algorithm 3. (Simplex)

Initialize with a vertex w associated with a basic index set $\mathbb{I}$; set $\mathbb{J} = \{1, \dots, M\} \setminus \mathbb{I}$.

① *Compute the reduced cost* $r = d_{\mathbb{J}} - A_{\mathbb{J}}^\top (A_{\mathbb{I}}^{-1})^\top d_{\mathbb{I}}$.

② *If* $r \geq 0$, *then w mimimizer; exit.*

③ *Choose* $j \in \mathbb{J}$ *such that* $r^{(j)} < 0$.

④ *Compute* $\Delta_j = A_{\mathbb{I}}^{-1} a_j$.

⑤ *If* $\Delta_j \leq 0$, *then unbounded problem; exit.*

⑥ *Compute*

$$\begin{aligned} & i_j \in \operatorname*{Argmin}_{i \in \widetilde{\mathbb{I}}_j} \frac{w^{(i)}}{\Delta_j^{(i)}} \quad \textit{where} \quad \widetilde{\mathbb{I}}_j = \left\{ i \in \mathbb{I} \;\middle|\; \Delta_j^{(i)} > 0 \right\} \\ & z^{(j)} = \frac{w^{(i_j)}}{\Delta_j^{(i_j)}}. \end{aligned}$$

⑦ *Update*

$$\begin{aligned} w_{\mathbb{I}} \leftarrow w_{\mathbb{I}} - z^{(j)} \Delta_j, \quad w^{(j)} \leftarrow z^{(j)} \\ \mathbb{I} \leftarrow (\mathbb{I} \setminus \{i_j\}) \cup \{j\}. \end{aligned}$$

and go to ①.

4.6.2 Convergence

Proposition 18

Assume that the standard problem is feasible and bounded. If none of the vertices of the feasible set of the standard problem is degenerate, then the following hold:

i) *the simplex algorithm generates a decreasing sequence of cost values.*

ii) *It terminates in a finite number of iterations and yields a solution to the problem.*

Proof. Each of the iterations of the simplex generates a vertex of the feasibility set.
Since it is not degenerate, $z^{(j)} > 0$ and the cost strictly decreases (see (4.10)).
Each cost value is thus associated to a distinct vertex. The number of vertices being finite (at most C_M^K), the number of iterations is finite.

When the algorithm stops, either an unbounded problem is obtained or a solution to the standard problem is obtained. □

Remark 26.

i) *If w is degenerate, it is possible that $z^{(j)} = 0$. Thus, we may enter in a cycle (oscillate between several values of j). To avoid this, we have to implement an anti-cycling rule. A popular one is the* ***Bland rule****: we choose*

- *the smallest j such that $r_j < 0$*
- *and the smallest i_j in* $\operatorname*{Argmin}_{i \in \widetilde{\mathbb{I}}_j} \frac{w^{(i)}}{\Delta_j^{(i)}}$.

ii) *In the worst case, the complexity of the simplex is exponential as a function of M, but it has usually a polynomial complexity.*

4.6.3 Example of use

We want to

$$\underset{(x^{(1)},x^{(2)})\in[0,+\infty[^2}{\text{maximize}} \quad x^{(1)} + x^{(2)} \quad \text{s.t.} \quad \begin{cases} x^{(1)} + 3x^{(2)} \leq 9 \\ 2x^{(1)} + x^{(2)} \leq 8. \end{cases} \tag{4.11}$$

i) We introduce slack variables $s^{(1)} = x^{(3)}$ and $s^{(2)} = x^{(4)}$ and reformulate the problem under standard form:

$$\underset{x=(x^{(1)},x^{(2)},x^{(3)},x^{(4)})\in[0,+\infty[^4}{\text{minimize}} \quad f(x) = -x^{(1)} - x^{(2)} \quad \text{s.t.} \quad \begin{cases} x^{(1)} + 3x^{(2)} + x^{(3)} = 9 \\ 2x^{(1)} + x^{(2)} + x^{(4)} = 8. \end{cases}$$

A traditional way of implementing Algorithm 3 consists of updating a table having the following form:

$x^\top$	
A	b
$d^\top$	f

Table 4.2: Generic table used to implement simplex algorithm.

where f corresponds to the function value. In our case, this table reads:

$x^{(1)}$	$x^{(2)}$	$x^{(3)}$	$x^{(4)}$	
1	3	1	0	9
2	1	0	1	8
-1	-1	0	0	f

Table 4.3: Table associated with Problem (4.11).

We initialize the simplex by choosing the basic index set $\mathbb{I} = \{3, 4\}$ for which the associated vertex is simply

$$w = [0 \;\; 0 \;\; 9 \;\; 8]^\top.$$

$x^{(1)}$	$x^{(2)}$	$x^{(3)}$	$x^{(4)}$	
1	3	1	0	9
2	1	0	1	8
-1	-1	0	0	f
		*	*	

Table 4.4: Initialization.

ii) First iteration of the simplex

We have $\mathbb{J} = \{1, 2\}$ and the reduced cost is $r = d_\mathbb{J} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$. By applying Bland rule, we choose $j = 1$.

We calculate $\Delta_j = A_\mathbb{I}^{-1} a_j = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, which shows that $\widetilde{\mathbb{I}}_j = \{3, 4\}$.

We deduce that $z^{(j)} = \min\left\{\frac{9}{1}, \frac{8}{2}\right\} = 4$, hence $i_j = 4$.

This leads to the new basic index set $\mathbb{I} = \{1, 3\}$ associated with the new vertex

$$w = [4 \;\; 0 \;\; (9 - 4 \times 1) \;\; 0]^\top = [4 \;\; 0 \;\; 5 \;\; 0]^\top.$$

iii) Second iteration of the simplex

To make the inversion of $A_\mathbb{I}$ easy, we transform it in a permutation matrix by linear combinations of the first two rows of the table.

The boxed element is called the **pivot** in Table 4.5.

We can do the same for the last line to express the cost as a function of $x_\mathbb{J}$.

$x^{(1)}$	$x^{(2)}$	$x^{(3)}$	$x^{(4)}$				$x^{(1)}$	$x^{(2)}$	$x^{(3)}$	$x^{(4)}$		
1	3	1	0	9	(1)		0	$\frac{5}{2}$	1	$-\frac{1}{2}$	5	(1)-(2)/2
$\boxed{2}$	1	0	1	8	(2)	$\rightsquigarrow$	$\boxed{1}$	$\frac{1}{2}$	0	$\frac{1}{2}$	4	(2)/2
-1	-1	0	0	f	(3)		0	$-\frac{1}{2}$	0	$\frac{1}{2}$	$f+4$	(3)+(2)/2
*		*					*		*			

Table 4.5: After the first iteration.

Then, we calculate the quantities required in Algorithm 3:

$r = d_\mathbb{J} = \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix} \quad \Rightarrow j = 2$

$\Delta_j = A_\mathbb{I}^{-1} a_j = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{5}{2} \\ \frac{1}{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ 5 \end{bmatrix} \quad \Rightarrow \quad \widetilde{\mathbb{I}}_j = \{1, 3\}$

$z^{(j)} = \min \left\{ \frac{4}{1/2}, \frac{5}{5/2} \right\} = 2 \quad \Rightarrow \quad i_j = 3.$
This leads to the new basic index set $\mathbb{I} = \{1, 2\}$ associated with the new vertex

$$w = [(4 - 2 \times \frac{1}{2}) \quad 2 \quad 0 \quad 0]^\top = [3 \;\; 2 \;\; 0 \;\; 0]^\top. \tag{4.12}$$

iv) Third iteration of the simplex

$$\begin{array}{cccc|cc}
x^{(1)} & x^{(2)} & x^{(3)} & x^{(4)} & & \\
\hline
0 & \boxed{\frac{5}{2}} & 1 & -\frac{1}{2} & 5 & (1) \\
1 & \frac{1}{2} & 0 & \frac{1}{2} & 4 & (2) \\
\hline
0 & -\frac{1}{2} & 0 & \frac{1}{2} & f+4 & (3) \\
* & * & & & &
\end{array}
\quad \rightsquigarrow \quad
\begin{array}{cccc|cc}
x_1 & x_2 & x_3 & x_4 & & \\
\hline
0 & \boxed{1} & \frac{2}{5} & -\frac{1}{5} & 2 & \frac{2}{5}(1) \\
1 & 0 & -\frac{1}{5} & \frac{3}{5} & 3 & (2) - \frac{1}{5}(1) \\
\hline
0 & 0 & \frac{1}{5} & \frac{2}{5} & f+5 & (3) + \frac{1}{5}(1) \\
* & * & & & &
\end{array}$$

Table 4.6: After the second iteration.

$r = d_{\mathbb{J}} = \frac{1}{5} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \geq 0.$
Hence, w in (4.12) is a solution to the LP problem and the optimal cost value is such that

$$\begin{bmatrix} 0 & 0 & \frac{1}{5} & \frac{2}{5} \end{bmatrix} w = f(w) + 5,$$

that is $f(w) = -5$.

4.6.4 Initialization of the simplex

In the previous example, it was easy to find a vertex to initialize the simplex algorithm, but this is not always the case. A **two-phase technique** can be used to circumvent this issue.

Suppose that the standard problem is feasible and we want to find $z \in [0, +\infty[^M$ such that $Az = b$. This technique consists of considering the optimization problem:

$$\underset{z \in [0,+\infty[^M, v = (v^{(j)})_{1 \leq j \leq K} \in [0,+\infty[^K}{\text{minimize}} \quad \sum_{j=1}^{K} v^{(j)} \quad \text{s.t.} \quad Az + Dv = b$$

where $D = \text{Diag}(\text{sign}(b^{(1)}), \ldots, \text{sign}(b^{(K)}))$ and v is an **artificial variable**.[1]

- This LP problem can be solved by the simplex algorithm initialized at $z = 0$ and $v = Db = [|b^{(1)}|, \ldots, |b^{(K)}|]^\top$. This corresponds to a vertex of the polyhedron $\{(z, w) \in [0, +\infty[^M \times [0, +\infty[^K | \; Az + Dv = b\}$ since rank $D = K$.

[1] For every $\xi \in \mathbb{R}$, $\text{sign}(\xi) = 1$ if $\xi \geq 0$, -1 is $\xi < 0$.

- Every solution to the problem is such that $v = 0$ and the simplex will return a basic solution to $[A \;\; D]\begin{bmatrix} z \\ v \end{bmatrix} = b$ with $z \in [0, +\infty[^M$.
  Therefore, z is a vertex of the feasible set of the original standard problem.

Example 20. *We want to*

$$\underset{(x^{(1)},x^{(2)})\in[0,+\infty[^2}{\text{minimize}} \quad x^{(1)} + 3x^{(2)} \quad s.t. \quad \begin{cases} x^{(1)} + 2x^{(2)} \geq 12 \\ 2x^{(1)} + 3x^{(2)} \geq 20. \end{cases}$$

i) *We introduce slack variables $x^{(3)}$ and $x^{(4)}$ and reformulate the problem under standard form:*

$$\underset{(x^{(1)},x^{(2)},x^{(3)},x^{(4)})\in[0,+\infty[^4}{\text{minimize}} \quad x^{(1)} + 3x^{(2)} \quad s.t. \quad \begin{cases} x^{(1)} + 2x^{(2)} - x^{(3)} = 12 \\ 2x^{(1)} + 3x^{(2)} - x^{(4)} = 20. \end{cases} \tag{4.13}$$

ii) *First phase: We introduce artificial variables $v^{(1)} = x^{(5)}$ and $v^{(2)} = x^{(6)}$:*

$$\underset{x=(x^{(1)},\ldots,x^{(6)})\in[0,+\infty[^6}{\text{minimize}} \quad x^{(5)} + x^{(6)} \quad s.t. \quad \begin{cases} x^{(1)} + 2x^{(2)} - x^{(3)} + x^{(5)} = 12 \\ 2x^{(1)} + 3x^{(2)} - x^{(4)} + x^{(6)} = 20. \end{cases} \tag{4.14}$$

A vertex is

$$w = [0 \; 0 \; 0 \;\; 0 \; 12 \; 20]^\top,$$

associated with the basic index set $\mathbb{I} = \{5, 6\}$ *The initial table for the simplex method is given by*

$x^{(1)}$	$x^{(2)}$	$x^{(3)}$	$x^{(4)}$	$x^{(5)}$	$x^{(6)}$		
1	2	-1	0	1	0	12	(1)
2	3	0	-1	0	1	20	(2)
0	0	0	0	1	1	f	(3)
				*	*		

Table 4.7: Initial table for the two-phase technique.

We modify the last row to make the cost only dependent on $x_{\mathbb{J}}$.

$x^{(1)}$	$x^{(2)}$	$x^{(3)}$	$x^{(4)}$	$x^{(5)}$	$x^{(6)}$		
1	2	-1	0	1	0	12	(1)
2	3	0	-1	0	1	20	(2)
-3	-5	1	1	0	0	$f - 32$	(3) – (1) – (2)
				*	*		

Table 4.8: Simplified initial table for the two-phase technique.

The first iteration of the simplex algorithm for solving Problem (4.14) yields:

$$r = d_{\mathbb{J}} = \begin{bmatrix} -3 \\ -5 \\ 1 \\ 1 \end{bmatrix} \quad \Rightarrow \quad j = 1$$

$$\Delta_j = A_{\mathbb{I}}^{-1} a_j = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \Rightarrow \quad \widetilde{\mathbb{I}}_j = \{5, 6\}$$

$$z^{(j)} = \min\left\{\tfrac{12}{1}, \tfrac{20}{2}\right\} = 10 \quad \Rightarrow \quad i_j = 6.$$

The new basic index set is $\mathbb{I} = \{1, 5\}$ *and the associated new vertex is*

$$w = [10\ \ 0\ \ 0\ \ 0\ \ (12 - 10 \times 1)\ \ 0]^\top = [10\ \ 0\ 0\ \ 0\ \ 2\ \ 0]^\top.$$

iii) *First phase: second iteration of the simplex*

$x^{(1)}$	$x^{(2)}$	$x^{(3)}$	$x^{(4)}$	$x^{(5)}$	$x^{(6)}$		
1	2	-1	0	1	0	12	(1)
$\boxed{2}$	3	0	-1	0	1	20	(2)
-3	-5	1	1	0	0	$f - 32$	(3)
*				*			

$\rightsquigarrow$

$x^{(1)}$	$x^{(2)}$	$x^{(3)}$	$x^{(4)}$	$x^{(5)}$	$x^{(6)}$		
0	$\frac{1}{2}$	-1	$\frac{1}{2}$	1	$-\frac{1}{2}$	2	$(1) - (2)/2$
$\boxed{1}$	$\frac{3}{2}$	0	$-\frac{1}{2}$	0	$\frac{1}{2}$	10	$(2)/2$
0	$-\frac{1}{2}$	1	$-\frac{1}{2}$	0	$\frac{3}{2}$	$f - 2$	$(3) + \frac{3}{2}(2)$
*				*			

Table 4.9: After the first iteration of the two-phase technique.

The second iteration yields:

$$r = d_{\mathbb{J}} = \begin{bmatrix} -\frac{1}{2} \\ 1 \\ -\frac{1}{2} \\ \frac{3}{2} \end{bmatrix} \quad \Rightarrow \quad j = 2$$

$$\Delta_j = A_{\mathbb{I}}^{-1} a_j = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{3}{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 3 \\ 1 \end{bmatrix} \quad \Rightarrow \quad \widetilde{\mathbb{I}}_j = \{1, 5\}$$

$$z^{(j)} = \min\left\{\frac{10}{3/2}, \frac{2}{1/2}\right\} = \min\left\{\tfrac{20}{3}, 4\right\} = 4 \quad \Rightarrow \quad i_j = 5.$$

The new basic index set is $\mathbb{I} = \{1, 2\}$ *and the associated new vertex is*

$$w = [(10 - 4 \times \frac{3}{2})\ \ 4\ \ 0\ \ 0\ \ 0\ \ 0]^\top = [4\ \ 4\ 0\ \ 0\ \ 0\ \ 0]^\top.$$

iv) *First phase: third iteration of the simplex*

$x^{(1)}$	$x^{(2)}$	$x^{(3)}$	$x^{(4)}$	$x^{(5)}$	$x^{(6)}$		
0	$\boxed{\frac{1}{2}}$	-1	$\frac{1}{2}$	1	$-\frac{1}{2}$	2	(1)
1	$\frac{3}{2}$	0	$-\frac{1}{2}$	0	$\frac{1}{2}$	10	(2)
0	$-\frac{1}{2}$	1	$-\frac{1}{2}$	0	$\frac{3}{2}$	$f-2$	(3)
*	*						

$\rightsquigarrow$

$x^{(1)}$	$x^{(2)}$	$x^{(3)}$	$x^{(4)}$	$x^{(5)}$	$x^{(6)}$		
0	$\boxed{1}$	-2	1	2	-1	4	$2 \times (1)$
1	0	3	-2	-3	2	7	$(2) - 3 \times (1)$
0	0	0	0	1	1	f	$(3) + (1)$
*	*						

Table 4.10: After the second iteration of the two-phase technique.

We have $r = d_{\mathbb{J}} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \geq 0.$

Thus $[4\ 4\ 0\ 0]^\top$ *is an initial vertex for the original LP problem.*

v) *Second phase: first iteration of the simplex*
We come back to Problem (4.13) *for which we have shown that*

$$w = [4\ 4\ 0\ 0]^\top$$

is a vertex of the feasible set associated with the basic index set $\mathbb{I} = \{1, 2\}$.

$x^{(1)}$	$x^{(2)}$	$x^{(3)}$	$x^{(4)}$		
1	2	-1	0	12	(1)
2	3	0	-1	20	(2)
1	3	0	0	f	(3)
*	*				

$\rightsquigarrow$

$x^{(1)}$	$x^{(2)}$	$x^{(3)}$	$x^{(4)}$		
0	1	-2	1	4	$(1') = 2 \times (1) - (2)$
1	0	3	-2	4	$(2') = (1) - 2 \times (1')$
0	0	3	-1	$f - 16$	$(3) - 3 \times (1') - (2')$
*	*				

Table 4.11: Original LP problem after the first phase.

The first iteration of the simplex for Problem (4.13) *yields:*

$r = d_{\mathbb{J}} = \begin{bmatrix} 3 \\ -1 \end{bmatrix} \quad \Rightarrow \quad j = 4$

$$\Delta_j = A_{\mathbb{I}}^{-1} a_j = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \quad \Rightarrow \quad \widetilde{\mathbb{I}}_j = \{2\}$$

$z_j = \frac{4}{1} = 4, \qquad i_j = 2.$

The new basic index set is $\mathbb{I} = \{1, 4\}$ *and the associated new vertex is*

$$w = [(4 - 4 \times (-2) \;\; 0 \;\; 0 \;\; 4]^\top = [12 \;\; 0 \;\; 0 \;\; 4]^\top. \tag{4.15}$$

vi) *Second phase: second iteration of the simplex*

$x^{(1)}$	$x^{(2)}$	$x^{(3)}$	$x^{(4)}$		
0	1	-2	$\boxed{1}$	4	(1)
1	0	3	-2	4	(2)
0	0	3	-1	$f - 16$	(3)
*			*		

$\rightsquigarrow$

$x^{(1)}$	$x^{(2)}$	$x^{(3)}$	$x^{(4)}$		
0	1	-2	$\boxed{1}$	4	(1)
1	2	-1	0	12	$(2) + 2 \times (1)$
0	1	1	0	$f - 12$	$(3) + (1)$
*			*		

Table 4.12: Original LP problem after the first iteration of the simplex.

We have $r = d_{\mathbb{J}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \geq 0.$

Hence, w *given by* (4.15) *is a solution to the standard LP problem and the optimal cost value is* $f(w) = 12$.

4.6.5 Sensitivity analysis

As we have seen in Section 4.6.1, the variable w generated by the simplex is a solution to the LP problem if w is a vertex of the admissible domain of the standard problem and the reduced cost r is greater than or equal to 0, that is

$$\begin{cases} w_{\mathbb{I}} = A_{\mathbb{I}}^{-1} b \geq 0 \\ d_{\mathbb{J}} - A_{\mathbb{J}}^\top (A_{\mathbb{I}}^{-1})^\top d_{\mathbb{I}} \geq 0 \end{cases}$$

One interesting question is to know whether $\mathbb{I}$ remains the optimal basic index set if the problem parameters are perturbed.
The answer to this question is provided below, in three situations.

<u>Case 1</u>: If b is perturbed by $\delta_b \in \mathbb{R}^K$, $\mathbb{I}$ is optimal if and only if

$$\begin{aligned} & A_{\mathbb{I}}^{-1}(b + \delta_b) \geq 0 \\ \Leftrightarrow \quad & A_{\mathbb{I}}^{-1} \delta_b \geq -A_{\mathbb{I}}^{-1} b. \end{aligned}$$

Case 2: If $d_{\mathbb{J}}$ is perturbed by $\delta_{d_{\mathbb{J}}} \in \mathbb{R}^{M-K}$, $\mathbb{I}$ is optimal if and only if

$$\begin{aligned} & d_{\mathbb{J}} + \delta_{d_{\mathbb{J}}} - A_{\mathbb{J}}^\top (A_{\mathbb{I}}^{-1})^\top d_{\mathbb{I}} \geq 0 \\ \Leftrightarrow \quad & \delta_{d_{\mathbb{J}}} \geq A_{\mathbb{J}}^\top (A_{\mathbb{I}}^{-1})^\top d_{\mathbb{I}} - d_{\mathbb{J}} = -r. \end{aligned}$$

Case 3: If $d_{\mathbb{I}}$ is perturbed by $\delta_{d_{\mathbb{I}}} \in \mathbb{R}^{K}$, $\mathbb{I}$ is optimal if and only if

$$\begin{aligned} & d_{\mathbb{J}} - A_{\mathbb{J}}^\top (A_{\mathbb{I}}^{-1})^\top (d_{\mathbb{I}} + \delta_{d_{\mathbb{I}}}) \geq 0 \\ \Leftrightarrow \quad & A_{\mathbb{J}}^\top (A_{\mathbb{I}}^{-1})^\top \delta_{d_{\mathbb{I}}} \leq d_{\mathbb{J}} - A_{\mathbb{J}}^\top (A_{\mathbb{I}}^{-1})^\top d_{\mathbb{I}} = r. \end{aligned}$$

Chapter 5

Integer linear programming

5.1 Motivation example

We consider the following problem in drug logistics:
A health organization wants to distribute g_t vaccine doses against some disease at week $t \in \{1, \ldots, T\}$. Vaccines can be bought from P suppliers with unit cost $(c_{p,t})_{1 \leq p \leq P, 1 \leq t \leq T}$. Supplier $p \in \{1, \ldots, P\}$ can only deliver a minimum quantity $m_{p,t}$ and a maximum quantity $M_{p,t}$ of doses. The health organization has a storage facility allowing to keep at most s doses from one week to the other (any further excess is wasted).
We want to know what is the minimal cost strategy.

To provide an answer, we introduce the following variable. For every $p \in \{1, \ldots, P\}$ and $t \in \{1, \ldots, T\}$,

- $z_{p,t}$ is binary decision variable indicating if doses are bought or not from supplier p at week t
- $n_{p,t}$ designates the number of doses bought from supplier p at week t
- q_t is the number of stored doses at the end of week t.

By gathering all these variables, we end up with a vector of variables x of dimension $N = (2P + 1)T$.

Then, the cost function is given by

$$f(x) = \sum_{t=1}^{T} \sum_{p=1}^{P} c_{p,t} n_{p,t},$$

which corresponds to a linear function.

The constraints to handle read

$$(\forall t \in \{1,\dots,T\}) \quad \begin{cases} q_{t-1} + \sum_{p=1}^{P} n_{p,t} \geq q_t + g_t \\ q_t \leq s \\ (\forall p \in \{1,\dots,P\}) \;\; m_{p,t} z_{p,t} \leq n_{p,t} \leq M_{p,t} z_{p,t} \end{cases}$$

by setting $q_0 = 0$. They are thus linear constraints.

This results in the following optimization formulation:

$$\begin{aligned} &\underset{x \in \mathbb{R}^{T(2P+1)}}{\text{minimize}} \; \sum_{t=1}^{T} \sum_{p=1}^{P} c_{p,t} n_{p,t} \\ &\text{s.t} \;\; (\forall t \in \{1,\dots,T\}) \\ &\quad \begin{cases} q_{t-1} + \displaystyle\sum_{p=1}^{P} n_{p,t} \geq q_t + g_t \\ q_t \leq s \\ q_t \in \mathbb{N} \\ (\forall p \in \{1,\dots,P\}) \;\; \begin{cases} m_{p,t} z_{p,t} \leq n_{p,t} \leq M_{p,t} z_{p,t} \\ n_{p,t} \in \mathbb{N} \\ z_{p,t} \in \{0,1\}, \end{cases} \end{cases} \end{aligned}$$

which can be reexpressed as

$$\begin{aligned} &\underset{x \in \mathcal{N}}{\text{minimize}} \; \sum_{t=1}^{T} \sum_{p=1}^{P} c_{p,t} n_{p,t} \\ &\text{s.t} \;\; (\forall t \in \{1,\dots,T\}) \\ &\quad \begin{cases} q_{t-1} + \displaystyle\sum_{p=1}^{P} n_{p,t} \geq q_t + g_t \\ q_t \leq s \\ (\forall p \in \{1,\dots,P\}) \;\; m_{p,t} z_{p,t} \leq n_{p,t} \leq M_{p,t} z_{p,t} \end{cases} \end{aligned}$$

with $\mathcal{N} = \{0,1\}^{PT} \times \mathbb{N}^{PT} \times \mathbb{N}^{T}$. We see in particular that the binary nature of the first PT unknowns play a crucial role in this formulation.

5.2 Canonical problem

The previous example is an instance of the more general class of problems formulated below.

Problem 6

Let $L \in \mathbb{R}^{K\times N}$, $b \in \mathbb{R}^K$, and $c \in \mathbb{R}^N$.
Let $\mathcal{N}$ be a nonempty subset of $\mathbb{N}^N$.
We consider the following **integer linear programming** (ILP) problem:

$$\underset{x\in\mathcal{N}}{\text{minimize}} \quad \langle c \mid x\rangle \quad \text{s.t.} \quad Lx \geq b.$$

Definition 36

i) The feasibility set is
$$\mathcal{A}_{\mathcal{N}} = \big\{x \in \mathcal{N} \;\big|\; Lx \geq b\big\}.$$

ii) The problem is **feasible** if $\mathcal{A}_{\mathcal{N}} \neq \varnothing$.

iii) The problem is (lower) **bounded** if $\mu_{\mathcal{N}} = \inf_{x\in\mathcal{A}_{\mathcal{N}}} \langle c \mid x\rangle > -\infty$.

Remark 27.

i) *Often $\mathcal{N}$ is finite and the problem is thus bounded. If, in addition, it is feasible, then it admits a solution.*

ii) *$\mathcal{N}$ is nonconvex (when it does not reduce to a singleton). This implies that an ILP problem is nonconvex.*

5.3 Binary linear programming problems

5.3.1 A general property

Definition 37

Problem 6 with $\mathcal{N} = \{0,1\}^N$ is a **binary linear programming** problem.

Proposition 19

Any integer linear programming problem over a finite set $\mathcal{N}$ is equivalent to a binary linear programming problem

Proof. If $\mathcal{N}$ is finite, we can write

$$\mathcal{N} = \{e_1, \dots, e_B\}$$

where $(\forall b \in \{1,\ldots,B\})\ e_b \in \mathbb{N}^N$. Therefore, $x \in \mathcal{N}$ if and only if there exists $\widetilde{x} = (\widetilde{x}^{(b)})_{1\le b\le B} \in \{0,1\}^B$ such that

$$x = \sum_{b=1}^{B} \widetilde{x}^{(b)} e_b$$

with

$$\sum_{b=1}^{B} \widetilde{x}^{(b)} = 1 \quad \Leftrightarrow \quad \langle \mathbf{1} \mid \widetilde{x} \rangle = 1$$

and $\mathbf{1} = [1,\ldots,1]^\top \in \mathbb{R}^B$. Thus, the problem can be recast as

$$\underset{\widetilde{x}\in\{0,1\}^B}{\text{minimize}} \quad \langle \widetilde{c} \mid \widetilde{x} \rangle \quad \text{s.t.} \quad \widetilde{L}\widetilde{x} \ge \widetilde{b}$$

where $\widetilde{L} = \begin{bmatrix} Le_1 & \cdots & Le_B \\ 1 & \cdots & 1 \\ -1 & \cdots & -1 \end{bmatrix}$, $\widetilde{b} = \begin{bmatrix} b \\ 1 \\ -1 \end{bmatrix}$, and $\widetilde{c} = \begin{bmatrix} \langle c \mid e_1 \rangle \\ \vdots \\ \langle c \mid e_B \rangle \end{bmatrix}$. □

5.3.2 Knapsack problem

Definition 38

A **knapsack problem** is a binary linear programming problem with $K = 1$, $L \le 0$, $b \le 0$, and $c \le 0$.

Example 21. *We want to fill a knapsack with N possible objects by maximizing the value of its contents. We define a vector $x \in \{0,1\}^N$ whose i-th component $x^{(i)}$ with $i \in \{1,\ldots,N\}$ indicates whether the i-th object is present ($x^{(i)} = 1$) or not ($x^{(i)} = 0$). We define a vector $\overline{c} \in [0,+\infty[^N$ whose components $(\overline{c}^{(i)})_{1\le i\le N}$ of $\overline{c}$ correspond to the value of each possible object. In addition, we have a limitation $\overline{b}$ on the global weight of the contents of the knapsack. To take into account this constraint, we define a vector $\ell \in [0,+\infty[^N$ whose components $(\ell^{(i)})_{1\le i\le N}$ correspond to the weights of each object.*

The optimization problem can be formulated as

$$\underset{x=(x_i)_{1\le i\le N}\in\{0,1\}^N}{\text{maximize}} \quad \sum_{i=1}^{N} \overline{c}^{(i)} x^{(i)} \quad s.t. \quad \sum_{i=1}^{N} \ell^{(i)} x^{(i)} \le \overline{b}.$$

This is an instance of Problem 6 where $L = -\ell^\top \le 0$, $b = -\overline{b} \le 0$, and $c = -\overline{c} \le 0$.

Proposition 20

Any feasible binary linear programming problem with $K = 1$ is equivalent to a knapsack problem.

Proof. The binary LP reads

$$\underset{(x^{(i)})_{1\le i\le N}\in\{0,1\}^N}{\text{minimize}} \quad \sum_{i=1}^{N} c^{(i)}x^{(i)} \quad \text{s.t.} \quad \sum_{i=1}^{N} \ell^{(i)}x^{(i)} \ge b.$$

For every $j \in \{1,\dots,N\}$,

- if $c^{(j)} \le 0$ and $\ell^{(j)} \ge 0$, setting $x^{(j)} = 1$ leads to a lower value of the cost while giving more freedom in the choice of the $(x^{(i)})_{i\neq j}$ So it is the best choice for the j-th optimized variable.
- If $c^{(j)} > 0$ and $\ell^{(j)} \le 0$, by symmetry, $x^{(j)} = 0$ is the optimal choice.

Let $\mathbb{I}$ be the indices of the remaining components to be optimized.

If $\mathbb{I} \neq \varnothing$, the binary LP reduces to

$$\underset{(x^{(i)})_{i\in\mathbb{I}}\in\{0,1\}^{|\mathbb{I}|}}{\text{minimize}} \quad \sum_{i\in\mathbb{I}} c^{(i)}x^{(i)} \quad \text{s.t.} \quad \sum_{i\in\mathbb{I}} \ell^{(i)}x^{(i)} \ge b' = b - \sum_{i\in\{1,\dots,N\}\setminus\mathbb{I}} \ell^{(i)}x^{(i)}.$$

We have $\mathbb{I} = \mathbb{I}_+ \cup \mathbb{I}_-$ where $\mathbb{I}_+ = \{i \in \mathbb{I} \mid \ell^{(i)} > 0, c^{(i)} > 0\}$ and $\mathbb{I}_- = \{i \in \mathbb{I} \mid \ell^{(i)} < 0, c^{(i)} \le 0\}$. So, we have to

$$\begin{aligned} &\underset{(x^{(i)})_{i\in\mathbb{I}}\in\{0,1\}^{|\mathbb{I}|}}{\text{minimize}} \quad \sum_{i\in\mathbb{I}_-} c^{(i)}x^{(i)} + \sum_{i\in\mathbb{I}_+}(-c^{(i)})(1-x^{(i)}) \\ &\text{s.t.} \quad \sum_{i\in\mathbb{I}_-} \ell^{(i)}x^{(i)} + \sum_{i\in\mathbb{I}_+}(-\ell^{(i)})(1-x^{(i)}) \ge b' - \sum_{i\in\mathbb{I}_+}\ell^{(i)} = \widetilde{b}. \end{aligned}$$

Since the problem is feasible, $\widetilde{b} \le 0$, and we thus obtain a knapsack problem with respect to $\widetilde{x} = (\widetilde{x}^{(i)})_{1\le i\le N} \in \{0,1\}^{|\mathbb{I}|}$ where

$$(\forall i \in \mathbb{I}) \quad \widetilde{x}^{(i)} = \begin{cases} x^{(i)} & \text{if } i \in \mathbb{I}_- \\ 1 - x^{(i)} & \text{if } i \in \mathbb{I}_+. \end{cases}$$

□

5.3.3 Some combinatorial problems

Definition 39

Let $(S_i)_{1\le i\le N}$ be nonempty subsets of a set E and let $\mathbb{I} \subset \{1,\dots,N\}$.

- $(S_i)_{i\in\mathbb{I}}$ is a **cover** of E if $\cup_{i\in\mathbb{I}} S_i = E$
- $(S_i)_{i\in\mathbb{I}}$ is a **packing** of E if $(\forall (i,j) \in \mathbb{I}^2)\ i \neq j \Rightarrow S_i \cap S_j = \varnothing$
- $(S_i)_{i\in\mathbb{I}}$ is a **partition** of E if it is a coverage and a packing of E.

Example 22. *Let $E = \{a, b, c, d, e\}$, $N = 4$, and*

$$\begin{aligned} S_1 &= \{a, b, c\} \\ S_2 &= \{d, e\} \\ S_3 &= \{a, d\} \\ S_4 &= \{e\} \end{aligned}$$

Then $S_1 \cup S_3 \cup S_4$ is a cover of E, $S_1 \cup S_4$ is a packing of E, and $S_1 \cup S_2$ is a partition of E.

Proposition 21

Using the same notation as in Definition 39, assume that $\operatorname{card} E = K$ and let $L = (L_{j,i})_{1 \le j \le K, 1 \le i \le N} \in \{0,1\}^{K \times N}$ be such that, for every $(i, j) \in \{1, \dots, N\} \times \{1, \dots, K\}$,

$$L_{j,i} = \begin{cases} 1 & \text{if the } j\text{-th element of } E \text{ belongs to } S_i \\ 0 & \text{otherwise.} \end{cases} \tag{5.1}$$

Let $x = (x^{(i)})_{1 \le i \le N} \in \{0,1\}^N$ be such that

$$(\forall i \in \{1, \dots, N\}) \quad x^{(i)} = \begin{cases} 1 & \text{if } i \in \mathbb{I} \\ 0 & \text{otherwise.} \end{cases} \tag{5.2}$$

Then

- $(S_i)_{i \in \mathbb{I}}$ *is a cover of E if $Lx \ge \mathbf{1}$*
- $(S_i)_{i \in \mathbb{I}}$ *is a packing of E if $Lx \le \mathbf{1}$*
- $(S_i)_{i \in \mathbb{I}}$ *is a partition of E if $Lx = \mathbf{1}$.*

Proof. This follows from the fact that the j-th component of Lx gives the number of sets $(S_i)_{\in \mathbb{I}}$ containing the j-th element of E. □

Example 23. *In Example 22, we have*

$$L = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}.$$

If $\mathbb{I} = \{1, 3, 4\}$, then

$$x = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix} \quad \Rightarrow \quad Lx = \begin{bmatrix} 2 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$

If $\mathbb{I} = \{1, 4\}$, *then*

$$x = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad \Rightarrow \quad Lx = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}.$$

If $\mathbf{I} = \{1, 2\}$, *then*

$$x = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad \Rightarrow \quad Lx = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$

We may be interesting in the following problems.

Problem 7. (Optimal cover)

Assume that, for every $i \in \{1, \ldots, N\}$, selecting S_i has a cost $c^{(i)} > 0$.
We want to find a cover with minimum global cost.

Problem 8. (Optimal packing)

Assume that, for every $i \in \{1, \ldots, N\}$, selecting S_i has a cost $c^{(i)} < 0$.
We want to find a packing with minimum global cost.

Problem 9. (Optimal partition)

Assume that, for every $i \in \{1, \ldots, N\}$, selecting S_i has a cost $c^{(i)}$.
We want to find a partition with minimum global cost.

Proposition 22

Let $c = (c^{(i)})_{1 \le i \le N}$ *and let* $L = (L_{j,i})_{1 \le j \le K, 1 \le i \le N}$ *be given by* (5.1).

i) *Problem* 7 *is formulated as*

$$\underset{x \in \{0,1\}^N}{\text{minimize}} \quad \langle c \mid x \rangle \quad \text{s.t.} \quad Lx \ge \mathbf{1}. \tag{5.3}$$

ii) *Problem* 8 *is formulated as*

$$\underset{x \in \{0,1\}^N}{\text{minimize}} \quad \langle c \mid x \rangle \quad \text{s.t.} \quad Lx \le \mathbf{1}.$$

iii) *Problem* 9 *is formulated as*

$$\underset{x \in \{0,1\}^N}{\text{minimize}} \quad \langle c \mid x \rangle \quad \text{s.t.} \quad Lx = \mathbf{1}.$$

Proof. Finding the optimal subset $\mathbb{I} \subset \{1, \ldots, N\}$ is equivalent to finding the optimal binary vector $x = (x^{(i)})_{1 \le i \le N}$ defined by (5.2). According to Proposition 21, the optimal cover problem is thus formulated as

$$\underset{x=(x^{(i)})_{1\le i\le N} \in \{0,1\}^N}{\text{minimize}} \quad \sum_{i=1}^{N} c^{(i)} x^{(i)} \quad \text{s.t.} \quad Lx \ge \mathbf{1}.$$

which can be rewritten as (5.3). The optimal packing / partition formulations are similarly obtained. □

5.4 Continuous relaxation

5.4.1 Basic properties

We will be interested in a continuous relaxation of Problem 6 as defined below.

Problem 10

Let $L \in \mathbb{R}^{K\times N}$, $b \in \mathbb{R}^K$, and $c \in \mathbb{R}^N$.
Let S be a nonempty closed convex such that $\mathcal{N} \subset S$.
We consider the following convex problem:

$$\underset{x\in S}{\text{minimize}} \quad \langle c \mid x \rangle \quad \text{s.t.} \quad Lx \ge b.$$

Proposition 23

Let

$$\mathcal{A}_{\mathcal{N}} = \{x \in \mathcal{N} \mid Lx \ge b\}, \quad \mu_{\mathcal{N}} = \inf_{x\in\mathcal{A}_{\mathcal{N}}} \langle c \mid x \rangle$$
$$\mathcal{A}_S = \{x \in S \mid Lx \ge b\}, \quad \mu_S = \inf_{x\in\mathcal{A}_S} \langle c \mid x \rangle .$$

Then,

i) $\mathcal{A}_{\mathcal{N}} \subset \mathcal{A}_S$

ii) *if* $\mathcal{A}_S = \varnothing$*, the ILP problem is unfeasible*

iii) $\mu_S \le \mu_{\mathcal{N}}$

iv) *if* $c \in \mathbb{Z}^N$*,* $\lceil \mu_S \rceil \le \mu_{\mathcal{N}}$ *(*$\lceil\cdot\rceil$*:* round up*).*

Proof.

i) $\mathcal{N} \subset S \Rightarrow \mathcal{A}_{\mathcal{N}} \subset \mathcal{A}_S$.

ii) $\mathcal{A}_S = \varnothing \Rightarrow \mathcal{A}_{\mathcal{N}} = \varnothing$.

iii) This follows from i).

iv) We deduce from iii) that

$$\lceil \mu_S \rceil \leq \lceil \mu_{\mathcal{N}} \rceil = \mu_{\mathcal{N}}.$$

□

Proposition 24

Let $\widehat{x}$ be a solution to Problem 10.
$\widehat{x} \in \mathcal{N}$ if and only if $\widehat{x}$ is a solution to Problem 6.

Proof. If $\widehat{x} \in \mathcal{N}$, it follows from iii) that

$$\mu_{\mathcal{N}} \leq \langle c \mid \widehat{x} \rangle = \mu_S \leq \mu_{\mathcal{N}},$$

which shows that $\langle c \mid \widehat{x} \rangle = \mu_{\mathcal{N}}$.
The converse property obviously holds. □

Remark 28. *Often, we choose $S = [0, +\infty[^N$ (or, for binary problems, $S = [0, 1]^N$).*

5.4.2 Total unimodularity

Definition 40

i) A square matrix is **unimodular** if its determinant is either equal to 1 or -1.

ii) A matrix $L \in \mathbb{R}^{K \times N}$ is **totally unimodular** (TUM) if every non singular square submatrix extracted from L (i.e. any square matrix obtained by deleting some rows and/or some columns of L) is unimodular.

Remark 29. *A matrix $L \in \mathbb{R}^{K \times N}$ is thus TUM if and only if every square submatrix T extracted from L is such that* $\det T \in \{-1, 0, 1\}$.

Property 17. *Let L be a $K \times N$ TUM matrix. Then the following hold.*

i) $L \in \{-1, 0, 1\}^{K \times N}$.

ii) *Any submatrix extracted from L is TUM.*

iii) *$L^\top$ is TUM.*

iv) *Adding a row/column with at most one nonzero component equal to 1 or -1 results in a TUM matrix.*

v) *Changing the sign of a row/column of L results in a TUM matrix.*

Proof.

i) This follows from Remark 29 by considering matrices T of dimension 1×1.

ii) This follows from Definition 40.

iii) This follows from Remark 29 and the fact that, for every square matrix T, $\det T^\top = \det T$.

iv) Because of Property iii), we can focus on the insertion of vector

$$[0, \dots, 0, \underbrace{\epsilon}_{j\text{-th column}}, 0, \dots, 0], \quad \epsilon \in \{-1, 1\}$$

in the i-th row of matrix $L \in \{-1, 0, 1\}^{K \times N}$. Let L' denote the resulting matrix of dimension $(K + 1) \times N$. Any square submatrix T extracted from L' containing elements of the added row will be of the form

$$T = \begin{bmatrix} T_1 \\ 0^\top \\ T_2 \end{bmatrix}$$

or

$$T = \begin{bmatrix} T_{1,1} & t_1 & T_{1,2} \\ 0^\top & \epsilon & 0^\top \\ T_{2,1} & t_2 & T_{2,2} \end{bmatrix}.$$

In the former case, $\det T = 0$ while, in the latter,

$$|\det T| = \left[\det \begin{bmatrix} T_{1,1} & T_{1,2} \\ T_{2,1} & T_{2,2} \end{bmatrix}\right| \in \{0, 1\}.$$

since $\begin{bmatrix} T_{1,1} & T_{1,2} \\ T_{2,1} & T_{2,2} \end{bmatrix}$ is a square submatrix from L.

v) Because of the multilinearity of the determinant, changing the sign of a row (resp. column) of L induces a sign change of the determinant of any square submatrix extracted from L containing elements of this row (resp. column).

□

We now provide a sufficient condition for a matrix to be TUM.

Proposition 25

Let $L = (L_{j,i})_{1 \le j \le K, 1 \le i \le N} \in \{-1, 0, 1\}^{K \times N}$.
Assume that

i) *each column of L contains at most 2 nonzero elements;*

ii) *there exist two disjoint subsets of row indices* $\mathbb{K}_1$ *and* $\mathbb{K}_2$ *such that* $\{1,\dots,K\} = \mathbb{K}_1 \cup \mathbb{K}_2$ *and, for every column* $i \in \{1,\dots,N\}$ *having two nonzero elements,*

$$\sum_{j\in\mathbb{K}_1} L_{j,i} = \sum_{j\in\mathbb{K}_2} L_{j,i}$$

(with $\sum_{j\in\varnothing}(\cdot) = 0$*).*

Then L *is TUM.*

Proof. Assume that i) and ii) hold and that $L \in \{-1,0,1\}^{K\times N}$ is not TUM. Let T be a square submatrix of L with minimum size such that $\det T \notin \{-1,0,1\}$.
Any column of T is non null (otherwise, $\det T = 0$).
Any column of T contains 2 nonzero elements (otherwise, T would not be of minimum size).
We have $T = (L_{j,i})_{j\in\mathbb{J},i\in\mathbb{I}}$ where $\mathbb{I} \subset \{1,\dots,N\}$ and $\mathbb{J} \subset \{1,\dots,K\}$. For every $i \in \mathbb{I}$,

$$\sum_{j\in\mathbb{K}_1} L_{j,i} = \sum_{j\in\mathbb{K}_2} L_{j,i}.$$

Since the i-th column of T contains two nonzero components and i) holds, for every $j \notin \mathbb{J}$, $L_{j,i} = 0$, and consequently,

$$\sum_{j\in\mathbb{K}_1\cap\mathbb{J}} L_{j,i} = \sum_{j\in\mathbb{K}_2\cap\mathbb{J}} L_{j,i}.$$

This shows that $\sum_{j\in\mathbb{K}_1\cap\mathbb{J}} \ell_j = \sum_{j\in\mathbb{K}_2\cap\mathbb{J}} \ell_j$, where $(\ell_j)_{j\in\mathbb{J}}$ are the columns of matrix T. Due to this linear dependence, $\det T = 0$, which contradicts our assumption. □

Remark 30. *Condition ii) means that, for each column having two nonzero elements,*

- *if they are of the same sign, then the row index of one of them should be in* $\mathbb{K}_1$*, and the other in* $\mathbb{K}_2$
- *if they have different signs, then their row indices should be both in* $\mathbb{K}_1$ *or both in* $\mathbb{K}_2$*.*

Example 24.

i) $L = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ *is not TUM since* $\det(L) = 2$.

ii) $L = \begin{bmatrix} 1 & 1 & -1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ *is TUM since Proposition 25 applies with*

$$\mathbb{K}_1 = \{1,2,3,4\}, \quad \mathbb{K}_2 = \varnothing.$$

iii) $L = \begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & -1 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$ *is TUM since Proposition 25 applies with*

$$\mathbb{K}_1 = \{1, 3\}, \quad \mathbb{K}_2 = \{2, 4\}.$$

5.4.3 Bipartite graphs

Definition 41

A graph is **bipartite** if its set of nodes can be divided into two sets $\mathbb{V}_1$ and $\mathbb{V}_2$ such that any edge of the graph links a node in $\mathbb{V}_1$ to a node in $\mathbb{V}_2$.

Definition 42

Let N be the number of edges and K the number of nodes.
$L = (L_{j,i})_{1 \le j \le K, 1 \le i \le N}$ is the **undirected incidence matrix** of the graph if, for every $i \in \{1, \ldots, N\}$ and $j \in \{1, \ldots, K\}$,

$$L_{j,i} = \begin{cases} 1 & \text{if the } i\text{-th edge connects the } j\text{-th node} \\ 0 & \text{otherwise.} \end{cases}$$

Example 25. *Let us consider the following graph:*

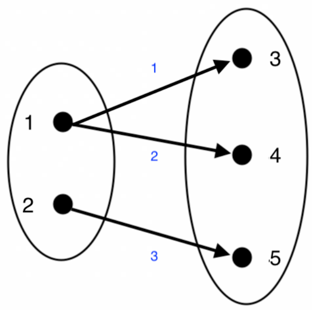

Figure 5.1: Bipartite graph.

Its undirected incidence matrix is

$$L = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Proposition 26

The undirected incidence matrix of a bipartite graph is TUM.

Proof. Each column of the undirected incidence matrix contains 2 nonzero elements. Condition ii) in Proposition 25 is satisfied by choosing $\mathbb{K}_1$ (resp. $\mathbb{K}_2$) as the index set of the nodes in $\mathbb{V}_1$ (resp. $\mathbb{V}_2$). □

Example 26. *Assume that the bipartite graph models compatible relations between two sets of individuals we would like to pair. Let* $x = (x^{(i)})_{1\leq i\leq N}$ *be such that* $x^{(i)} = 1$ *if the individuals linked by the* i*-th edge are paired,* 0 *otherwise. Assume that the selection of the* i*-th link results in a satisfaction score* $\overline{c}^{(i)}$*. Let* $\overline{c} = (\overline{c}^{(i)})_{1\leq i\leq N}$*.*
An ***optimal matching*** *is a solution to*

$$\underset{x\in\{0,1\}^N}{\text{maximize}} \quad \langle \overline{c} \mid x\rangle \quad \textit{s.t.} \quad Lx \leq \mathbf{1}.$$

5.4.4 Exact LP relaxation

Proposition 27

Let L *be a* $K\times N$ *TUM matrix, let* $b \in \mathbb{Z}^K$*, and* $c \in \mathbb{R}^N$*.*
Any solution $\widehat{x}$ *delivered by the simplex to the LP problem*

$$\underset{x\in[0,+\infty[^N}{\text{minimize}} \quad \langle c \mid x\rangle \quad \textit{s.t.} \quad Lx \geq b.$$

is such that $\widehat{x} \in \mathbb{N}^N$*.*

Proof. The standard form of the LP problem reads

$$\underset{z\in[0,+\infty[^M}{\text{minimize}} \quad \langle d \mid z\rangle \quad \text{s.t.} \quad Az = b.$$

where $M = N + K$, $A = [L \quad -\mathrm{Id}]$, $z = \begin{bmatrix} x \\ s \end{bmatrix} \in \mathbb{R}^M$, and $d = \begin{bmatrix} c \\ 0 \end{bmatrix} \in \mathbb{R}^M$. According to Property 17iv), A is TUM. The simplex can be applied if $\operatorname{rank}(A) = K$.

If a solution exists, the optimal basic index set $\mathbb{I}$ returned by the simplex is associated with a solution $\widehat{z}$ such that

$$\widehat{z}_{\mathbb{I}} = A_{\mathbb{I}}^{-1} b \quad \text{and} \quad \widehat{z}_{\mathbb{J}} = 0 \quad \text{with} \quad \mathbb{J} = \{1, \ldots, M\} \setminus \mathbb{I}.$$

$A_{\mathbb{I}}$ is a matrix composed of the columns of A with indices in $\mathbb{I}$.
Since A is TUM, we deduce from Property 17ii) that $A_{\mathbb{I}}$ is TUM.
According to Cramer's rule,

$$A_{\mathbb{I}}^{-1} = \frac{1}{\det A_{\mathbb{I}}} \operatorname{co}(A_{\mathbb{I}})^\top$$

where $\operatorname{co}(A_{\mathbb{I}})$ is the matrix of cofactors of $A_{\mathbb{I}}$ which are up to a sign change equal to the determinants of minors of matrix $A_{\mathbb{I}}$.
Hence, the elements of $A_{\mathbb{I}}^{-1}$ are in $\{-1, 0, 1\}$ and $\widehat{z} \in \mathbb{Z}^M \cap [0, +\infty[^N$. □

Corollary 3

Let L be a $K \times N$ TUM matrix, let $b \in \mathbb{Z}^K$, and $c \in \mathbb{R}^N$.
Any solution $\widehat{x}$ delivered by the simplex to the LP problem

$$\underset{x \in [0, +\infty[^N}{\text{minimize}} \quad \langle c \mid x \rangle \quad s.t. \quad Lx = b.$$

is such that $\widehat{x} \in \mathbb{N}^N$.

5.5 Branch and bound

5.5.1 Principles

Proposition 28

Using the same notation as in Proposition 23, let $(\mathcal{N}_k)_{1 \le k \le n}$ be a partition of $\mathcal{N}$ and let $(S_k)_{1 \le k \le n}$ be a packing of $\mathbb{R}^N$ such that

$$(\forall k \in \{1, \ldots, n\}) \quad \mathcal{N}_k \subset S_k.$$

Then, $\min\{\mu_{S_1}, \ldots, \mu_{S_n}\} \le \mu_{\mathcal{N}}$.
In addition, if there exists $k \in \{1, \ldots, n\}$ such that

i) $\mu_{S_k} = \min\{\mu_{S_1}, \ldots, \mu_{S_n}\}$

ii) $\exists \widehat{x}_k \in \mathcal{N} \cap \underset{x \in \mathcal{A}_{S_k}}{\operatorname{Argmin}} \ \langle c \mid x \rangle,$

then $\widehat{x}_k$ is a solution to the ILP problem.

Proof. Since $(\mathcal{N}_k)_{1\le k\le n}$ is a partition of $\mathcal{N}$, $\mu_{\mathcal{N}} = \min\{\mu_{\mathcal{N}_1},\dots,\mu_{\mathcal{N}_n}\}$.
In addition, for every $k \in \{1,\dots,n\}$, $\mathcal{N}_k \subset S_k \Rightarrow \mu_{S_k} \le \mu_{\mathcal{N}_k}$. Hence, $\min\{\mu_{S_1},\dots,\mu_{S_n}\} \le \mu_{\mathcal{N}}$. If Conditions i) and ii) hold, then $\widehat{x}_k \in \mathcal{A}_{\mathcal{N}}$ and $\mu_{S_k} = \langle c \mid \widehat{x}_k\rangle \le \mu_{\mathcal{N}}$. So, we necessarily have $\langle c \mid \widehat{x}_k\rangle = \mu_{\mathcal{N}}$ and $\widehat{x}_k$ is a solution to the ILP. □

The idea of the numerical approach we will present consists of building iteratively sets $(S_k)_{1\le k\le n}$ until Conditions i) and ii) are met. To do so, we set

$$(\forall k \in \{1,\dots,n\}) \quad \mathcal{N}_k = S_k \cap \mathcal{N}.$$

At each iteration, a lower bound of $\mu_{\mathcal{N}}$ is obtained by computing μ_{S_k} with $k \in \{1,\dots,n\}$. For this **bounding** operation to be efficient, the computation of $(\mu_{S_k})_{1\le k\le n}$ should be easy. To meet this requirement, we choose $(S_k)_{1\le k\le n}$ as polyhedrons, which allows us to compute $(\mu_{S_k})_{1\le k\le n}$ by linear programming.

5.5.2 Algorithm

In the following, we will assume that

i) $\mathcal{N} = \mathcal{Q}^N$ where $\mathcal{Q} \subset \mathbb{N}$ and $\operatorname{card} \mathcal{Q} \ge 2$

ii) the LP relaxation of the ILP problem on $[0,+\infty[^N$ is bounded.

The algorithm involves a **branching** process. At iteration $n \ge 1$ of the algorithm, we split one set $S_{n-1,\ell}$ in two sets $S_{n,\ell}$ and $S_{n,\ell+1}$ while keeping the others, that is

$$S_{n,k} = \begin{cases} S_{n-1,k} & \text{if } k < \ell \\ S_{n-1,k-1} & \text{if } k > \ell + 1. \end{cases}$$

When $n > 1$, an heuristic choice for ℓ is made. Among the sets for which a computed LP solution $\widehat{x}_{n-1,\ell} \notin \mathcal{N}$, we look for the most promising one, i.e. the one with the smallest value for $\mu_{S_{n-1,\ell}}$. Let $\widehat{x}^{(i)}_{n-1,\ell}$ be the component of $\widehat{x}_{n-1,\ell}$ which is the farthest from an element of $\mathcal{Q}$. We set

$$S_{n,\ell} = \big\{x \in S_{n-1,\ell} \;\big|\; x^{(i)} \le \lfloor \widehat{x}^{(i)}_{n-1,\ell} \rfloor\big\}$$
$$S_{n,\ell+1} = \big\{x \in S_{n-1,\ell} \;\big|\; x^{(i)} \ge \lceil \widehat{x}^{(i)}_{n-1,\ell} \rceil\big\}$$

where $\lfloor \widehat{x}^{(i)}_{n-1,\ell} \rfloor < \lceil \widehat{x}^{(i)}_{n-1,\ell} \rceil$ are the two elements of $\mathcal{Q}$ closest to $\widehat{x}^{(i)}_{n-1,\ell}$.

Algorithm 4. (Branch and Bound)

① *Intialization: Set $S_{0,1} = [0,+\infty[^N$.*
If the LP problem is unfeasible, then the ILP one is unfeasible too; exit.
Compute a solution $\widehat{x}_{0,1}$ to the LP problem.

If $\widehat{x}_{0,1} \in \mathcal{N}$*, then* $\widehat{x}_{0,1}$ *is a solution to the ILP; exit.*
Set $n = 1$*.*

② *Branch: Split* $S_{n-1,\ell}$ *into* $S_{n,\ell}$ *and* $S_{n,\ell+1}$*.*

③ *For every* $k \in \{\ell, \ell+1\}$*,*

- *Solve the associated LP problem.*
- *If it is unfeasible, then the solution does not belongs to* $S_{n,k}$*, and* $\mathcal{N}_{n,k} = S_{n,k} \cap \mathcal{N}$ *can be discarded.*
- *If it is feasible, compute a solution* $\widehat{x}_{n,k}$ *to the LP problem and the associated bound* $\mu_{S_{n,k}}$*.*

④ *If all the sets* $(\mathcal{N}_{n,k})_k$ *have been discarded, the ILP problem is unfeasible; exit.*

⑤ *For every* k*,*

- *if* $\mu_{S_{n,k}} = \min_{k'} \mu_{S_{n,k'}}$ *and* $\widehat{x}_{n,k} \in \mathcal{N}$*, then* $\widehat{x}_{n,k}$ *is a solution to the ILP problem; exit*

⑥ $n \leftarrow n+1$*; goto* ②*.*

Remark 31.

i) *If the initial LP relaxation is unbounded and* $\mathcal{Q}$ *is finite, an initial LP relaxation on* $[0, \max \mathcal{Q}]^N$ *can be performed.*

ii) *If* $\mathcal{Q}$ *is finite, in the worst case, the algorithm stops after a complete enumeration of* $\mathcal{N}$*.*

iii) *The algorithm actually builds a dyadic tree where the leaves, at a given iteration, represent the sets* $(S_{n,k})_k$*. Discarding a domain is then equivalent to prune the tree.*

5.5.3 Use of an upper bound

The following remark can be useful to reduce the complexity of Algorithm 4.

At iteration n, assume that an upper bound $\overline{\mu}_n$ is available on $\mu_{\mathcal{N}}$. Then, for every k associated with a feasible LP problem, if $\mu_{S_{n,k}} > \overline{\mu}_n$, then no solution to the ILP problem can belong to $S_{n,k}$ and this set can be discarded.

The **upper bounding** process is implemented as follows:

- For every (n, k) for which the LP problem is feasible, compute the following upper bound on $\mu_{\mathcal{N}}$:

$$\overline{\mu}_{n,k} = \langle c \mid \widetilde{x}_{n,k} \rangle$$

if $\widetilde{x}_{n,k} \in \mathcal{A}_{\mathcal{N}}$ can be (easily) found by rounding $\widehat{x}_{n,k}$.
Otherwise, set $\overline{\mu}_{n,k} = +\infty$.

- At iteration $n \geq 1$, set $\overline{\mu}_n = \min\{\overline{\mu}_{n-1}, \min_k \overline{\mu}_{n,k}\}$ with $\overline{\mu}_0 = \overline{\mu}_{0,1}$.

5.5.4 Example

We want to

$$\underset{(x^{(1)},x^{(2)})\in\mathbb{N}^2}{\text{minimize}} \quad x^{(1)} - 4x^{(2)} \quad \text{s.t} \quad \begin{cases} -5x^{(1)} + 5x^{(2)} \leq 11 \\ 5x^{(1)} + 5x^{(2)} \leq 49 \end{cases}$$

Note that we necessarily have $5x^{(1)} \leq 49$ and $10x^{(2)} \leq 60$, so that $(x^{(1)}, x^{(2)}) \in \{0,\dots,9\} \times \{0,\dots,6\}$.

i) Initialization of branch and bound
$S_{0,1} = [0, +\infty[^2$
By solving the associated LP problem by the simplex method, we get

$$\widehat{x}_{0,1} = \begin{bmatrix} 3.8 \\ 6 \end{bmatrix}$$

and the associated lower bound is $\mu_{S_{0,1}} = 3.8 - 4 \times 6 = -20.2$.
On the other hand, rounding $\widehat{x}_{0,1}$ would not obviously deliver a feasible point. So, we set $\overline{\mu}_{0,1} = +\infty$ as an upper bound.

ii) Iteration $n = 1$
We branch $(0, 1)$ and split with respect to the first component:

- $S_{1,1} = \{(x^{(1)}, x^{(2)}) \in [0, +\infty[^2 \mid x^{(1)} \leq 3\}$
  $\Rightarrow \widehat{x}_{1,1} = \begin{bmatrix} 3 \\ 5.2 \end{bmatrix}$ and $\mu_{S_{1,1}} = -17.8$.

  By rounding down the second component, we get a feasible point $\widetilde{x}_{1,1} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$, so leading to the upper bound $\overline{\mu}_{1,1} = 3 - 4 \times 5 = -17$.
- $S_{1,2} = \{(x^{(1)}, x^{(2)}) \in [0, +\infty[^2 \mid x^{(1)} \geq 4\}$
  $\Rightarrow \widehat{x}_{1,2} = \begin{bmatrix} 4 \\ 5.8 \end{bmatrix}$ and $\mu_{S_{1,2}} = -19.2$.

  We can choose $\widetilde{x}_{1,2} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$, so leading to the upper bound $\overline{\mu}_{1,2} = -16$.

The upper bound is now $\overline{\mu}_1 = \min\{\overline{\mu}_{1,1}, \overline{\mu}_{1,2}\} = -17$.

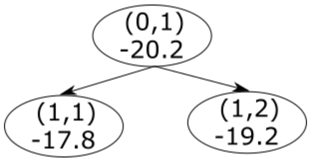

Figure 5.2: First iteration of Branch and Bound.

iii) Iteration $n = 2$
The most promising domain being $S_{1,2}$, we branch $(1, 2)$.

- $S_{2,2} = \{(x^{(1)}, x^{(2)}) \in [0, +\infty[^2 \mid x^{(1)} \geq 4, x^{(2)} \leq 5\}$
$\Rightarrow \widehat{x}_{2,2} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$ and $\mu_{S_{2,2}} = -16$.
Since $\widetilde{x}_{2,2} = \widehat{x}_{2,2}$, $\overline{\mu}_{2,2} = -16$.
Since $\mu_{S_{2,2}} > \overline{\mu}_1$, $S_{2,2}$ can be however discarded.
- $S_{2,3} = \{(x^{(1)}, x^{(2)}) \in [0, +\infty[^2 \mid x^{(1)} \geq 4, x^{(2)} \geq 6\}$
$\rightsquigarrow$ unfeasible.

The upper bound is $\overline{\mu}_2 = \min\{\overline{\mu}_1, \overline{\mu}_{2,2}\} = -17$.

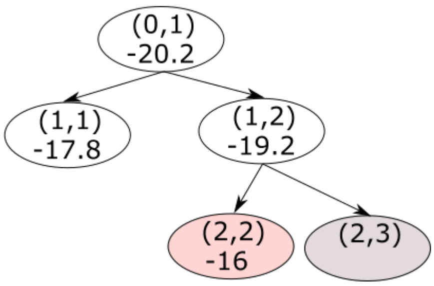

Figure 5.3: Second iteration of Branch and Bound.

iv) Iteration $n = 3$
The only possible choice is to branch $(1, 1)$.

- $S_{3,1} = \{(x^{(1)}, x^{(2)}) \in [0, +\infty[^2 \mid x^{(1)} \leq 3, x^{(2)} \leq 5\}$
$\Rightarrow \widehat{x}_{3,1} = \begin{bmatrix} 2.8 \\ 5 \end{bmatrix}$ and $\mu_{S_{3,1}} = -17.2$.
No obvious feasible rounding: $\overline{\mu}_{3,1} = +\infty$ (actually $(3, 5)$ would work).
- $S_{3,2} = \{(x^{(1)}, x^{(2)}) \in [0, +\infty[^2 \mid x^{(1)} \leq 3, x^{(2)} \geq 6\}$
$\rightsquigarrow$ unfeasible.

The upper bound is unchanged $\overline{\mu}_3 = -17$.

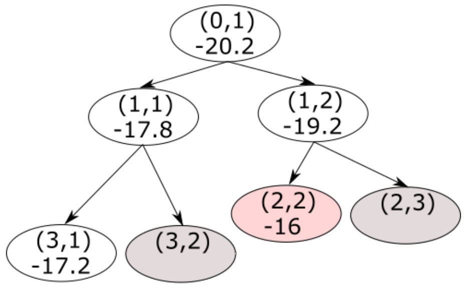

Figure 5.4: Third iteration of Branch and Bound.

v) Iteration $n = 4$
The only possible choice is to branch $(3, 1)$.

- $S_{4,1} = \{(x^{(1)}, x^{(2)}) \in [0, +\infty[^2 \mid x^{(1)} \leq 2, x^{(2)} \leq 5\}$
  $\Rightarrow \widehat{x}_{4,1} = \begin{bmatrix} 2 \\ 4.2 \end{bmatrix}$ and $\mu_{S_{4,1}} = -14.8$.
  Since $\mu_{S_{4,1}} > \overline{\mu}_3$, $S_{4,1}$ can be discarded.
- $S_{4,2} = \{(x^{(1)}, x^{(2)}) \in [0, +\infty[^2 \mid x^{(1)} = 3, x^{(2)} \leq 5\}$
  $\Rightarrow \widehat{x}_{4,2} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$ and $\mu_{S_{4,2}} = -17$.
  Since we have reached the minimum lower bound for an element in $\mathbb{N}^2$, we have obtained an optimal solution to the ILP problem.

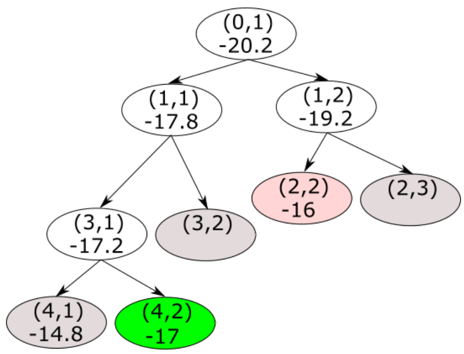

Figure 5.5: Fourth iteration of Branch and Bound.

5.6 Mixed problems

Branch and bound can also be applied to solve MILP problems as defined below.

Problem 11

Let $L_1 \in \mathbb{R}^{K \times N_1}$, $L_2 \in \mathbb{R}^{K \times N_2}$, $b \in \mathbb{R}^K$, $c_1 \in \mathbb{R}^{N_1}$, and $c_2 \in \mathbb{R}^{N_2}$.
Let $\mathcal{N}$ be a nonempty subset of $\mathbb{N}^{N_2}$.
We consider the following **mixed integer linear programming** (MILP) problem:

$$\underset{x_1 \in [0,+\infty[^{N_1}, x_2 \in \mathcal{N}}{\text{minimize}} \quad \langle c_1 \mid x_1 \rangle + \langle c_2 \mid x_2 \rangle \quad \text{s.t.} \quad L_1 x_1 + L_2 x_2 \geq b.$$

Chapter 6

Lagrange multipliers method

6.1 Motivating example

Let $u \in \mathbb{R}^N$ be such that $\|u\| = 1$ and let $\delta \in [0, +\infty[$.
We define the affine hyperplane

$$A = \{x_1 \in \mathbb{R}^N \mid \langle u \mid x_1 \rangle = \delta\}.$$

Let $\varphi \colon \mathbb{R}^N \to \mathbb{R}$ be a lower-semicontinuous function. We define the closed set

$$S = \{x_2 \in \mathbb{R}^N \mid \varphi(x_2) \leq 1\}.$$

The problem we want to solve is to determine the distance $d(A, S)$ between A and S. This problem can be recast as the following optimization one:

$$\begin{aligned}
&\underset{(x_1,x_2)\in(\mathbb{R}^N)^2}{\text{minimize}} \quad \|x_1 - x_2\|^2 \\
&\text{s.t.} \quad \begin{cases} \langle u \mid x_1 \rangle = \delta \\ \varphi(x_2) \leq 1. \end{cases}
\end{aligned}$$

A first question is: How to solve such problem? A second one is: Which results can we expect when φ is nonconvex?

Remark 32. *If S is nonempty and bounded, the existence of a solution to the optimization problem is guaranteed.*

Proof. The problem is equivalent to

$$\underset{(x_1,x_2)\in(\mathbb{R}^N)^2}{\text{minimize}} \quad \underbrace{\|x_1 - x_2\| + \iota_A(x_1) + \iota_S(x_2)}_{\psi(x_1,x_2)}.$$

For every $(x_1, x_2) \in (\mathbb{R}^N)^2$, either $x_2 \notin S \Rightarrow \psi(x_1, x_2) = +\infty$
or $x_2 \in S$ and there exists $\rho \in\,]0, +\infty[$ such that $\|x_2\| \leq \rho$. In this case,

$$\psi(x_1, x_2) = \|x_1 - x_2\| + \iota_A(x_1) \geq \|x_1\| - \|x_2\| + \iota_A(x_1) \geq \|x_1\| - \rho + \iota_A(x_1).$$

Since $x_1 \mapsto \|x_1\| - \rho + \iota_A(x_1)$ is coercive, we deduce that, for every $M \in \,]0,+\infty[$, there exists $\eta(= M+\rho)$ such that

$$\|x_1\| \geq \eta \quad \Rightarrow \quad \psi(x_1,x_2) \geq M.$$

Therefore, $\|x_1\| + \|x_2\| \geq \eta + \rho \Rightarrow \psi(x_1,x_2) \geq M$, which shows that ψ is coercive. As A and S are nonempty and closed, ψ is also proper and lower-semicontinuous and it thus admits a minimizer. □

6.2 Constrained optimization problem

Problem 12

Let $\mathcal{H}$ be a Hilbert space. Let $f\colon \mathcal{H} \to \,]-\infty,+\infty]$ be proper.
Let $(m,q) \in \mathbb{N}^2$. For every $i \in \{1,\ldots,m\}$, let $g_i\colon \mathcal{H} \to \mathbb{R}$ and for every $j \in \{1,\ldots,q\}$, let $h_j\colon \mathcal{H} \to \mathbb{R}$.
Let

$$\begin{aligned} C = \{x \in \mathcal{H} \mid (\forall i \in \{1,\ldots,m\})\ \ & g_i(x) = 0 \\ (\forall j \in \{1,\ldots,q\})\ \ & h_j(x) \leq 0\}. \end{aligned}$$

We want to:

$$\text{Find} \quad \widehat{x} \in \operatorname*{Argmin}_{x \in C} \ f(x).$$

Remark 33. *A vector $x \in \mathcal{H}$ is said to be **feasible** if $x \in \operatorname{dom} f \cap C$.*

Definition 43

The **Lagrange function** (or Lagrangian) associated with Problem 12 is defined as

$$(\forall x \in \mathcal{H})(\forall \nu = (\nu^{(i)})_{1\leq i\leq m} \in \mathbb{R}^m)(\forall \lambda = (\lambda^{(j)})_{1\leq j\leq q} \in [0,+\infty[^q)$$
$$\mathcal{L}(x,\nu,\lambda) = f(x) + \sum_{i=1}^{m} \nu^{(i)} g_i(x) + \sum_{j=1}^{q} \lambda^{(j)} h_j(x). \tag{6.1}$$

The vectors ν and λ are called **Lagrange multipliers**.

Remark 34.

i) $\operatorname{dom} \mathcal{L} = \operatorname{dom} f \times \mathbb{R}^m \times [0,+\infty[^q$.

ii) *When $q = 0$ (**only equality constraints**), the Lagrange function simplifies to*

$$(\forall x \in \mathcal{H})(\forall \nu = (\nu^{(i)})_{1\leq i\leq m} \in \mathbb{R}^m) \quad \mathcal{L}(x,\nu) = f(x) + \sum_{i=1}^{m} \nu^{(i)} g_i(x). \tag{6.2}$$

iii) *When* $m = 0$ *(**only inequality constraints**), the Lagrange function simplifies to*

$$(\forall x \in \mathcal{H})(\forall \lambda = (\lambda^{(j)})_{1 \le j \le q} \in [0, +\infty[^q) \quad \mathcal{L}(x, \lambda) = f(x) + \sum_{j=1}^{q} \lambda^{(j)} h_j(x). \tag{6.3}$$

6.3 Lagrange duality

Definition 44

The **primal Lagrange function** is

$$(\forall x \in \mathcal{H}) \quad \overline{\mathcal{L}}(x) = \sup_{\nu \in \mathbb{R}^m, \lambda \in [0, +\infty[^q} \mathcal{L}(x, \nu, \lambda).$$

The **dual Lagrange function** is

$$(\forall (\nu, \lambda) \in \mathbb{R}^m \times [0, +\infty[^q) \quad \underline{\mathcal{L}}(\nu, \lambda) = \inf_{x \in \mathcal{H}} \mathcal{L}(x, \nu, \lambda)$$

Proposition 29

i) *For every* $x \in \mathcal{H}$

$$\overline{\mathcal{L}}(x) = \begin{cases} f(x) & \text{if } x \in C \\ +\infty & \text{otherwise.} \end{cases}$$

ii) $-\underline{\mathcal{L}}$ *is convex and l.s.c.*

Proof.

i) For every $(x, \nu, \lambda) \in C \times \mathbb{R}^m \times [0, +\infty[^q$,

$$\mathcal{L}(x, \nu, \lambda) = f(x) + \sum_{i=1}^{m} \underbrace{\nu^{(i)} g_i(x)}_{=0} + \sum_{j=1}^{q} \underbrace{\lambda^{(j)} h_j(x)}_{\le 0} \le f(x)$$

and $\overline{\mathcal{L}}(x) = \sup_{\nu \in \mathbb{R}^m, \lambda \in [0, +\infty[^q} \mathcal{L}(x, \nu, \lambda) = f(x)$.

For every $(x, \nu, \lambda) \in \mathcal{H} \times \mathbb{R}^m \times [0, +\infty[^q$, if $x \notin C$, $(\exists i \in \{1, \dots, m\})\ g_i(x) \neq 0$ or $(\exists j \in \{1, \dots, q\})\ h_j(x) > 0$.

In the latter case, by letting $\lambda^{(j)} \to +\infty$ in (6.1), we deduce that

$$\overline{\mathcal{L}}(x) = \sup_{\nu \in \mathbb{R}^m, \lambda \in [0, +\infty[^q} \mathcal{L}(x, \nu, \lambda) = +\infty.$$

A similar reasoning can be followed in the former case.

ii) For every $(\nu, \lambda) \in \mathbb{R}^m \times [0, +\infty[^q$,

$$-\underline{\mathcal{L}}(\nu, \lambda) = \sup_{x \in \operatorname{dom} f} (-\mathcal{L}(x, \nu, \lambda)).$$

$-\underline{\mathcal{L}}$ is thus the supremum of a set of affine functions.

□

Proposition 30. (Weak Lagrange duality)

For every $(x, \nu, \lambda) \in \mathcal{H} \times \mathbb{R}^m \times [0, +\infty[^q$,

$$\underline{\mathcal{L}}(\nu, \lambda) \leq \overline{\mathcal{L}}(x).$$

In addition,

$$\sup_{\nu \in \mathbb{R}^m, \lambda \in [0,+\infty[^q} \underline{\mathcal{L}}(\nu, \lambda) \leq \mu = \inf_{x \in C} f(x).$$

Proof. We have, for every $(x, \nu, \lambda) \in \mathcal{H} \times \mathbb{R}^m \times [0, +\infty[^q$,

$$\begin{aligned} & \inf_{x'} \mathcal{L}(x', \nu, \lambda) \leq \mathcal{L}(x, \nu, \lambda) \leq \sup_{\nu', \lambda'} \mathcal{L}(x, \nu', \lambda') \\ \Rightarrow \quad & \underline{\mathcal{L}}(\nu, \lambda) \leq \overline{\mathcal{L}}(x). \end{aligned}$$

We deduce from Proposition 29i) that, for every $x \in C$, $\underline{\mathcal{L}}(\nu, \lambda) \leq \overline{\mathcal{L}}(x) = f(x)$, which yields the last inequality. □

6.4 Saddle points

Definition 45

$(\widehat{x}, \widehat{\nu}, \widehat{\lambda}) \in \mathcal{H} \times \mathbb{R}^m \times [0, +\infty[^q$ is a **saddle point** of $\mathcal{L}$ if

$$\big(\forall (x, \nu, \lambda) \in \mathcal{H} \times \mathbb{R}^m \times [0, +\infty[^q\big) \qquad \mathcal{L}(\widehat{x}, \nu, \lambda) \leq \mathcal{L}(\widehat{x}, \widehat{\nu}, \widehat{\lambda}) \leq \mathcal{L}(x, \widehat{\nu}, \widehat{\lambda}).$$

Remark 35. *If* $(\widehat{x}, \widehat{\nu}, \widehat{\lambda})$ *is a saddle point of* $\mathcal{L}$*, then it follows from the right inequality that* $\widehat{x} \in \operatorname{dom} f$*.*

Theorem 20

$(\widehat{x}, \widehat{\nu}, \widehat{\lambda}) \in \mathcal{H} \times \mathbb{R}^m \times [0, +\infty[^q$ is a saddle point of $\mathcal{L}$ if and only if

$$\begin{aligned}
&(\forall x \in \mathcal{H}) \quad \overline{\mathcal{L}}(\widehat{x}) \leq \overline{\mathcal{L}}(x) \\
&(\forall (\nu, \lambda) \in \mathbb{R}^m \times [0, +\infty[^q) \quad \underline{\mathcal{L}}(\nu, \lambda) \leq \underline{\mathcal{L}}(\widehat{\nu}, \widehat{\lambda}) \\
&\overline{\mathcal{L}}(\widehat{x}) = \underline{\mathcal{L}}(\widehat{\nu}, \widehat{\lambda}).
\end{aligned}$$

Proof. If $(\widehat{x}, \widehat{\nu}, \widehat{\lambda})$ is a saddle point of $\mathcal{L}$ then, for every $(x', \nu', \lambda') \in \mathcal{H} \times \mathbb{R}^m \times [0, +\infty[^q$,

$$\begin{aligned}
& \mathcal{L}(\widehat{x}, \nu', \lambda') \leq \mathcal{L}(\widehat{x}, \widehat{\nu}, \widehat{\lambda}) \leq \mathcal{L}(x', \widehat{\nu}, \widehat{\lambda}) \\
\Rightarrow \quad & \sup_{\nu', \lambda'} \mathcal{L}(\widehat{x}, \nu', \lambda') \leq \inf_{x'} \mathcal{L}(x', \widehat{\nu}, \widehat{\lambda}) \\
\Leftrightarrow \quad & \overline{\mathcal{L}}(\widehat{x}) \leq \underline{\mathcal{L}}(\widehat{\nu}, \widehat{\lambda}) \\
\Rightarrow \quad & \inf_x \overline{\mathcal{L}}(x) \leq \overline{\mathcal{L}}(\widehat{x}) \leq \underline{\mathcal{L}}(\widehat{\nu}, \widehat{\lambda}) \leq \sup_{\nu, \lambda} \underline{\mathcal{L}}(\nu, \lambda).
\end{aligned}$$

In addition, it follows from Proposition 30 that

$$\sup_{\nu, \lambda} \underline{\mathcal{L}}(\nu, \lambda) \leq \inf_x \overline{\mathcal{L}}(x).$$

Therefore, $\inf_x \overline{\mathcal{L}}(x) = \sup_{\nu, \lambda} \underline{\mathcal{L}}(\nu, \lambda)$.
We deduce that $\overline{\mathcal{L}}(\widehat{x}) = \inf_x \overline{\mathcal{L}}(x)$, $\underline{\mathcal{L}}(\widehat{\nu}, \widehat{\lambda}) = \sup_{\nu, \lambda} \underline{\mathcal{L}}(\nu, \lambda)$, and $\overline{\mathcal{L}}(\widehat{x}) = \underline{\mathcal{L}}(\widehat{\nu}, \widehat{\lambda})$.

Conversely, if the last condition holds, then

$$\begin{aligned}
\mathcal{L}(\widehat{x}, \widehat{\nu}, \widehat{\lambda}) \leq \sup_{\nu, \lambda} \mathcal{L}(\widehat{x}, \nu, \lambda) &= \overline{\mathcal{L}}(\widehat{x}) \\
&= \underline{\mathcal{L}}(\widehat{\nu}, \widehat{\lambda}) = \inf_x \mathcal{L}(x, \widehat{\nu}, \widehat{\lambda}).
\end{aligned}$$

Similarly,

$$\begin{aligned}
\mathcal{L}(\widehat{x}, \widehat{\nu}, \widehat{\lambda}) \geq \inf_x \mathcal{L}(x, \widehat{\nu}, \widehat{\lambda}) &= \underline{\mathcal{L}}(\widehat{\nu}, \widehat{\lambda}) \\
&= \overline{\mathcal{L}}(\widehat{x}) = \sup_{\nu, \lambda} \mathcal{L}(\widehat{x}, \nu, \lambda).
\end{aligned}$$

In conclusion,

$$\sup_{\nu, \lambda} \mathcal{L}(\widehat{x}, \nu, \lambda) \leq \mathcal{L}(\widehat{x}, \widehat{\nu}, \widehat{\lambda}) \leq \inf_x \mathcal{L}(x, \widehat{\nu}, \widehat{\lambda})$$

that is, for every $(x, \nu, \lambda) \in \mathcal{H} \times \mathbb{R}^m \times [0, +\infty[^q$,

$$\mathcal{L}(\widehat{x}, \nu, \lambda) \leq \mathcal{L}(\widehat{x}, \widehat{\nu}, \widehat{\lambda}) \leq \mathcal{L}(x, \widehat{\nu}, \widehat{\lambda}).$$

$\square$

Corollary 4

Let $\widehat{x} \in C \cap \operatorname{dom} f$, *let* $\widehat{\nu} \in \mathbb{R}^m$, *and let* $\widehat{\lambda} \in [0,+\infty[^q$ *be such that*

$$\underline{\mathcal{L}}(\widehat{\nu},\widehat{\lambda}) = f(\widehat{x}). \tag{6.4}$$

Then $(\widehat{x},\widehat{\nu},\widehat{\lambda})$ *is a saddle point of the Lagrange function.*

Proof. From the second part of the proof of Theorem 20, it appears that a sufficient condition for $(\widehat{x},\widehat{\nu},\widehat{\lambda})$ to a saddle point of $\mathcal{L}$ is

$$\underline{\mathcal{L}}(\widehat{\nu},\widehat{\lambda}) = \overline{\mathcal{L}}(\widehat{x}). \tag{6.5}$$

In addition, according to Proposition 29i), for every $x \in C$, $\overline{\mathcal{L}}(x) = f(x)$.

Therefore (6.5) is equivalent to (6.4). □

We now provide a sufficient condition for a constrained minimum.

Proposition 31. (Strong Lagrange duality)

Assume that Problem 12 admits a feasible point.
If $(\widehat{x},\widehat{\nu},\widehat{\lambda}) \in \mathcal{H} \times \mathbb{R}^m \times [0,+\infty[^q$ *is a saddle point of* $\mathcal{L}$, *then* $\widehat{x}$ *is a minimizer of* f *over* C. *In addition,*

$$\mu = f(\widehat{x}) = \underline{\mathcal{L}}(\widehat{\nu},\widehat{\lambda})$$

and the ***complementary slackness*** *condition holds:*

$$(\forall j \in \{1,\dots,q\}) \qquad \widehat{\lambda}^{(j)} h_j(\widehat{x}) = 0.$$

Proof. We know that $\operatorname{dom} f \cap C \neq \varnothing$.
We have

$$(\forall x \in C) \quad \overline{\mathcal{L}}(\widehat{x}) \leq \overline{\mathcal{L}}(x) = f(x).$$

This implies that $\overline{\mathcal{L}}(\widehat{x}) < +\infty$, hence $\widehat{x} \in C$, which yields

$$(\forall x \in C) \quad \overline{\mathcal{L}}(\widehat{x}) = f(\widehat{x}) \leq f(x).$$

As a consequence of Theorem 20, $\mu = f(\widehat{x}) = \overline{\mathcal{L}}(\widehat{x}) = \underline{\mathcal{L}}(\widehat{\nu},\widehat{\lambda})$.

Since $\widehat{x} \in C$, $(\forall i \in \{1,\dots,m\})$ $g_i(\widehat{x}) = 0$ and

$$(\forall j \in \{1,\dots,q\}) \;\; h_j(\widehat{x}) \leq 0 \quad \Rightarrow \quad \widehat{\lambda}^{(j)} h_j(\widehat{x}) \leq 0. \tag{6.6}$$

In addition, since

$$\mathcal{L}(\widehat{x},\widehat{\nu},0) \leq \mathcal{L}(\widehat{x},\widehat{\nu},\widehat{\lambda}),$$

we have

$$\sum_{j=1}^{m} \widehat{\lambda}^{(j)} h_j(\widehat{x}) \geq 0,$$

which, together with (6.6), implies that the complementary slackness condition holds. □

6.5 Convex case

6.5.1 Main result

Theorem 21

In Problem 12, assume that f is a convex function, $(g_i)_{1\le i\le m}$ are affine functions and $(h_j)_{1\le j\le q}$ are convex functions. Assume that the Slater condition holds, i.e. there exists $\overline{x}\in \operatorname{int}(\operatorname{dom} f)$ such that

$$\begin{aligned}(\forall i\in\{1,\dots,m\}) &\qquad g_i(\overline{x})=0\\ (\forall j\in\{1,\dots,q\}) &\qquad h_j(\overline{x})<0.\end{aligned}$$

$\widehat{x}$ is a minimizer of f over C if and only if there exist $\widehat{\nu}\in\mathbb{R}^m$ and $\widehat{\lambda}\in[0,+\infty[^q$ such that $(\widehat{x},\widehat{\nu},\widehat{\lambda})$ is a saddle point of the Lagrangian.

Proof. Assume that Slater condition holds and that $\widehat{x}$ is a minimizer of f on C. Let us show that there exists $\widehat{\nu}\in\mathbb{R}^m$ and $\widehat{\lambda}\in[0,+\infty[^q$ such that $f(\widehat{x})=\underline{\mathcal{L}}(\widehat{\nu},\widehat{\lambda})$.
For every $i\in\{1,\dots,m\}$, since g_i is an affine function, there exists $v_i\in\mathcal{H}$ such that

$$(\forall x\in\mathcal{H})\quad g_i(x)=g_i(\overline{x})+\langle v_i\mid x-\overline{x}\rangle=\langle v_i\mid x-\overline{x}\rangle\,.$$

Assume first that the vectors $(v_i)_{1\le i\le m}$ are linearly independent.
Let

$$C_1=\Big\{(u^{(0)},u^{(1)},\dots,u^{(m)},u^{(m+1)},\dots,u^{(m+q)})\in\mathbb{R}^{m+q+1}\;\Big|\;(\exists x\in\mathcal{H})\;\begin{array}{l}f(x)\le u^{(0)}\\ (\forall i\in\{1,\dots,m\})\;\; g_i(x)=u^{(i)}\\ (\forall j\in\{1,\dots,q\})\;\; h_j(x)\le u^{(m+j)}\end{array}\Big\}$$

$$C_2=\Big\{(u^{(0)},u^{(1)},\dots,u^{(m)},u^{(m+1)},\dots,u^{(m+q)})\in\mathbb{R}^{m+q+1}\;\Big|\;\begin{array}{l}u^{(0)}<f(\widehat{x})\\ (\forall i\in\{1,\dots,m\})\;\; u^{(i)}=0\\ (\forall j\in\{1,\dots,q\})\;\; u^{(m+j)}\le 0\end{array}\Big\}.$$

Since f and $(h_j)_{1\le i\le q}$ are convex and $(g_i)_{1\le i\le m}$ are affine, C_1 is convex. As a consequence of Slater condition, it is nonempty.
C_2 is convex and nonempty. In addition, $C_1\cap C_2=\varnothing$ since there does not exist $x\in C$ such that $f(x)<f(\widehat{x})$. According to separation theorem in $\mathbb{R}^{m+q+1}$, there exists $a=(a^{(0)},a^{(1)},\dots,a^{(m)},a^{(m+1)},\dots,a^{(m+q)})\in\mathbb{R}^{m+q+1}\setminus\{0\}$ such that

$$\inf_{u\in C_1}\langle a\mid u\rangle\ge\sup_{u\in C_2}\langle a\mid u\rangle=\sup_{u=(u^j)_{0\le j\le m+q}\in C_2}\Big(a^{(0)}u^{(0)}+\sum_{j=1}^{q}a^{(j+m)}u^{(j+m)}\Big).$$

If one of the components $a^{(0)}$ or $(a^{(j+m)})_{1\le j\le q}$ would be negative, the right-hand side term would be $+\infty$, which is prohibited.

Since these components belong to $[0,+\infty[^q$, $\sup_{u\in C_2} \langle a \mid u\rangle = a^{(0)} f(\widehat{x})$. In addition,

$$(\forall x \in \operatorname{dom} f) \quad \big(f(x), g_1(x), \ldots, g_m(x), h_1(x), \ldots, h_q(x)\big) \in C_1.$$

Hence,

$$\begin{aligned} &\inf_{x\in\mathcal{H}} a^{(0)} f(x) + \sum_{i=1}^m a^{(i)} g_i(x) + \sum_{j=1}^q a^{(m+i)} h_j(x) \\ =& \inf_{x\in\operatorname{dom} f} a^{(0)} f(x) + \sum_{i=1}^m a^{(i)} g_i(x) + \sum_{j=1}^q a^{(m+j)} h_j(x) \geq \sup_{u\in C_2} \langle a \mid u\rangle = a^{(0)} f(\widehat{x}). \end{aligned} \tag{6.7}$$

If $a^{(0)} = 0$, then

$$\inf_{x\in\operatorname{dom} f} \sum_{i=1}^m a^{(i)} g_i(x) + \sum_{j=1}^q a^{(m+j)} h_j(x) \geq 0. \tag{6.8}$$

This implies that $\displaystyle\sum_{j=1}^q a^{(m+j)} h_j(\overline{x}) \geq 0$. Since $(\forall j \in \{1,\ldots,q\})$ $a^{(m+j)} \geq 0$ and $h_j(\overline{x}) < 0$, then $(\forall j \in \{1,\ldots,q\})$ $a^{(m+j)} = 0$. We deduce from (6.8) that

$$(\forall x \in \operatorname{dom} f) \quad \sum_{i=1}^m a^{(i)} g_i(x) = \sum_{i=1}^m a^{(i)} \langle v_i \mid x - \overline{x}\rangle = \left\langle \sum_{i=1}^m a^{(i)} v_i \mid x - \overline{x} \right\rangle \geq 0.$$

Since $\overline{x} \in \operatorname{int}(\operatorname{dom} f)$, $\sum_{i=1}^m a^{(i)} v_i = 0$ and, since $(v_i)_{1\leq i\leq m}$ are linearly independent vectors, $(\forall i \in \{1,\ldots,m\})$ $a^{(i)} = 0$, which is impossible since $a \neq 0$.
Hence $a^{(0)} > 0$ and, by setting

$$\begin{aligned} (\forall i \in \{1,\ldots,m\}) \quad \widehat{\nu}^{(i)} &= \frac{a^{(i)}}{a^{(0)}} \\ (\forall j \in \{1,\ldots,q\}) \quad \widehat{\lambda}^{(j)} &= \frac{a^{(m+j)}}{a^{(0)}} \geq 0, \end{aligned}$$

(6.7) yields

$$\underline{\mathcal{L}}(\widehat{\nu},\widehat{\lambda}) = \inf_{x\in\operatorname{dom} f} f(x) + \sum_{i=1}^m \widehat{\nu}^{(i)} g_i(x) + \sum_{j=1}^q \widehat{\lambda}^{(j)} h_j(x) \geq f(\widehat{x}).$$

By using Proposition 30, $\sup_{\nu\in\mathbb{R}^m,\lambda\in[0,+\infty[^q} \underline{\mathcal{L}}(\nu,\lambda) \leq f(\widehat{x})$, hence $\underline{\mathcal{L}}(\widehat{\nu},\widehat{\lambda}) = f(\widehat{x})$.
If $(v_i)_{1\leq i\leq m}$ are linearly dependent, let $(v_i)_{i\in\mathbb{I}\subset\{1,\ldots,m\}}$ be a maximum size subfamily of linearly independent vectors. Then the same equality holds by setting $(\forall i \in \{1,\ldots,m\}\setminus\mathbb{I})$ $\widehat{\nu}_i = 0$.
The fact that $(\widehat{x},\widehat{\nu},\widehat{\lambda})$ is a saddle point of $\mathcal{L}$ follows from Corollary 4.

The converse property follows from Proposition 31. □

Remark 36. *Under the assumptions of the above theorem, if $\widehat{x}$ is a minimizer of f over C then $\mathcal{L}(\cdot,\widehat{\nu},\widehat{\lambda})$ is a convex function which is minimum at $\widehat{x}$. This optimality condition is often used to calculate $\widehat{x}$, in conjunction with the equality constraints and the complementary slackness condition.*

6.5.2 Example

We come back to the example introduced in Section 6.1 when S is the **ellipsoid** defined as

$$S = \{x_2 \in \mathbb{R}^N \mid x_2^\top Q x_2 \le 1\},$$

where Q be a positive definite symmetric matrix of size $N \times N$. We have thus to solve an instance of Problem 12 where

- $C = A \times S$, $\mathcal{H} = (\mathbb{R}^N)^2$;
- the cost function is
$$\big(\forall x = (x_1, x_2) \in \mathcal{H}\big) \qquad f(x) = \|x_1 - x_2\|^2; \tag{6.9}$$
- we have $m = 1$ equality constraint:
$$\big(\forall x = (x_1, x_2) \in \mathcal{H}\big) \qquad g_1(x) = u^\top x_1 - \delta;$$
- we have $q = 1$ inequality constraint:
$$\big(\forall x = (x_1, x_2) \in \mathcal{H}\big) \qquad h_1(x) = x_2^\top Q x_2 - 1.$$

Since S is nonempty and bounded, it follows from Remark 32 that a solution to the constrained optimization problem exists.

Let us first check that the assumptions of Theorem 21 are satisfied:

- f is convex and $\operatorname{dom} f = \mathcal{H}$;
- g_1 is affine;
- h_1 is convex;
- the Slater condition holds since there exists $\overline{x} = (\overline{x}_1, \overline{x}_2) \in \operatorname{int}(\operatorname{dom} f) = \mathcal{H}$ such that
$$\begin{aligned} u^\top \overline{x}_1 - \delta &= 0 \\ \overline{x}_2^\top Q \overline{x}_2 &< 1 \end{aligned}$$
(choose $\overline{x}_1 = \delta u$ and $\overline{x}_2 = 0$).

The Lagrange function is

$$(\forall x = (x_1, x_2) \in \mathcal{H})(\forall (\nu, \lambda) \in \mathbb{R} \times [0, +\infty[)$$
$$\mathcal{L}(x, \nu, \lambda) = \|x_1 - x_2\|^2 + \nu(u^\top x_1 - \delta) + \lambda(x_2^\top Q x_2 - 1). \tag{6.10}$$

If $\widehat{x} = (\widehat{x}_1, \widehat{x}_2) \in \mathcal{H}$ is a minimizer of f over C, then there exists $(\widehat{\nu}, \widehat{\lambda}) \in \mathbb{R} \times [0, +\infty[$ such that $(\widehat{x}, \widehat{\nu}, \widehat{\lambda})$ is a saddle point of $\mathcal{L}$.

The dual Lagrange function is

$$(\forall \nu, \lambda) \in \mathbb{R} \times [0, +\infty[\quad \underline{\mathcal{L}}(\nu, \lambda) = \inf_{x \in \mathcal{H}} \mathcal{L}(x, \nu, \lambda).$$

If $\lambda = \nu = 0$, then $\underline{\mathcal{L}}(\nu, \lambda) = 0$.
If $\lambda = 0$ and $\nu \neq 0$, then $\underline{\mathcal{L}}(\nu, \lambda) = -\infty$ (set $x_2 = x_1 = \alpha \,\mathrm{sign}(\nu) u$ with $\alpha \to -\infty$).
If $\lambda > 0$, $\mathcal{L}(\cdot, \nu, \lambda)$ has a minimizer at (x_1, x_2) such that

$$\begin{aligned}
& \nabla_{\mathrm{x}} \mathcal{L}(x, \nu, \lambda) = 0 \\
\Leftrightarrow \quad & \begin{cases} 2(x_1 - x_2) + \nu u = 0 \\ x_2 - x_1 + \lambda Q x_2 = 0 \end{cases} \\
\Leftrightarrow \quad & \begin{cases} x_1 = -\dfrac{\nu}{2\lambda} Q^{-1} u - \dfrac{\nu}{2} u \\ x_2 = -\dfrac{\nu}{2\lambda} Q^{-1} u. \end{cases}
\end{aligned} \tag{6.11}$$

This leads to

$$\underline{\mathcal{L}}(\nu, \lambda) = -\frac{\nu^2}{4} - \frac{\nu^2}{4\lambda} u^\top Q^{-1} u - \delta \nu - \lambda.$$

We know that $\underline{\mathcal{L}}$ is a concave function which is maximum at $(\widehat{\nu}, \widehat{\lambda})$, and

$$\underline{\mathcal{L}}(\widehat{\nu}, \widehat{\lambda}) = \|\widehat{x}_1 - \widehat{x}_2\|^2 = d(A, S)^2.$$

If $\widehat{\lambda} > 0$, then

$$\begin{aligned}
& \nabla \underline{\mathcal{L}}(\widehat{\nu}, \widehat{\lambda}) = 0 \\
\Leftrightarrow \quad & \begin{cases} -\dfrac{\widehat{\nu}}{2} - \dfrac{\widehat{\nu}}{2\widehat{\lambda}} u^\top Q^{-1} u = \delta \\ \dfrac{\widehat{\nu}^2}{4\widehat{\lambda}^2} u^\top Q^{-1} u = 1. \end{cases}
\end{aligned}$$

Since $\delta \in \mathbb{R}$, these conditions imply that $\widehat{\nu} < 0$ and are also equivalent to

$$\begin{aligned}
& \begin{cases} \dfrac{\widehat{\nu}}{2\widehat{\lambda}} = -\dfrac{1}{\sqrt{u^\top Q^{-1} u}} \\ \widehat{\nu} = 2(\sqrt{u^\top Q^{-1} u} - \delta) \end{cases} \\
\Leftrightarrow \quad & \begin{cases} \widehat{\nu} = -2\eta \\ \widehat{\lambda} = \eta \sqrt{u^\top Q^{-1} u}, \end{cases}
\end{aligned} \tag{6.12}$$

where $\eta = \delta - \sqrt{u^\top Q^{-1} u}$. This is only possible if $\eta > 0$, and we have then

$$\underline{\mathcal{L}}(\widehat{\nu}, \widehat{\lambda}) = -\eta^2 - 2\widehat{\lambda} - \delta \widehat{\nu} = \eta^2.$$

In summary,

i) if $\eta > 0$, the optimal Lagrange multipliers $(\widehat{\nu}, \widehat{\lambda})$ are given by (6.12), and $d(A, S) = \eta$.

ii) if $\eta \leq 0$, then $\widehat{\nu} = \widehat{\lambda} = 0$ and $d(A,S) = 0$, which means that $A \cap S \neq \varnothing$.

Since $\widehat{x}$ is a minimizer of $\mathcal{L}(\cdot, \widehat{\nu}, \widehat{\lambda})$, it follows from (6.11) that, in Case i),

$$\begin{cases} \widehat{x}_1 = -\dfrac{\widehat{\nu}}{2\widehat{\lambda}} Q^{-1}u - \dfrac{\widehat{\nu}}{2} u \\ \widehat{x}_2 = -\dfrac{\widehat{\nu}}{2\widehat{\lambda}} Q^{-1}u \end{cases}$$

$$\Leftrightarrow \quad \begin{cases} \widehat{x}_1 = \dfrac{Q^{-1}u}{\sqrt{u^\top Q^{-1}u}} + \eta u \\ \widehat{x}_2 = \dfrac{Q^{-1}u}{\sqrt{u^\top Q^{-1}u}}. \end{cases}$$

A unique minimizer is thus found.

6.5.3 Special cases

When there are only equality constraints, we obtain the following result.

Corollary 5

Assume that f is a convex function and $(g_i)_{1 \leq i \leq m}$ are affine functions. Assume that the Slater condition holds, i.e. there exists $\overline{x} \in \operatorname{int}(\operatorname{dom} f)$ such that

$$(\forall i \in \{1, \ldots, m\}) \qquad g_i(\overline{x}) = 0.$$

$\widehat{x}$ is a minimizer of f over C if and only if there exists $\widehat{\nu} \in \mathbb{R}^m$ such that $(\widehat{x}, \widehat{\nu})$ is a saddle point of the Lagrangian (6.2).

When there are only inequality constraints, the following holds.

Corollary 6

Assume that f is a convex function, and $(h_j)_{1 \leq j \leq q}$ are convex functions. Assume that the Slater condition holds, i.e. there exists $\overline{x} \in \operatorname{dom} f$ such that

$$(\forall j \in \{1, \ldots, q\}) \qquad h_j(\overline{x}) < 0.$$

$\widehat{x}$ is a minimizer of f over C if and only if there exists $\widehat{\lambda} \in [0, +\infty[^q$ such that $(\widehat{x}, \widehat{\lambda})$ is a saddle point of the Lagrangian (6.3).

6.6 Differentiable case

6.6.1 Necessary condition for a local minimizer

Proposition 32

In Problem 12, assume that f, $(g_i)_{1\le i\le m}$, and $(h_j)_{1\le j\le q}$ are continuously differentiable on $\mathcal{H}=\mathbb{R}^N$.
*If $\widehat{x}$ is a local minimizer of f over C then the **Fritz-John conditions** hold, i.e. there exists a nonzero vector $(\alpha_0,\alpha_1,\dots,\alpha_m,\alpha_{m+1},\dots,\alpha_{m+q})\in[0,+\infty[\times\mathbb{R}^m\times[0,+\infty[^q$ such that*

$$\alpha_0\nabla f(\widehat{x})+\sum_{i=1}^m\alpha_i\nabla g_i(\widehat{x})+\sum_{j=1}^q\alpha_{m+j}\nabla h_j(\widehat{x})=0 \tag{6.13}$$

$$(\forall j\in\{1,\dots,q\})\qquad \alpha_{m+j}h_j(\widehat{x})=0. \tag{6.14}$$

Proof. Let $J(\widehat{x})=\{j\in\{1,\dots,q\}\mid h_j(\widehat{x})=0\}$ and let $\overline{J}(\widehat{x})=\{1,\dots,q\}\setminus J(\widehat{x})$.
For every $j\in\overline{J}(\widehat{x})$, $h_j(\widehat{x})<0$. Since functions $(h_j)_{1\le j\le q}$ are continuous, there exists $\rho\in\,]0,+\infty[$ such that $\widehat{x}$ is a global minimizer of f on $B(\widehat{x},\rho)\cap C$, $B(\widehat{x},\rho)$ being the open ball centered at $\widehat{x}$ and with radius ρ, and

$$(\forall x\in B(\widehat{x},\rho))(\forall j\in\overline{J}(\widehat{x}))\quad h_j(x)<0.$$

For every $\eta\in\,]0,+\infty[$, define

$$(\forall x\in\mathbb{R}^N)\quad f_\eta(x)=f(x)+\|x-\widehat{x}\|^2+\frac{\eta}{2}\Big(\sum_{i=1}^m(g_i(x))^2+\sum_{j\in J(\widehat{x})}\max\{0,h_j(x)\}^2\Big).$$

Let $\epsilon\in]0,\rho[$. Let us first show that

$$(\exists\eta_\epsilon\in\,]0,+\infty[)(\forall x\in\mathbb{R}^N)\quad \|x-\widehat{x}\|=\epsilon\Rightarrow f_{\eta_\epsilon}(x)>f(\widehat{x}). \tag{6.15}$$

If this property does not hold, we can build a sequence $(\eta_n)_{n\in\mathbb{N}}$ converging to $+\infty$ and a sequence $(x_n)_{n\in\mathbb{N}}$ such that $(\forall n\in\mathbb{N})$ $\|x_n-\widehat{x}\|=\epsilon$ and $f_{\eta_n}(x_n)\le f(\widehat{x})$, i.e.

$$f(x_n)+\epsilon^2+\frac{\eta_n}{2}\Big(\sum_{i=1}^m(g_i(x_n))^2+\sum_{j\in J(\widehat{x})}\max\{0,h_j(x_n)\}^2\Big)\le f(\widehat{x}). \tag{6.16}$$

Since $(x_n)_{n\in\mathbb{N}}$ is bounded, it has a cluster point $\widetilde{x}$. Since f, $(g_i)_{1\le i\le m}$, and $(h_j)_{1\le i\le q}$ are continuous and $\eta_n\to+\infty$, we deduce that

$$\sum_{i=1}^m(g_i(\widetilde{x}))^2+\sum_{j\in J(\widehat{x})}\max\{0,h_j(\widetilde{x})\}^2=0\quad\Rightarrow\begin{cases}(\forall i\in\{1,\dots,m\})\ g_i(\widetilde{x})=0\\(\forall j\in J(\widehat{x}))\ h_j(\widetilde{x})\le 0.\end{cases}$$

As

$$\widetilde{x}\in B(\widehat{x},\rho)\quad\Rightarrow\quad(\forall j\in\overline{J}(\widehat{x}))\ \ h_j(\widetilde{x})<0,$$

this shows that $\widetilde{x} \in C$. By taking the limit in (6.16), we deduce that

$$f(\widetilde{x}) \leq f(\widehat{x}) - \epsilon^2,$$

which contradicts the fact that $\widehat{x}$ minimizes f over $B(\widehat{x}, \rho) \cap C$. This proves the validity of (6.15).

Since $\overline{B}(\widehat{x}, \epsilon)$ is a compact set and f_{η_ϵ} is continuous, it follows from Weierstrass theorem that f_{η_ϵ} admits a minimizer $\widehat{x}_\epsilon$ on $\overline{B}(\widehat{x}, \epsilon)$.
We have thus $f(\widehat{x}_\epsilon) \leq f(\widehat{x})$ and we deduce from (6.15) that $\widehat{x}_\epsilon \in B(\widehat{x}, \epsilon)$. Since f, $(g_i)_{1 \leq i \leq m}$ and $(h_j)_{1 \leq j \leq q}$ are differentiable, a necessary first-order optimality condition is

$$\begin{aligned}
&\nabla f_{\eta_\epsilon}(\widehat{x}_\epsilon) = 0 \\
\Leftrightarrow\ &\nabla f(\widehat{x}_\epsilon) + 2(\widehat{x}_\epsilon - \widehat{x}) + \eta_\epsilon\Big(\sum_{i=1}^m g_i(\widehat{x}_\epsilon)\nabla g_i(\widehat{x}_\epsilon) + \sum_{j \in J(\widehat{x})} \max\{0, h_j(\widehat{x}_\epsilon)\}\nabla h_j(\widehat{x}_\epsilon)\Big) = 0.
\end{aligned}$$

Let $a_\epsilon = (1, a_\epsilon^{(1)}, \ldots, a_\epsilon^{(m)}, a_\epsilon^{(m+1)}, \ldots, a_\epsilon^{(m+q)}) \in \mathbb{R}^{m+q+1}$ with

$$\begin{aligned}
(\forall i \in \{1, \ldots, m\}) \quad & a_\epsilon^{(i)} = \eta_\epsilon g_i(\widehat{x}_\epsilon) \\
(\forall j \in \{1, \ldots, q\}) \quad & a_\epsilon^{(j+m)} = \begin{cases} \eta_\epsilon \max\{0, h_j(\widehat{x}_\epsilon)\} & \text{if } j \in J(\widehat{x}) \\ 0 & \text{otherwise.} \end{cases}
\end{aligned}$$

We get

$$\nabla f(\widehat{x}_\epsilon) + 2(\widehat{x}_\epsilon - \widehat{x}) + \sum_{i=1}^m a_\epsilon^{(i)} \nabla g_i(\widehat{x}_\epsilon) + \sum_{j=1}^q a_\epsilon^{(j+m)} \nabla h_j(\widehat{x}_\epsilon) = 0.$$

Let $\alpha_\epsilon = a_\epsilon / \|a_\epsilon\|$. Consider now a sequence $(\epsilon_n)_{n \in \mathbb{N}}$ of $]0, \rho[$ converging to 0.
Then $\widehat{x}_{\epsilon_n} \to \widehat{x}$, which implies that $(\widehat{x}_{\epsilon_n} - \widehat{x})/\|a_{\epsilon_n}\| \to 0$ (since $\|a_{\epsilon_n}\| \geq 1$).
In addition, $(\alpha_{\epsilon_n})_{n \in \mathbb{N}}$ being bounded, there exists a subsequence $(\alpha_{\epsilon_{n_k}})_{k \in \mathbb{N}}$ converging to some $\alpha = (\alpha^{(i)})_{0 \leq i \leq m+q}$.
It follows from the continuity of ∇f, $(\nabla g_i)_{1 \leq i \leq m}$, and $(\nabla h_j)_{1 \leq j \leq q}$ that

$$\alpha^{(0)} \nabla f(\widehat{x}) + \sum_{i=1}^m \alpha^{(i)} \nabla g_i(\widehat{x}) + \sum_{j=1}^q \alpha^{(m+j)} \nabla h_j(\widehat{x}) = 0.$$

It can be finally observed that, for every $j \in \overline{J}(\widehat{x})$, $\alpha^{(j+m)} = 0$, which means that the complementarity condition is satisfied. □

Theorem 22. (Karush-Kuhn-Tucker)

In Problem 12, assume that f, $(g_i)_{1 \leq i \leq m}$, and $(h_j)_{1 \leq j \leq q}$ are continuously differentiable on $\mathcal{H} = \mathbb{R}^N$.
Assume that $\widehat{x}$ is a local minimizer of f over C satisfying the following Mangasarian-Fromovitz ***constraint qualification conditions****:*

i) *$\{\nabla g_i(\widehat{x}) \mid i \in \{1, \ldots, m\}\}$ is a family of linearly independent vectors;*

ii) *there exists* $z \in \mathbb{R}^N$ *such that*

$$\begin{aligned} (\forall i \in \{1,\dots,m\}) \qquad & \langle \nabla g_i(\widehat{x}) \mid z\rangle = 0 \\ (\forall j \in J(\widehat{x})) \qquad & \langle \nabla h_j(\widehat{x}) \mid z\rangle < 0 \end{aligned}$$

where $J(\widehat{x}) = \{j \in \{1,\dots,q\} \mid h_j(\widehat{x}) = 0\}$ *is the set of* ***active inequality constraints*** *at* $\widehat{x}$.

Then, there exist $\widehat{\nu} \in \mathbb{R}^N$ *and* $\widehat{\lambda} \in [0,+\infty[^q$ *such that* $\widehat{x}$ *is a critical point of* $\mathcal{L}(\cdot,\widehat{\nu},\widehat{\lambda})$ *and the complementary slackness condition holds.*

Proof. According to Proposition 32, there exists a nonzero vector $(\alpha_0,\alpha_1,\dots,\alpha_m,\alpha_{m+1},\dots,\alpha_{m+q}) \in [0,+\infty[\times\mathbb{R}^m\times[0,+\infty[^q$ such that (6.13) and (6.14) hold. The latter condition implies that

$$(\forall j \notin J(\widehat{x})) \quad \alpha_{m+j} = 0$$

and the first equality reduces to

$$\alpha_0 \nabla f(\widehat{x}) + \sum_{i=1}^m \alpha_i \nabla g_i(\widehat{x}) + \sum_{j\in J(\widehat{x})} \alpha_{m+j}\nabla h_j(\widehat{x}) = 0,$$

which yields

$$\begin{aligned} & \alpha_0 \langle \nabla f(\widehat{x}) \mid z\rangle + \sum_{i=1}^m \alpha_i \langle \nabla g_i(\widehat{x}) \mid z\rangle + \sum_{j\in J(\widehat{x})} \alpha_{m+j} \langle \nabla h_j(\widehat{x}) \mid z\rangle = 0 \\ \Leftrightarrow \quad & \alpha_0 \langle \nabla f(\widehat{x}) \mid z\rangle + \sum_{j\in J(\widehat{x})} \alpha_{m+j} \langle \nabla h_j(\widehat{x}) \mid z\rangle = 0. \end{aligned}$$

Let us suppose that $\alpha_0 = 0$.
Since, for every $j \in J(\widehat{x})$, $\langle \nabla h_j(\widehat{x}) \mid z\rangle < 0$, in the latter equality, we would have, $\alpha_{m+j} = 0$ which, in (6.6.1), would lead to

$$\sum_{i=1}^m \alpha_i \nabla g_i(\widehat{x}) = 0.$$

Since the vectors $(\nabla g_i(\widehat{x}))_{1\le i\le m}$ are linearly independent,

$$(\forall i \in \{1,\dots,m\}) \quad \alpha_i = 0.$$

In conclusion, the vector $(\alpha_i)_{0\le i\le m+q}$ would be zero, which is impossible.
This shows that $\alpha_0 > 0$.

By defining

$$\begin{aligned} (\forall i \in \{1,\dots,m\}) \quad & \widehat{\nu}_i = \alpha_i/\alpha_0, \\ (\forall j \in \{1,\dots,q\}) \quad & \widehat{\lambda}_j = \alpha_{m+j}/\alpha_0 \ge 0, \end{aligned}$$

(6.13) and (6.14) can be rewritten as

$$\begin{aligned}
&\nabla f(\widehat{x})+\sum_{i=1}^{m}\widehat{\nu}_i\nabla g_i(\widehat{x})+\sum_{j=1}^{q}\widehat{\lambda}_j\nabla h_j(\widehat{x})=0\\
&(\forall j\in\{1,\ldots,q\})\qquad \widehat{\lambda}_j h_j(\widehat{x})=0.
\end{aligned}$$

By setting $\widehat{\nu}=(\widehat{\nu}_i)_{1\le i\le m}\in\mathbb{R}^m$ and $\widehat{\lambda}=(\widehat{\lambda}_j)_{1\le j\le q}\in[0,+\infty[^q$, the first equality also reads

$$\nabla_x\mathcal{L}(\widehat{x},\widehat{\nu},\widehat{\lambda})=0.$$

□

Remark 37.

i) *A sufficient condition for Mangasarian-Fromovitz conditions to be satisfied is that* $\{\nabla g_i(\widehat{x})\mid i\in\{1,\ldots,m\}\}\cup\{\nabla h_j(\widehat{x})\mid j\in J(\widehat{x})\}$ *is a family of linearly independent vectors.*

   Proof. If this condition holds, matrix

   $$A=\begin{bmatrix}((\nabla g_i(\widehat{x}))^\top)_{1\le i\le m}\\((\nabla h_j(\widehat{x}))^\top)_{j\in J(\widehat{x})}\end{bmatrix}\in\mathbb{R}^{(m+|J(\widehat{x})|)\times N}$$

   has rank $m+|J(\widehat{x})|$. Let $\mathbf{1}$ be the unit vector of $\mathbb{R}^{|J(\widehat{x})|}$. Hence, the equation

   $$Az=\begin{bmatrix}0\\-\mathbf{1}\end{bmatrix},\qquad z\in\mathbb{R}^N,$$

   admits a solution. Such a solution satisfies ii). □

ii) *The KKT theorem is valid in the case when f, $(g_i)_{1\le i\le m}$, and $(h_j)_{1\le j\le q}$ are defined on a nonempty open set $D\subset\mathbb{R}^N$.*

6.6.2 Example

Formulation Let us consider the example introduced in Section 6.1 when set S is defined as

$$S=\left\{x_2\in[0,+\infty[^N\;\middle|\;p(x_2)\ge 1\right\}$$

where

$$\left(\forall x_2=(x_2^{(n)})_{1\le i\le N}\in\mathbb{R}^N\right)\quad p(x_2)=\prod_{n=1}^{N}x_2^{(n)}.$$

We have thus an instance of Problem 12 where

- $C=A\times S$, $\mathcal{H}=(\mathbb{R}^N)^2$

- The cost function is given by (6.9).
- We have $m = 1$ equality constraint:
$$\big(\forall x = (x_1, x_2) \in \mathcal{H}\big) \qquad g_1(x) = u^\top x_1 - \delta$$
where $\delta \in [0, +\infty[$ and $u \in \mathbb{R}^N$. We will assume that u has nonzero components of the same sign and $\|u\| = 1$.
- We have $q = N + 1$ inequality constraints:
$$\big(\forall x = (x_1, x_2) \in \mathcal{H}\big) \qquad \begin{aligned} &h_1(x) = 1 - p(x_2) \\ &h_j(x) = -x_2^{(j-1)}, \quad j \in \{2, \ldots, N+1\}. \end{aligned}$$

Use of KKT theorem Let us first check that the assumptions of Theorem 22 hold.

- f, g_1, and $(h_j)_{1 \le j \le N+1}$ are continuously differentiable functions.
- Mangasarian-Fromovitz conditions are satisfied. Indeed, if $\widehat{x} = (\widehat{x}_1, \widehat{x}_2)$ is a local minimizer of f over C, it belongs to C. Thus $p(\widehat{x}_2) \ge 1$, which implies that $\widehat{x}_2 \in \,]0, +\infty[^N$. Consequently, either $J(\widehat{x}) = \varnothing$ or $J(\widehat{x}) = \{1\}$.
If $J(\widehat{x}) = \varnothing$, the sufficient condition is met since $\nabla g_1(\widehat{x}) = (u, 0) \neq 0$.
If $J(\widehat{x}) = \{1\}$, since $\nabla h_1(\widehat{x}) = (0, -\nabla p(\widehat{x}_2))$ and
$$\nabla p(\widehat{x}_2) = \Big(\prod_{\substack{n'=1 \\ n' \neq n}}^{N} \widehat{x}_2^{(n')} \Big)_{1 \le n \le N} \neq 0,$$
then $\nabla g_1(\widehat{x})$ and $\nabla h_1(\widehat{x})$ are linearly independent.

For every $x = (x_1, x_2) \in \mathcal{H}$ and $(\nu, \lambda) \in \mathbb{R} \times [0, +\infty[^{N+1}$, the Lagrange function reads
$$\mathcal{L}(x, \nu, \lambda) = \|x_1 - x_2\|^2 + \nu(u^\top x_1 - \delta) + \lambda^{(1)}(1 - p(x_2)) - \sum_{j=2}^{N+1} \lambda^{(j)} x_2^{(j-1)}.$$

According to Theorem 22, If $\widehat{x} = (\widehat{x}_1, \widehat{x}_2) \in \mathcal{H}$ is a local minimizer of f over C then there exist $\widehat{\nu} \in \mathbb{R}$ and $\widehat{\lambda} = (\widehat{\lambda}^{(j)})_{1 \le j \le N+1} \in [0, +\infty[^{N+1}$ such that
$$\begin{aligned} &\nabla_x \mathcal{L}(\widehat{x}, \widehat{\nu}, \widehat{\lambda}) = 0 \\ \Leftrightarrow &\begin{cases} 2(\widehat{x}_1 - \widehat{x}_2) + \widehat{\nu} u = 0 \\ 2(\widehat{x}_2 - \widehat{x}_1) - \widehat{\lambda}^{(1)} \nabla p(\widehat{x}_2) - \displaystyle\sum_{j=2}^{N+1} \widehat{\lambda}^{(j)} e_{j-1} = 0, \end{cases} \end{aligned} \tag{6.17}$$
where $(e_n)_{1 \le n \le N}$ is the canonical basis of $\mathbb{R}^N$.
In addition, the complementarity slackness conditions are expressed as
$$\begin{aligned} &\widehat{\lambda}^{(1)} \big(1 - p(\widehat{x}_2)\big) = 0 \\ &(\forall j \in \{2, \ldots, N+1\}) \quad -\widehat{\lambda}^{(j)} \widehat{x}_2^{(j-1)} = 0. \end{aligned}$$

As we have seen that $\widehat{x}_2 \in \,]0,+\infty[^{N+1}$,

$$(\forall j \in \{2,\dots,N+1\}) \quad \widehat{\lambda}^{(j)} = 0.$$

Two cases can be distinguished:
Case 1: $\widehat{\lambda}^{(1)} = 0$. It follows from (6.17) that $\widehat{x}_1 = \widehat{x}_2$ and $\widehat{\nu} = 0$, that is $A \cap S \neq \varnothing$.
Case 2: $\widehat{\lambda}^{(1)} \neq 0$ and $p(\widehat{x}_2) = 1$.
Since $\nabla p(\widehat{x}_2) = (1/\widehat{x}_2^{(n)})_{1\le n\le N}$, (6.17) reads

$$\begin{cases} \widehat{x}_2 - \widehat{x}_1 = \dfrac{\widehat{\nu}}{2} u \\ (\forall n \in \{1,\dots,N\}) \quad \dfrac{1}{\widehat{x}_2^{(n)}} = \dfrac{\widehat{\nu}}{\widehat{\lambda}^{(1)}} u^{(n)}. \end{cases}$$

The positivity of the components of $\widehat{x}_2$ guarantees that $\widehat{\nu} \neq 0$ and $u = \operatorname{sign}(\widehat{\nu})|u|$ with $|u| > 0$. Using the constraints,

$$\begin{aligned} & p(\widehat{x}_2) = 1 \\ \Leftrightarrow \quad & \frac{\widehat{\lambda}^{(1)}}{\widehat{|\nu|}} = (p(|u|))^{1/N} \\ \Rightarrow \quad & \widehat{x}_2 = (p(|u|))^{1/N} (1/|u^{(n)}|)_{1\le n\le N}. \end{aligned}$$

and

$$\begin{aligned} & u^\top \widehat{x}_1 = \delta \\ \Leftrightarrow \quad & u^\top \widehat{x}_2 - \frac{\widehat{\nu}}{2} = \delta \\ \Leftrightarrow \quad & N(p(|u|))^{1/N} \operatorname{sign}(\widehat{\nu}) - \delta = \frac{\widehat{\nu}}{2}. \end{aligned}$$

If $u < 0$, then $\operatorname{sign}(\widehat{\nu}) = -1$ and $\frac{\widehat{\nu}}{2} = -N(p(|u|))^{1/N} - \delta$.
If $u > 0$, then $\operatorname{sign}(\widehat{\nu}) = 1$ and $\frac{\widehat{\nu}}{2} = N(p(u))^{1/N} - \delta$.
The latter case is possible only if $\delta < N(p(u))^{1/N}$.

In summary, by setting

$$\eta = (p(|u|))^{1/N}$$

and by assuming that there exists a global (hence local) minimizer $(\widehat{x}_1, \widehat{x}_2)$, one of the following cases arises:

i) $u > 0$, $\delta < N\eta$, and

$$\begin{cases} \widehat{x}_1 = \eta(1/u^{(n)})_{1\le n\le N} + (\delta - N\eta)u \\ \widehat{x}_2 = \eta(1/u^{(n)})_{1\le n\le N}. \end{cases} \tag{6.18}$$

We have then $d(A,S) = \|\widehat{x}_1 - \widehat{x}_2\| = N\eta - \delta$.

ii) $u < 0$ and

$$\begin{cases} \widehat{x}_1 = -\eta(1/u^{(n)})_{1\le n\le N} + (\delta + N\eta)u \\ \widehat{x}_2 = -\eta(1/u^{(n)})_{1\le n\le N}. \end{cases} \tag{6.19}$$

We have then $d(A,S) = \|\widehat{x}_1 - \widehat{x}_2\| = N\eta + \delta$.

iii) $A \cap S \neq \varnothing$ and $\widehat{x}_1 = \widehat{x}_2$. We have then $d(A,S) = 0$.

However, since Theorem 22 only provides necessary conditions, we need to check the validity of the obtained solution.

Deducing the solution An alternative expression of S is

$$S = \big\{x_2 = (x_2^{(n)})_{1\le n\le N} \in \mathbb{R}^N \mid \sum_{n=1}^{N} \psi(x^{(n)}) \le 0\big\},$$

where $\psi \in \Gamma_0(\mathbb{R})$ is defined as

$$(\forall \xi \in \mathbb{R}) \quad \psi(\xi) = \begin{cases} -\ln \xi & \text{if } \xi \in \,]0,+\infty[\\ +\infty & \text{otherwise.} \end{cases}$$

Since S is a lower level set of a convex function, it is a convex set and $C = A \times S$ is also convex.

A necessary and sufficient for $\widehat{x} = (\widehat{x}_1, \widehat{x}_2) \in \mathcal{H}$ to be minimizer of f over C is then given by Euler's inequality

$$\begin{aligned} (\forall x = (x_1,x_2) \in C) \qquad & \langle \nabla f(\widehat{x}) \mid x - \widehat{x}\rangle \ge 0 \\ \Leftrightarrow\ & \langle \widehat{x}_1 - \widehat{x}_2 \mid x_1 - \widehat{x}_1\rangle + \langle \widehat{x}_2 - \widehat{x}_1 \mid x_2 - \widehat{x}_2\rangle \ge 0. \end{aligned}$$

i) if $u > 0$ and $\delta < N\eta$, by using the expressions of $(\widehat{x}_1, \widehat{x}_2)$ in (6.18) and the fact that $(x_1, \widehat{x}_1) \in A^2 \Leftrightarrow \langle u \mid x_1\rangle = \langle u \mid \widehat{x}_1\rangle = \delta$,

$$\begin{aligned} &\langle \widehat{x}_1 - \widehat{x}_2 \mid x_1 - \widehat{x}_1\rangle + \langle \widehat{x}_2 - \widehat{x}_1 \mid x_2 - \widehat{x}_2\rangle \\ &= (\delta - N\eta)(\langle u \mid x_1 - \widehat{x}_1\rangle - \langle u \mid x_2 - \widehat{x}_2\rangle) \\ &= (N\eta - \delta)\,\langle u \mid x_2 - \widehat{x}_2\rangle \\ &= (N\eta - \delta)(\langle u \mid x_2\rangle - N\eta). \end{aligned}$$

By using the arithmetic-geometric mean inequality and the fact that $x_2 \in S \Leftrightarrow x_2 \in \,]0,+\infty[^N$ and $p(x_2) \ge 1$,

$$\frac{1}{N}\,\langle u \mid x_2\rangle \ge (p(u)p(x_2))^{1/N} \ge \eta,$$

which shows that Euler's inequality is satisfied.

ii) If $u < 0$, by proceeding similarly for the values of $(\widehat{x}_1, \widehat{x}_2)$ in (6.19) we obtain,

$$\begin{aligned} &\langle \widehat{x}_1 - \widehat{x}_2 \mid x_1 - \widehat{x}_1\rangle + \langle \widehat{x}_2 - \widehat{x}_1 \mid x_2 - \widehat{x}_2\rangle \\ &= -(N\eta + \delta)\,\langle u \mid x_2 - \widehat{x}_2\rangle \\ &= (N\eta + \delta)(\langle -u \mid x_2\rangle - N\eta) \ge 0. \end{aligned}$$

iii) If $u > 0$ and $\delta \geq N\eta$, it can be noticed that

$$\widehat{x}_1 = \eta(1/u^{(n)})_{1\leq n\leq N} + (\delta - N\eta)u \in A \cap \,]0, +\infty[^N$$

Then $p(\widehat{x}_1) \geq p(\eta(1/u^{(n)})_{1\leq n\leq N}) = 1$, which shows that $\widehat{x}_1 \in S$. Therefore $(\widehat{x}_1, \widehat{x}_1)$ is a solution to the problem associated with the cost value $d(A, S)^2 = 0$.

In conclusion,

i) if $u > 0$ and $\delta < N\eta$, then $(\widehat{x}_1, \widehat{x}_2)$ is given by (6.18) and $d(A, S) = \|\widehat{x}_1 - \widehat{x}_2\| = N\eta - \delta$;

ii) if $u < 0$, then $(\widehat{x}_1, \widehat{x}_2)$ is given by (6.19) and $d(A, S) = \|\widehat{x}_1 - \widehat{x}_2\| = N\eta + \delta$;

iii) otherwise, $A \cap S \neq \varnothing$ and $d(A, S) = 0$.

6.6.3 Special cases

When there are only equality constraints, we obtain the following result.

Corollary 7

Assume that f and $(g_i)_{1\leq i\leq m}$ are continuously differentiable on $\mathcal{H} = \mathbb{R}^N$.
Assume that $\widehat{x}$ is a local minimizer of f over C and
$\{\nabla g_i(\widehat{x}) \mid i \in \{1, \ldots, m\}\}$ is a family of linearly independent vectors.
Then, there exists $\widehat{\nu} \in \mathbb{R}^m$ such that $\widehat{x}$ is a critical point of $\mathcal{L}(\cdot, \widehat{\nu})$ as defined in (6.2).

When there are only equality constraints, we get.

Corollary 8

Assume that f and $(h_j)_{1\leq j\leq q}$ are continuously differentiable on $\mathcal{H} = \mathbb{R}^N$.
Assume that $\widehat{x}$ is a local minimizer of f over C and
there exists $z \in \mathbb{R}^N$ such that

$$(\forall j \in J(\widehat{x})) \qquad \langle \nabla h_j(\widehat{x}) \mid z\rangle < 0$$

where $J(\widehat{x}) = \{j \in \{1, \ldots, q\} \mid h_j(\widehat{x}) = 0\}$ is the set of active inequality constraints at $\widehat{x}$. Then, there exists $\widehat{\lambda} \in [0, +\infty[^q$ such that $\widehat{x}$ is a critical point of $\mathcal{L}(\cdot, \widehat{\lambda})$ as defined in (6.3). In addition, the complementary slackness condition holds.

Remark 38. *A sufficient condition for the qualification conditions to be satisfied is that $\{\nabla h_j(\widehat{x}) \mid j \in J(\widehat{x})\}$ is a family of linearly independent vectors.*

Chapter 7

Some iterative algorithms

7.1 Objective

In this chapter, we will be interested in the following problem which has many practical applications.

Problem 13

Let $\mathcal{H}$ be a Hilbert space.
Let $f\colon \mathcal{H} \to \mathbb{R}$ be Gâteaux differentiable.
Let C be a nonempty closed convex subset of $\mathcal{H}$.
We want to

$$\text{Find} \quad \widehat{x} \in \underset{x \in C}{\operatorname{Argmin}} \; f(x).$$

More precisely, we will show how to build a sequence $(x_n)_{n\in\mathbb{N}}$ converging to a solution to this problem.

7.2 Principle of first-order methods

The numerical strategies we will develop are grounded on the following ideas.

- If f is Fréchet-differentiable then, at iteration n, we have

$$(\forall x \in \mathcal{H}) \qquad f(x) = f(x_n) + \langle \nabla f(x_n) \mid x - x_n \rangle + o(\|x - x_n\|).$$

  So if $\|x_{n+1} - x_n\|$ is small enough and x_{n+1} is chosen such that

$$\langle \nabla f(x_n) \mid x_{n+1} - x_n \rangle < 0,$$

  then $f(x_{n+1}) < f(x_n)$.

- In particular, the **steepest descent direction** is given by

$$x_{n+1} - x_n = -\gamma_n \nabla f(x_n), \qquad \gamma_n \in \left]0, +\infty\right[.$$

- To secure that the solution belongs to C, we can add a projection step.
- A relaxation parameter λ_n can also be added.

We end up with the following iterative algorithm.

Algorithm 5. (Projected gradient descent)

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = x_n + \lambda_n \big(P_C(x_n - \gamma_n \nabla f(x_n)) - x_n\big)$$

where $\gamma_n \in \left]0, +\infty\right[$ *and* $\lambda_n \in]0, 1]$.

Remark 39.

i) *If, for every* $n \in \mathbb{N}$*, the relaxation parameter* λ_n *is set to 1, the following simplified form of the projected gradient algorithm is obtained*

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = P_C(x_n - \gamma_n \nabla f(x_n)).$$

ii) *When there is no constraint* ($C = \mathcal{H}$)*, the gradient algorithm is recovered:*

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = x_n - \gamma_n \nabla f(x_n).$$

Proposition 33

i) x *is a fixed point of the projected gradient iteration if and only if* $x \in C$ *and*

$$(\forall y \in C) \qquad \langle \nabla f(x) \mid y - x \rangle \geq 0.$$

ii) *When* f *is convex,* x *is a fixed point of the projected gradient iteration if and only if* x *is a global minimizer of* f *over* C.

Proof.

i) If x is a fixed point, then

$$\begin{aligned} & x = x + \lambda_n \big(P_C(x - \gamma_n \nabla f(x)) - x\big) \\ \Leftrightarrow \quad & x = P_C(x - \gamma_n \nabla f(x)). \end{aligned}$$

According to the characterization of the projection, for every $y \in C$,

$$\begin{aligned} & \langle x - \gamma_n \nabla f(x) - x \mid y - x \rangle \leq 0 \\ \Leftrightarrow \quad & \langle \nabla f(x) \mid y - x \rangle \geq 0. \end{aligned}$$

ii) According to Theorem 9, this condition is satisfied if and only if x is a global minimizer of f over C.

□

7.3 Convergence

7.3.1 Convex case

Proposition 34. (Cocoercity property)

Assume that f is convex and has a Lipschtzian gradient with constant $\beta \in \,]0,+\infty[$, i.e.

$$(\forall (x,y) \in \mathcal{H}^2) \quad \|\nabla f(x) - \nabla f(y)\| \leq \beta \|x-y\|.$$

Then

$$(\forall (x,y) \in \mathcal{H}^2) \quad \beta \langle \nabla f(x) - \nabla f(y) \mid x-y \rangle \geq \|\nabla f(x) - \nabla f(y)\|^2.$$

Proof. This follows from the Baillon-Haddad theorem. □

We will also need the following technical result

Lemma 4

Let $f \in \Gamma_0(\mathcal{H})$ be a Gâteaux differentiable. Let $(x_n)_{n\in\mathbb{N}}$ be a sequence of $\mathcal{H}$ such that $x_n \rightharpoonup \widehat{x}$ and $\nabla f(x_n) \to u$, where $(\widehat{x}, u) \in \mathcal{H}^2$. Then $u = \nabla f(\widehat{x})$.

Theorem 23

Assume that f is convex and has a Lipschtzian gradient with constant $\beta \in \,]0,+\infty[$.
Assume that $D = \operatorname{Argmin}_{x\in C} f(x) \neq \varnothing$.
Assume that $\inf_{n\in\mathbb{N}} \gamma_n > 0$, $\sup_{n\in\mathbb{N}} \gamma_n < 2/\beta$, $\inf_{n\in\mathbb{N}} \lambda_n > 0$, and $\sup_{n\in\mathbb{N}} \lambda_n \leq 1$.
Then the sequence $(x_n)_{n\in\mathbb{N}}$ generated by Algorithm 5

i) *is Fejér monotone with respect to D, i.e.*

$$(\forall x \in D)(\forall n \in \mathbb{N}) \quad \|x_{n+1} - x\| \leq \|x_n - x\|;$$

ii) *converges weakly to a minimizer of f over C.*

Proof.

i) Let $x \in D$. It follows from Proposition 33ii) that

$$x = P_C(x - \gamma_n \nabla f(x)).$$

Then, by using the nonexpansiveness of the projection onto C, for every $n \in \mathbb{N}$,

$$\begin{aligned}
&\|P_C(x_n - \gamma_n \nabla f(x_n)) - x\|^2 \\
&= \|P_C(x_n - \gamma_n \nabla f(x_n)) - P_C(x - \gamma_n \nabla f(x))\|^2 \\
&\leq \|x_n - \gamma_n \nabla f(x_n) - x + \gamma_n \nabla f(x)\|^2 \\
&= \|x_n - x\|^2 - 2\gamma_n \langle \nabla f(x_n) - \nabla f(x) \mid x_n - x \rangle + \gamma_n^2 \|\nabla f(x_n) - \nabla f(x)\|^2.
\end{aligned}$$

Using Proposition 34 leads to

$$\begin{aligned}
&\|P_C(x_n - \gamma_n \nabla f(x_n)) - x\|^2 \\
&\leq \|x_n - x\|^2 - 2\gamma_n \beta^{-1} \|\nabla f(x_n) - \nabla f(x)\|^2 + \gamma_n^2 \|\nabla f(x_n) - \nabla f(x)\|^2 \\
&= \|x_n - x\|^2 - \gamma_n(2\beta^{-1} - \gamma_n) \|\nabla f(x_n) - \nabla f(x)\|^2 \leq \|x_n - x\|^2.
\end{aligned}$$

We deduce that

$$\begin{aligned}
\|x_{n+1} - x\| &\leq (1 - \lambda_n)\|x_n - x\| + \lambda_n \|P_C(x_n - \gamma_n \nabla f(x_n)) - x\| \\
&\leq \|x_n - x\|.
\end{aligned}$$

This shows that $(x_n)_{n \in \mathbb{N}}$ is Fejér-monotone with respect to D.

ii) This is a technical consequence of the Fejér-monotonicity, whose details are given hereafter.

For every $n \in \mathbb{N}$, define

$$(\forall x \in \mathcal{H}) \quad R_n(x) = x - \gamma_n \nabla f(x).$$

Since P_C is firmly nonexpansive, it follows from Remark 10ii) that, for every $n \in \mathbb{N}$ and $x \in D$,

$$\begin{aligned}
&\|P_C(x_n - \gamma_n \nabla f(x_n)) - x\|^2 \\
&= \|P_C(R_n(x_n)) - P_C(R_n(x))\|^2 \\
&\leq \|R_n(x_n) - R_n(x)\|^2 - \|R_n(x_n) - R_n(x) - P_C(R_n(x_n)) + P_C(R_n(x))\|^2.
\end{aligned}$$

Similarly to the majorization we performed in the proof of i), by using the cocoercivity of ∇f, we have

$$\|R_n(x_n) - R_n(x)\|^2 \leq \|x_n - x\|^2 - \gamma_n(2\beta^{-1} - \gamma_n)\|\nabla f(x_n) - \nabla f(x)\|^2.$$

We deduce that

$$\begin{aligned}
&\|P_C(x_n - \gamma_n \nabla f(x_n)) - x\|^2 \\
&\leq \|x_n - x\|^2 - \gamma_n(2\beta^{-1} - \gamma_n)\|\nabla f(x_n) - \nabla f(x)\|^2 - \|R_n(x_n) - P_C(R_n(x_n)) + \gamma_n \nabla f(x)\|^2.
\end{aligned}$$

By using the convexity of the squared norm,

$$\begin{aligned}
&\|x_{n+1}-x\|^2\\
&\le (1-\lambda_n)\|x_n-x\|^2+\lambda_n\|P_C(x_n-\gamma_n\nabla f(x_n))-x\|^2\\
&\le \|x_n-x\|^2-\lambda_n\gamma_n(2\beta^{-1}-\gamma_n)\|\nabla f(x_n)-\nabla f(x)\|^2\\
&\quad-\lambda_n\|x_n-\gamma_n\nabla f(x_n)-P_C(x_n-\gamma_n\nabla f(x_n))+\gamma_n\nabla f(x)\|^2\\
&\le \|x_n-x\|^2-\underline{\lambda}\underline{\gamma}(2\beta^{-1}-\overline{\gamma})\|\nabla f(x_n)-\nabla f(x)\|^2\\
&\quad-\underline{\lambda}\|x_n-\gamma_n\nabla f(x_n)-P_C(x_n-\gamma_n\nabla f(x_n))+\gamma_n\nabla f(x)\|^2, \qquad (7.1)
\end{aligned}$$

where

$$\begin{aligned}
\underline{\gamma} &= \inf_{n\in\mathbb{N}}\gamma_n>0\\
\overline{\gamma} &= \sup_{n\in\mathbb{N}}\gamma_n<2/\beta \qquad (7.2)\\
\underline{\lambda} &= \inf_{n\in\mathbb{N}}\lambda_n>0.
\end{aligned}$$

Since $(x_n)_{n\in\mathbb{N}}$ is Fejèr monotone with respect to D, $(\|x_n-x\|)_{n\in\mathbb{N}}$ converges and it follows from (7.1) that

$$\begin{aligned}
&\|\nabla f(x_n)-\nabla f(x)\|^2\to 0\\
&\|x_n-\gamma_n\nabla f(x_n)-P_C(x_n-\gamma_n\nabla f(x_n))+\gamma_n\nabla f(x)\|^2\to 0.
\end{aligned}$$

Since $(\gamma_n)_{n\in\mathbb{N}}$ is bounded, this is equivalent to

$$\nabla f(x_n)\to\nabla f(x) \qquad (7.3)$$

$$x_n-p_n\to 0, \qquad (7.4)$$

where $p_n=P_C(x_n-\gamma_n\nabla f(x_n))$. In addition, from the characterization of the projection,

$$\begin{aligned}
(\forall y\in C)\qquad &\langle x_n-\gamma_n\nabla f(x_n)-p_n\mid y-p_n\rangle\le 0\\
\Leftrightarrow\ &\gamma_n^{-1}\langle x_n-p_n\mid y-p_n\rangle\le\langle\nabla f(x_n)\mid y-p_n\rangle\,. \qquad (7.5)
\end{aligned}$$

Let $\widehat{x}$ be a weak cluster point of $(x_n)_{n\in\mathbb{N}}$. There exists a subsequence $(x_{n_k})_{k\in\mathbb{N}}$ converging weakly to $\widehat{x}$. Because of (7.4), $x_{n_k}-p_{n_k}\to 0$ and $p_{n_k}\rightharpoonup\widehat{x}$, which entails that $(y-p_{n_k})_{k\in\mathbb{N}}$ is bounded. Since $\{\gamma_{n_k}\}_{k\in\mathbb{N}}\subset[\underline{\gamma},\overline{\gamma}]$,

$$\gamma_{n_k}^{-1}\langle x_{n_k}-p_{n_k}\mid y-p_{n_k}\rangle\to 0. \qquad (7.6)$$

According to (7.3) and Lemma 4, as $x_{n_k}\rightharpoonup\widehat{x}$ and $\nabla f(x_{n_k})\to\nabla f(x)$, $\nabla f(x)=\nabla f(\widehat{x})$. We thus deduce from (7.5) and (7.6) that

$$\langle\nabla f(\widehat{x})\mid y-\widehat{x}\rangle\ge 0.$$

This shows that Euler's inequality is satisfied by $\widehat{x}$ and it is thus a minimizer of f over C. Theorem 12 allows us to conclude that $(x_n)_{n\in\mathbb{N}}$ converges weakly to a point in D.

□

7.3.2 Convergence of function values

Proposition 35. (Descent lemma)

Assume that f is Gâteaux differentiable and has a β-Lipschtzian gradient with $\beta \in]0,+\infty[$. Then,

$$\big(\forall (x,y) \in \mathcal{H}^2\big) \quad f(y) \leq f(x) + \langle y-x \mid \nabla f(x)\rangle + \frac{\beta}{2}\|y-x\|^2.$$

Proof. For every $(x,y) \in \mathcal{H}^2$ and $t \in \mathbb{R}$, let $\varphi(t) = f(x+t(y-x))$.
φ is differentiable and $\varphi'(t) = \langle y-x \mid \nabla f(x+t(y-x))\rangle$. We have then

$$\begin{aligned}
& \varphi(1)-\varphi(0) = \int_0^1 \varphi'(t)dt \\
\Leftrightarrow\ & f(y)-f(x) = \int_0^1 \langle y-x \mid \nabla f(x+t(y-x))\rangle\, dt \\
\Leftrightarrow\ & f(y)-f(x)-\langle y-x \mid \nabla f(x)\rangle = \int_0^1 \langle y-x \mid \nabla f(x+t(y-x)) - \nabla f(x)\rangle\, dt.
\end{aligned}$$

In addition, according to the Cauchy-Schwarz inequality,

$$\begin{aligned}
& \langle y-x \mid \nabla f(x+t(y-x)) - \nabla f(x)\rangle \\
\leq\ & \|y-x\|\|\nabla f(x+t(y-x)) - \nabla f(x)\| \\
\leq\ & t\beta\|y-x\|^2.
\end{aligned}$$

This leads to $f(y) \leq f(x) + \langle y-x \mid \nabla f(x)\rangle + \frac{\beta}{2}\|y-x\|^2$. □

Theorem 24

Assume that f is Gâteaux differentiable and has a β-Lipschtzian gradient with $\beta \in]0,+\infty[$.
Assume that $\mu = \inf_{x\in C} f(x) > -\infty$.
Assume that $\sup_{n\in\mathbb{N}} \gamma_n < 2/\beta$, and $(\forall n \in \mathbb{N})\ \lambda_n = 1$.
Then the sequence $(x_n)_{n\in\mathbb{N}}$ generated by Algorithm 5 is such that

i) $\big(f(x_n)\big)_{n\in\mathbb{N}}$ *is a decaying convergent sequence*

ii) $\displaystyle\sum_{n=0}^{N-1} \|x_{n+1}-x_n\|^2 \leq +\infty.$

Proof.

i) Let $n \geq 1$. According to the descent lemma,

$$f(x_{n+1}) \leq f(x_n) + \langle \nabla f(x_n) \mid x_{n+1}-x_n\rangle + \frac{\beta}{2}\|x_{n+1}-x_n\|^2$$

Since $x_{n+1} = P_C(x_n - \gamma_n \nabla f(x_n))$ and $x_n \in C$, it follows from the characterization of P_C that

$$\begin{aligned} & \langle x_n - \gamma_n \nabla f(x_n) - x_{n+1} \mid x_n - x_{n+1}\rangle \le 0 \\ \Leftrightarrow \quad & \langle \nabla f(x_n) \mid x_{n+1} - x_n\rangle \le -\frac{1}{\gamma_n}\|x_{n+1} - x_n\|^2. \end{aligned}$$

Therefore,

$$f(x_{n+1}) \le f(x_n) + \Big(\frac{\beta}{2} - \gamma_n^{-1}\Big)\|x_{n+1} - x_n\|^2 \le f(x_n). \tag{7.7}$$

Since $(f(x_n))_{n\in\mathbb{N}}$ is a decaying sequence, lower bounded by μ, it converges.

ii) Let $\overline{\gamma}$ be defined by (7.2). We deduce from (7.7) that

$$(\forall n \in \mathbb{N}) \quad \Big(\overline{\gamma}^{-1} - \frac{\beta}{2}\Big)\|x_{n+1} - x_n\|^2 \le f(x_n) - f(x_{n+1}). \tag{7.8}$$

For every $N \in \mathbb{N} \setminus \{0\}$, we have then

$$0 \le \Big(\overline{\gamma}^{-1} - \frac{\beta}{2}\Big)\sum_{n=0}^{N-1}\|x_{n+1} - x_n\|^2 \le f(x_0) - f(x_N).$$

The summability result follows from the fact that $f(x_N)$ converges as $N \to +\infty$.

□

Remark 40.

i) *If f is convex, the same result holds when $(\forall n \in \mathbb{N})$ $\lambda_n \in]0,1]$.*
In addition, under the assumptions of Theorem 23, $f(x_n) \to \mu$.

Proof. We have, for every $n \in \mathbb{N}$,

$$x_{n+1} = (1 - \lambda_n)x_n + \lambda_n \widetilde{x}_n \tag{7.9}$$

where

$$\widetilde{x}_n = P_C(x_n - \gamma_n \nabla f(x_n)).$$

By proceeding as in the previous proof,

$$f(\widetilde{x}_n) \le f(x_n) + \Big(\frac{\beta}{2} - \gamma_n^{-1}\Big)\|\widetilde{x}_n - x_n\|^2.$$

By using the convexity of f, we deduce from (7.9) that

$$\begin{aligned} f(x_{n+1}) &\le (1 - \lambda_n)f(x_n) + \lambda_n f(\widetilde{x}_n) \\ &\le f(x_n) + \lambda_n\Big(\frac{\beta}{2} - \gamma_n^{-1}\Big)\|\widetilde{x}_n - x_n\|^2 \\ &\le f(x_n) + \frac{1}{\lambda_n}\Big(\frac{\beta}{2} - \gamma_n^{-1}\Big)\|x_{n+1} - x_n\|^2 \le f(x_n). \end{aligned}$$

Since $\lambda_n \le 1$, (7.8) is also satisfied.
If $\mathcal{H}$ is finite dimensional, it follows from Theorem 23 that $x_n \to \widehat{x}$. We deduce from the continuity of f that $\lim_{n\to+\infty} f(x_n) = \mu$. This property remains valid in the infinite dimensional case, but the proof is more technical. □

ii) *Suppose that the assumptions of Theorem 24 hold and* $(\gamma_n)_{n\in\mathbb{N}}$ *is a convergent sequence. Then any cluster point of* $(x_n)_{n\in\mathbb{N}}$ *satisfies Euler's inequality.*

Proof. Let $\widehat{x}$ be a cluster point of $(x_n)_{n\in\mathbb{N}}$. There exists a subsequence $(x_{n_k})_{k\in\mathbb{N}}$ of $(x_n)_{n\in\mathbb{N}}$ such that $x_{n_k} \to \widehat{x}$. According to Theorem 24ii),

$$P_C\big(x_{n_k} - \gamma_{n_k}\nabla f(x_{n_k})\big) - x_{n_k} = x_{n_k+1} - x_{n_k} \to 0.$$

Since $\lim_{n\to+\infty}\gamma_n = \gamma_\infty > 0$, P_C is continuous, and f is continuously differentiable,

$$P_C\big(\widehat{x} - \gamma_\infty\nabla f(\widehat{x})\big) = \widehat{x}.$$

We deduce from Proposition 33 that

$$(\forall y \in C) \qquad \langle \nabla f(\widehat{x}) \mid y - \widehat{x}\rangle \geq 0.$$

□

iii) *In the nonconvex case, there is no guarantee that the limit of* $\big(f(x_n)\big)_{n\in\mathbb{N}}$ *is* μ *since the iterates may get stuck in a spurious local minimum.*

7.4 Preconditioning

7.4.1 Metric change

Definition 46

Let $A \in \mathcal{B}(\mathcal{H},\mathcal{H})$ be a self-adjoint operator which is strongly positive, i.e. there exists $\alpha \in \,]0,+\infty[$ such that

$$(\forall x \in \mathcal{H}) \qquad \langle x \mid Ax\rangle \geq \alpha\|x\|^2.$$

The inner product induced by A is

$$(\forall (x,y) \in \mathcal{H}^2) \qquad \langle x \mid y\rangle_A = \langle x \mid Ay\rangle\,.$$

Remark 41.

i) *When* $\mathcal{H} = \mathbb{R}^N$, $A \in \mathbb{R}^{N\times N}$ *is strongly positive if and only if* A *is symmetric positive definite.*

ii) *If* $A \in \mathcal{B}(\mathcal{H},\mathcal{H})$ *is strongly positive, then* A *is invertible.*

Proposition 36

Let $A \in \mathcal{B}(\mathcal{H}, \mathcal{H})$ be a strongly positive self-adjoint operator.
Let $f : \mathcal{H} \to \mathbb{R}$ be a Gâteaux differentiable function.
In the Hilbert space $(\mathcal{H}, \langle \cdot \mid \cdot \rangle_A)$, the gradient of f is $\nabla_A f = A^{-1} \nabla f$.

Proof. The Gâteaux differential is such that

$$(\forall (x, y) \in \mathcal{H}^2) \qquad \begin{aligned} f'(x)y &= \langle \nabla f(x) \mid y \rangle \\ &= \big\langle A A^{-1} \nabla f(x) \mid y \big\rangle \\ &= \big\langle A^{-1} \nabla f(x) \mid Ay \big\rangle \\ &= \big\langle A^{-1} \nabla f(x) \mid y \big\rangle_A . \end{aligned}$$

□

7.4.2 Newton algorithm

In this section, we will focus on the unconstrained case when $C = \mathcal{H}$.

Proposition 37

Let $A \in \mathcal{B}(\mathcal{H}, \mathcal{H})$ be a strongly positive self-adjoint operator.
Let $f : \mathcal{H} \to \mathbb{R}$ be a Gâteaux differentiable function.
In the Hilbert space $(\mathcal{H}, \langle \cdot \mid \cdot \rangle_A)$, the gradient descent algorithm for minimizing f reads

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = x_n - \gamma_n A^{-1} \nabla f(x_n)$$

with $\gamma_n \in \,]0, +\infty[$.

Proof. In $(\mathcal{H}, \langle \cdot \mid \cdot \rangle_A)$, the gradient descent algorithm has the following form:

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = x_n - \gamma_n \nabla_A f(x_n).$$

The result then follows from Proposition 36. □

A more general form of algorithm is derived by replacing $\gamma_n^{-1} A$ by a variable metric operator $\widetilde{A}_n$ depending on iteration $n \in \mathbb{N}$.

Algorithm 6. (Quasi-Newton)

Let $(\widetilde{A}_n)_{n \in \mathbb{N}}$ be a sequence of strongly positive self-adjoint operators in $\mathcal{B}(\mathcal{H}, \mathcal{H})$.

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = x_n - \widetilde{A}_n^{-1} \nabla f(x_n).$$

Remark 42.

i) *By being more flexible, this algorithm may lead to a faster convergence by a suitable choice of* $(\widetilde{A}_n)_{n\in\mathbb{N}}$.

ii) *If* f *is twice Fréchet differentiable and its Hessian is strongly positive on* $\mathcal{H}$, *one can choose*

$$(\forall n \in \mathbb{N}) \quad \widetilde{A}_n = \nabla^2 f(x_n).$$

We then obtain ***Newton's method****.*

iii) *Alternatively, Newton's method can be derived from a second-order Taylor expansion of* f *around* x_n *at iteration* $n \in \mathbb{N}$*:*

$$x_{n+1} = \underset{x\in\mathcal{H}}{\operatorname{argmin}} \; f(x_n) + \langle \nabla f(x_n) \mid x - x_n\rangle + \frac{1}{2}\left\langle x - x_n \mid \nabla^2 f(x_n)(x - x_n)\right\rangle.$$

Indeed, since $\nabla^2 f(x_n)$ *is strongly positive, the second-order approximation defines a strictly convex function which admits a unique minimizer satisfying the first-order optimality condition:*

$$\nabla f(x_n) + \nabla^2 f(x_n)(x_{n+1} - x_n) = 0.$$

7.5 Related algorithms

We now briefly describe two examples of optimization approaches which make use of the first-order strategies described in the previous sections.

7.5.1 Uzawa algorithm

Problem 14

Let $\mathcal{L}\colon \mathcal{H} \times \mathbb{R}^q \to \mathbb{R}$ be differentiable with respect to its second argument. We want to find a saddle point of $\mathcal{L}$ over $\mathcal{H} \times [0, +\infty[^q$.

The proposed method called **Uzawa algorithm** alternates between a minimization with respect to the first variable and a projected gradient ascent step with respect to the second one. This second step uses stepsizes $(\gamma_n)_{n\in\mathbb{N}}$ and relaxation parameters $(\rho_n)_{n\in\mathbb{N}}$.

Algorithm 7

$$
\begin{array}{l}
\textit{Set } \lambda_0 \in [0,+\infty[^q \\
\textit{For } n = 0,1,\ldots \\
\left\lfloor
\begin{array}{l}
\textit{Set } \gamma_n \in \,]0,+\infty[,\ \rho_n \in]0,1] \\
x_n \in \operatorname{Argmin} \mathcal{L}(\cdot,\lambda_n) \\
\lambda_{n+1} = \lambda_n + \rho_n\big(P_{[0,+\infty[^q}(\lambda_n + \gamma_n \nabla_\lambda \mathcal{L}(x_n,\lambda_n)) - \lambda_n\big).
\end{array}
\right.
\end{array}
$$

7.5.2 DC programming

Problem 15

Let $f \in \Gamma_0(\mathcal{H})$ and let $g \in \Gamma_0(\mathcal{H})$.
We want to minimize the difference of convex functions $f - g$.

We say that we have to solve a DC (for difference of convex functions) problem. This one is equivalent to

$$
\begin{aligned}
& \underset{x\in\mathcal{H}}{\text{minimize}} \;\; f(x) - \sup_{v\in\mathcal{H}}(\langle x \mid v\rangle - g^*(v)) \\
\Leftrightarrow \quad & \underset{(x,v)\in\mathcal{H}^2}{\text{minimize}} \;\; f(x) - \langle x \mid v\rangle + g^*(v). \qquad (7.10)
\end{aligned}
$$

Although the resulting cost function is not convex, when one of its two variables is given, we obtain a convex function with respect to the other one. If f and g^* are Gâteaux differentiable, (7.10) suggests to alternate between gradient descent steps with respect to each of the variables.

Algorithm 8

$$
\begin{array}{l}
\textit{Set } (x_0,v_0) \in \mathcal{H}^2 \\
\textit{For } n = 0,1,\ldots \\
\left\lfloor
\begin{array}{l}
\textit{Set } \gamma_n \in \,]0,+\infty[,\ \mu_n \in \,]0,+\infty[\\
x_{n+1} = x_n - \gamma_n(\nabla f(x_n) - v_n) \\
v_{n+1} = v_n - \mu_n(\nabla g^*(v_n) - x_{n+1}).
\end{array}
\right.
\end{array}
$$

Appendix A

Result of convex geometry

Theorem 25. (Hyperplane separation)

Let C be a convex set of a real Hilbert space $\mathcal{H}$ and let $y \in \mathcal{H}$.

i) *Assume that C is nonempty closed and $y \notin C$. There exists a closed half-space D such that $C \subset D$ and $y \notin D$, i.e. there exists $v \in \mathcal{H} \setminus \{0\}$ and $\alpha \in \mathbb{R}$ such that*

$$(\forall x \in C) \quad \langle v \mid x \rangle \geq \alpha > \langle v \mid y \rangle .$$

ii) *Assume that C has a nonempty interior and $y \notin \operatorname{int}(C)$. There exists a closed hyperplane containing y which delimits a closed half-space including C, i.e. there exists $v \in \mathcal{H} \setminus \{0\}$ such that*

$$(\forall x \in C) \quad \langle v \mid x \rangle \geq \langle v \mid y \rangle .$$

Proof.

i) Let $\widehat{y}$ be the projection of y onto C. We have

$$(\forall x \in C) \quad \langle x - \widehat{y} \mid y - \widehat{y} \rangle \leq 0.$$

Set $v = \widehat{y} - y$. Since $y \notin C$, $v \neq 0$. Then

$$\begin{aligned}(\forall x \in C) \quad \langle v \mid x \rangle &\geq \langle v \mid \widehat{y} \rangle \\ &= \langle v \mid \widehat{y} - y \rangle + \langle v \mid y \rangle \\ &= \|v\|^2 + \langle v \mid y \rangle .\end{aligned}$$

We deduce that

$$(\forall x \in C) \quad \langle v \mid x \rangle \geq \alpha = \inf_{x' \in C} \langle v \mid x' \rangle \geq \|v\|^2 + \langle v \mid y \rangle > \langle v \mid y \rangle .$$

ii) This is a known geometrical consequence of the Hahn-Banach theorem.

$\square$

Appendix B

Proof of Lemma 3

Step 1: When $f \in \Gamma_0(\mathcal{H})$, there exists $x_0 \in \mathcal{H}$ such that $f(x_0) < +\infty$. Let $\eta_0 \in]-\infty, f(x_0)[$, then $(x_0, \eta_0) \notin \operatorname{epi} f$. In addition $\operatorname{epi} f$ is a closed convex set of $\mathcal{H} \times \mathbb{R}$. By invoking Theorem 25ii), in the Hilbert space $\mathcal{H} \times \mathbb{R}$, there thus exist $(v_0, \nu_0) \in \mathcal{H} \times \mathbb{R} \setminus \{(0,0)\}$ and $\alpha_0 \in \mathbb{R}$ such that

$$(\forall (x,\eta) \in \operatorname{epi} f) \quad \langle v_0 \mid x \rangle + \nu_0 \eta \geq \alpha_0 > \langle v_0 \mid x_0 \rangle + \nu_0 \eta_0.$$

In particular, if $x = x_0$, and $\eta \geq f(x_0)$, then $\langle v_0 \mid x_0 \rangle + \nu_0 \eta > \langle v_0 \mid x_0 \rangle + \nu_0 \eta_0$, which implies that $\nu_0 > 0$ (if $\nu_0 < 0$, the left-hand side can made abitrary small when $\eta \to +\infty$ and if $\nu_0 = 0$, the strict inequality is impossible). By defining $\widetilde{v}_0 = -v_0/\nu_0$ and $\widetilde{\alpha}_0 = \alpha_0/\nu_0$, it follows that

$$(\forall (x,\eta) \in \operatorname{epi} f) \quad - \langle \widetilde{v}_0 \mid x \rangle + \eta \geq \widetilde{\alpha}_0. \tag{B.1}$$

and consequently, by setting $\eta = f(x)$,

$$(\forall x \in \operatorname{dom} f) \quad f(x) \geq \widetilde{\alpha}_0 + \langle \widetilde{v}_0 \mid x \rangle .$$

The inequality obviously remains valid if $x \notin \operatorname{dom} f$. This shows that f is lower bounded by an affine function.

Step 2: Let $\widetilde{f} = \sup_{a \in \mathcal{A}_f} a$. To establish that $f = \widetilde{f}$, we will show that $\operatorname{epi}(f) = \operatorname{epi}(\widetilde{f})$.
For every $a \in \mathcal{A}_f$, $a \leq f \Rightarrow \widetilde{f} \leq f$. Consequently, $\operatorname{epi}(f) \subset \operatorname{epi}(\widetilde{f})$.
Let us show that the converse inclusion holds or, by contraposition, let us assume that $(x_1, \eta_1) \notin \operatorname{epi}(f)$ and show that $(x_1, \eta_1) \notin \operatorname{epi}(\widetilde{f})$.
As previously, there exist $(v_1, \nu_1) \in \mathcal{H} \times \mathbb{R} \setminus \{(0,0)\}$ and $\alpha_1 \in \mathbb{R}$ such that

$$(\forall (x,\eta) \in \operatorname{epi} f) \quad \langle v_1 \mid x \rangle + \nu_1 \eta \geq \alpha_1 > \langle v_1 \mid x_1 \rangle + \nu_1 \eta_1$$

and $\nu_1 \geq 0$.
Case when $\nu_1 > 0$: f is lower-bounded by $a = \widetilde{\alpha}_1 + \langle \widetilde{v}_1 \mid \cdot \rangle$ with $\widetilde{v}_1 = -v_1/\nu_1$ and $\widetilde{\alpha}_1 = \alpha_1/\nu_1$, and $(x_1, \eta_1) \notin \operatorname{epi}(a)$.
Case when $\nu_1 = 0$: Then

$$(\forall (x,\eta) \in \operatorname{epi} f) \quad \langle v_1 \mid x \rangle \geq \alpha_1 > \langle v_1 \mid x_1 \rangle .$$

Let $v_2 = \delta v_1 - \widetilde{v}_0$ with $\delta \in [0, +\infty[$. Then, it follows from (B.1) that

$$(\forall (x,\eta) \in \operatorname{epi} f) \quad \langle v_2 \mid x \rangle + \eta \geq \widetilde{\alpha}_0 + \delta \alpha_1.$$

Thus, f is lower-bounded by $a = \widetilde{\alpha}_0 + \delta\alpha_1 - \langle v_2 \mid \cdot \rangle$ and

$$\begin{aligned} a(x_1) > \eta_1 &\Leftrightarrow \widetilde{\alpha}_0 + \delta\alpha_1 - \langle v_2 \mid x_1 \rangle > \eta_1 \\ &\Leftrightarrow \delta > \frac{\eta_1 - \widetilde{\alpha}_0 - \langle \widetilde{v}_0 \mid x_1 \rangle}{\alpha_1 - \langle v_1 \mid x_1 \rangle}. \end{aligned}$$

So, when δ satisfies the above condition, $(x_1, \eta_1) \notin \operatorname{epi}(a)$.
In both cases, this shows that $(x_1, \eta_1) \notin \operatorname{epi}(\widetilde{f})$.

Step 3: Assume now that $\mathcal{H}$ is of dimension $N \in \mathbb{N} \setminus \{0\}$ and $f \colon \mathcal{H} \to]-\infty, +\infty]$ is convex. Let $\overline{x} \in \operatorname{int}(\operatorname{dom} f)$. There exists a closed ball centered at $\overline{x}$ included in $\operatorname{dom} f$. Because of the equivalence of norms in finite dimension, we can consider an ℓ_1 ball:

$$\overline{B}_1(\overline{x}, \rho) = \Big\{x = (x^{(i)})_{1\leq i\leq N} \in \mathcal{H} \;\Big|\; \sum_{i=1}^N |x^{(i)} - \overline{x}^{(i)}| \leq \rho\Big\},$$

where $\rho \in \ ]0, +\infty[$ and $(x^{(i)})_{1\leq i\leq N}$ denote the components of x in some orthonormal basis $(e_i)_{1\leq i\leq N}$ of $\mathcal{H}$. $\overline{B}_1(\overline{x}, \rho)$ is an hypercube which contains $(\overline{x} \pm \rho e_i)_{1\leq i\leq N}$. In addition, for every $x \in \overline{B}_1(\overline{x}, \rho)$, there exist $(\lambda_i)_{1\leq i\leq N} \in]0, 1[^N$ with $\sum_{i=1}^N \lambda_i \leq 1$ and $(\zeta_i)_{1\leq i\leq N} \in \{-1, 1\}^N$ such that

$$\begin{aligned} & x - \overline{x} = \rho \sum_{i=1}^N \lambda_i \zeta_i e_i \\ \Leftrightarrow \quad & x = \sum_{i=1}^N \lambda_i(\rho\zeta_i e_i + \overline{x}) + \Big(1 - \sum_{i=1}^N \lambda_i\Big)\overline{x} \\ \Rightarrow \quad & f(x) \leq \sum_{i=1}^N \lambda_i f(\overline{x} + \zeta_i \rho e_i) + \Big(1 - \sum_{i=1}^N \lambda_i\Big) f(\overline{x}). \end{aligned}$$

Let $M = \sup_{1\leq i\leq N} f(\overline{x} \pm \rho e_i) < +\infty$. For every $i \in \{1, \ldots, N\}$, $f(\overline{x}) \leq \big(f(\overline{x} - \rho e_i) + f(\overline{x} + \rho e_i)\big)/2 \leq M$ and it follows that $f(x) \leq M$.

Let $B_1(\overline{x}, \rho) = \operatorname{int}(\overline{B}_1(\overline{x}, \rho))$. Since we have shown that $(\forall x \in B_1(\overline{x}, \rho) \subset \operatorname{dom} f)\ f(x) \leq M$, $B_1(\overline{x}, \rho) \times]M, +\infty[$ is an open subset of $\operatorname{epi}(f)$.
On the other hand, $(\overline{x}, f(\overline{x})) \notin \operatorname{int}(\operatorname{epi}(f))$.
By applying Theorem 25ii) to the convex set $\operatorname{epi}(f)$, there exists $(v, \nu) \in \mathcal{H} \times \mathbb{R} \setminus \{(0,0)\}$ such that

$$(\forall (x,\eta) \in \operatorname{epi} f) \quad \langle v \mid x \rangle + \nu\eta \geq \langle v \mid \overline{x} \rangle + \nu f(\overline{x}).$$

Once again, $\nu \geq 0$. If $\nu = 0$, then

$$\begin{aligned} & (\forall x \in \overline{B}_1(\overline{x}, \rho)) \quad \langle v \mid x \rangle \geq \langle v \mid \overline{x} \rangle \\ \Rightarrow \quad & (\forall i \in \{1, \ldots, N\}) \quad \langle v \mid \pm\rho e_i \rangle \geq 0 \\ \Leftrightarrow \quad & v = 0 \end{aligned}$$

which is impossible. Hence $\nu > 0$, and by setting $\widetilde{v} = -v/\nu$ and $\eta = f(x)$, we conclude that

$$(\forall x \in \operatorname{dom} f) \quad f(x) \geq a(x) = f(\overline{x}) + \langle \widetilde{v} \mid x - \overline{x} \rangle\,.$$

Appendix C

English/French glossary

English	**Français**
bipartite graph	graphe biparti
branch and bound	séparation et évaluation
closed form solution	solution explicite
closure	adhérence
convex hull	enveloppe convexe
cost function	fonction objectif, coût
countable	dénombrable
cover	couverture / recouvrement
edge of a graph	arrête d'un graphe
extended real line	droite réelle achevée
extreme point	point extrême ou extrémal
feasible problem	problème réalisable ou faisable
feasibility	compatibilité (ou faisabilité)
feasibility set	ensemble admissible
firmly nonexpansive	fermement contractant
increment	accroissement
inner product	produit scalaire
knapsack	sac à dos
linearly independent	linéairement indépendant (formant une famille libre)
lower semi-continuous (l.s.c.)	semi-continue inférieurement (s.c.i.)
mean value theorem	théorème des accroissements finis
minimizer	minimiseur
nonexpansive	contractant
pair	couple
pay-off matrix	matrice de gain
positive definite	définie positive
positive semidefinite	semi-définie positive
power set	ensemble des parties
reward function	fonction de gain / récompense
slack variable	variable d'écart
strongly convex	fortement convexe
support function	fonction d'appui
undirected graph	graphe non orienté
vertex / vertices	sommet(s)